Core Finance
Week 11

MGMP 543

Prof. Benedict Guttman-Kenney

Spring 2025

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The Penultimate Week! Today’s Topics

- Final Exam
- Options / Derivatives Recap (Week 10, Quiz 9)
- Great Recession

Final Exam

Final Exam Logistics

- **Review Session:** Reese will be going over practice questions in his zoom office hour Sunday 4 May at 4pm CT.
  - He will share practice questions in advance.
  - The zoom will be recorded and posted on canvas along with the solutions.
- **Exam is released straight after Week 12 class finishes.**
- **Exam is due 10pm Monday 12 May for Section 01 (Monday class), and due 10pm Wednesday 14 May for Section 02 (Wednesday class).**
- **Open Book:** You may use any textbook, your own notes, Excel, and any materials posted to Canvas.
- **You may not receive assistant from any person or website. Failure to abide by this will result in a zero. AI tools must not be used for this or any other graded assignments.**
- I will post an updated "Cheat Sheet" from the Midterm Exam with main formulas.

Final Exam Structure

- Exam covers all material weeks 1 to 10. More emphasis on material weeks 5 to 10.
- **3 hour exam**. 180 points. 45 questions.
  - **Part I: 15 Questions are True/False (2 points each).**
  - **Part II: 10 questions (4 points each) are True/False, but also requiring a very brief (1-2) sentence explanation.**
  - **Part III: 18 questions (4-5 points each) are calculations or require a short response no more than a few sentences** of varying difficulty. Some of these questions are building off each other. Some are standalone questions. None of these questions require long essays.
  - **Part IV: 2 longer calculation questions (12 points each)**. For full marks on these questions, you need to show your workings.
- **The last exam question lets you upload an excel or another document for partial credit. You can also upload your Excel via the ‘Final Exam Upload Workings’ assignment.**

Week 10 Recap

What Did We Learn Last Week?

What Did We Learn Last Week?

- **Uncertainty in Free Cash Flows**
  - Probability weighting cashflows (assigning weights to different outcomes)
  - Sensitivity analysis (changing assumptions)
  - Scenario analysis (considering different economic outcomes)
- **Real options:** ability to modify projects is beneficial.
  - Option to expand/contract, abandon, delay/speed-up.
  - Projects that allow more flexibility are more valuable especially if there is more uncertainty.
- **Financial Options**
  - Contract giving owner the right, not obligation, to buy/sell (Exercise the Option) at a fixed price (Strike/Exercise Price) on/before a given day (Expiration Date). Writer of option has to buy/sell if owner exercises option.
  - European/American Options
  - Call Options: Option to buy (Call Me, May Buy). Value to owner: $C = \max(S_T - K, 0)$
  - Put Options: Option to sell. Value to owner: $P = \max(K - S_T, 0)$
  - Put-Call Parity: $C + PV(K) = P + S$
- **Derivatives**
  - Futures: standardized contracts obligating to buy/sell asset at given price on given day.
  - Forwards: bespoke contracts obligating to buy/sell asset asset at given price on given day.
  - Swaps: contract agreeing exchange of cash flows related to an asset.
- "Long" position ("bullish") if want price to go up. "Short" position ("Bearish") if want price to go down.

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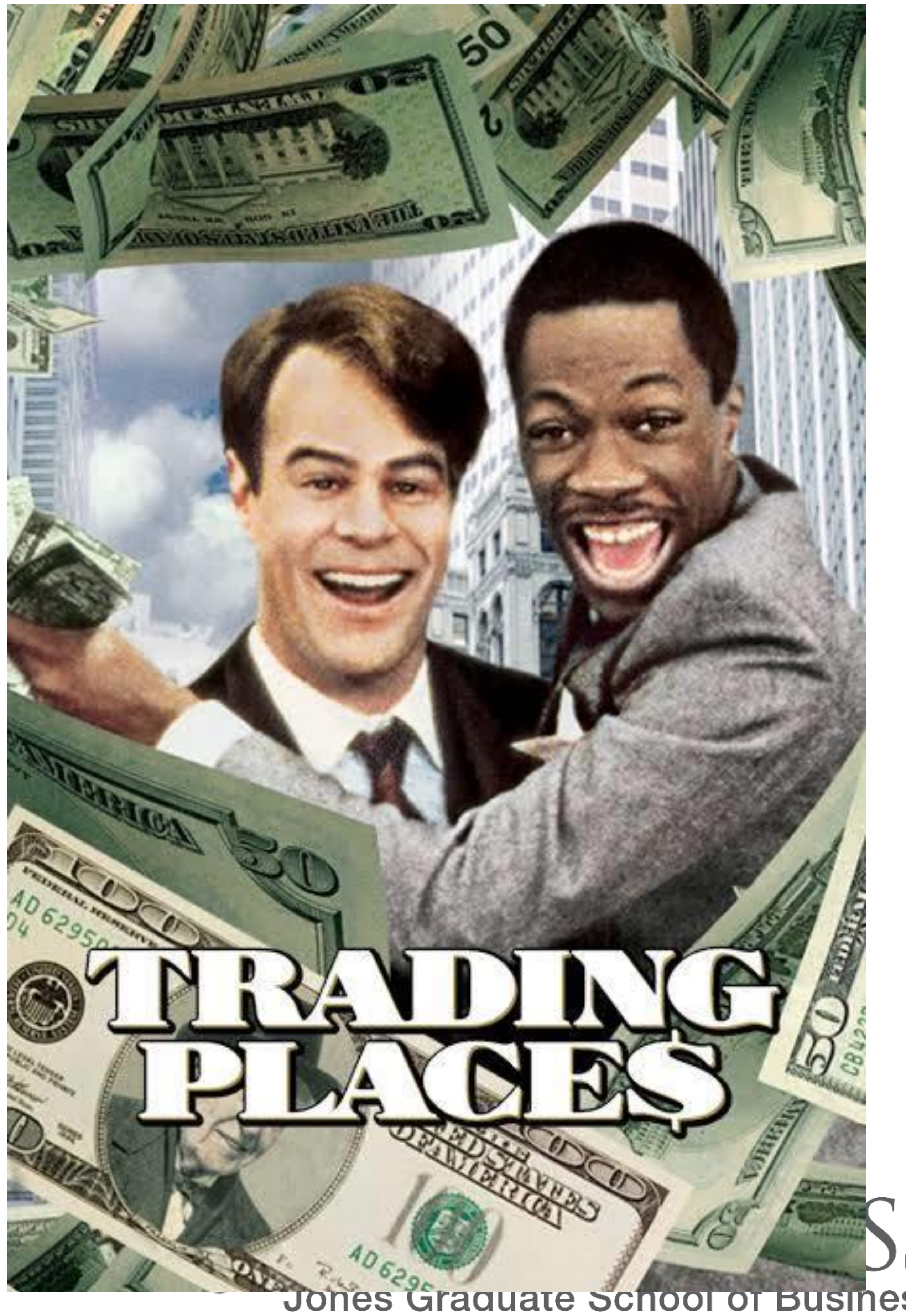
Examples Trading Derivatives

https://www.youtube.com/watch?v=d80ahvRSV8E



The Last Quiz:
Quiz 9

Q2. An option that cannot be exercised any time before its expiration date is called a(n)

- European Option
- Put Option
- Call Option
- American Option

Q2. An option that cannot be exercised any time before its expiration date is called a(n)

- **European Option**
- Put Option
- Call Option
- American Option

Q3. A put option gives the owner the right

- And the obligation to sell an asset at a given price
- But not the obligation to buy an asset at a given price
- But not the obligation to sell an asset at a given price
- And the obligation to buy an asset at a given price

Q3. A put option gives the owner the right

- And the obligation to sell an asset at a given price
- But not the obligation to buy an asset at a given price
- **But not the obligation to sell an asset at a given price**
- And the obligation to buy an asset at a given price

Q4. Suppose you buy a call and lend the present value of its exercise price. You could match the payoffs of this strategy by

- Buying the underlying stock and selling a put
- Selling a put and lending the present value of the exercise price
- Buying the underlying stock and selling a call
- Buying the underlying stock and buying a put

Q4. Suppose you buy a call and lend the present value of its exercise price. You could match the payoffs of this strategy by

- Buying the underlying stock and selling a put
- Selling a put and lending the present value of the exercise price
- Buying the underlying stock and selling a call
- **Buying the underlying stock and buying a put**

Q5. Suppose an investor buys one share of stock and a put option on the stock. What will be the value of her investment on the final exercise date if the stock price is below the exercise price? (Ignore transaction costs.)

- The value of two shares of stock
- The value of one share of stock plus the exercise price
- The exercise price
- The value of one share of stock minus the exercise price

Q5. Suppose an investor buys one share of stock and a put option on the stock. What will be the value of her investment on the final exercise date if the stock price is below the exercise price? (Ignore transaction costs.)

- The value of two shares of stock
- The value of one share of stock plus the exercise price
- **The exercise price**
- The value of one share of stock minus the exercise price

Working Out
Options Strategies

Reminder: Payoffs of Different Assets & Options

Steps to Working Out Options Strategies

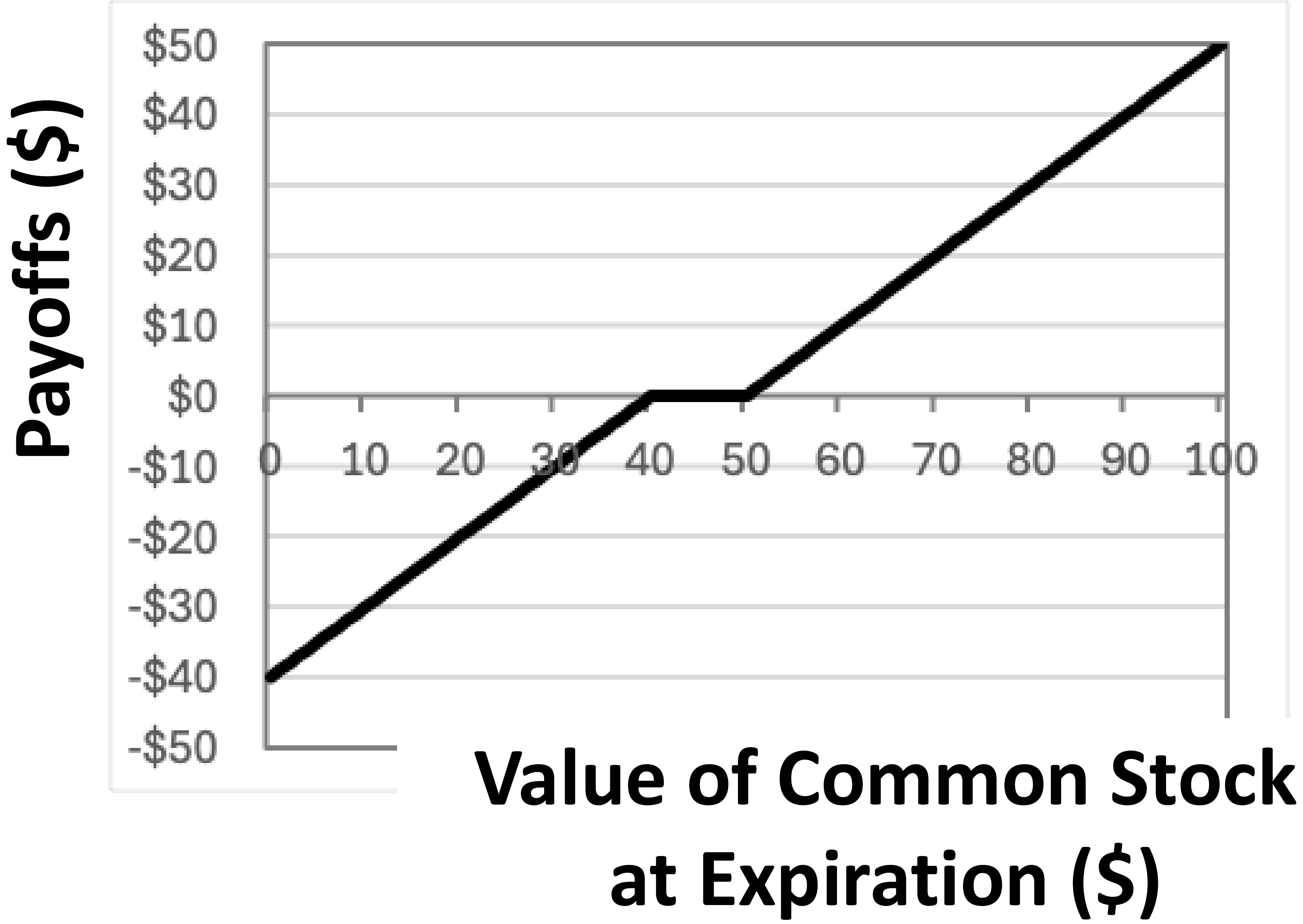
1. Find kink points
2. Assess slopes

Example 1

Steps

1. Find kink points
2. Assess slopes

Find strategy that recreates the payoff below:



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Example 1

Steps

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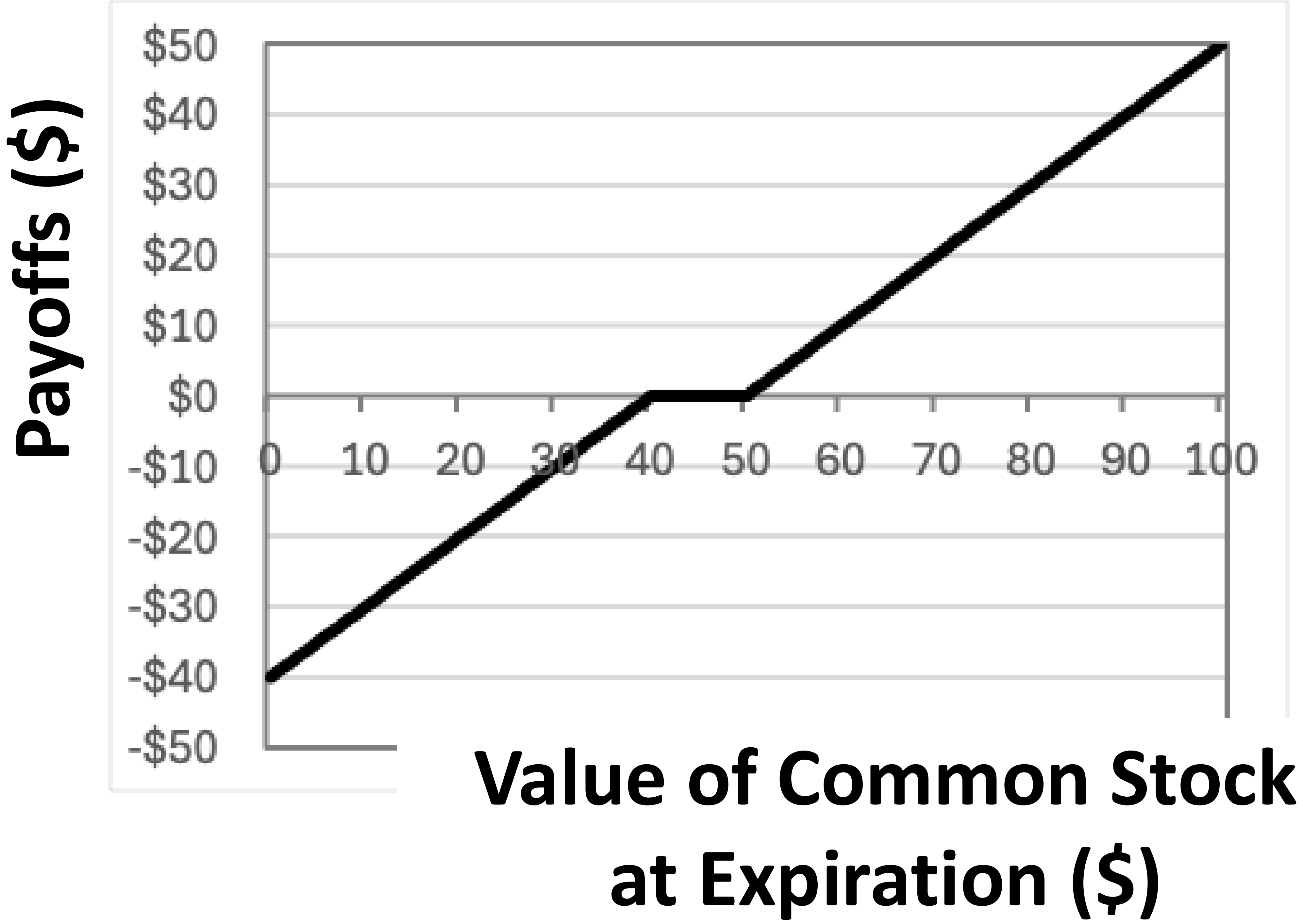


Step 1:
Kink Point 1 at $40
Kink Point 2 at $50

Step 2:
Slope is +1 from $0 to $40
Slope is 0 from $40 to $50
Slope is +1 from $50 onwards

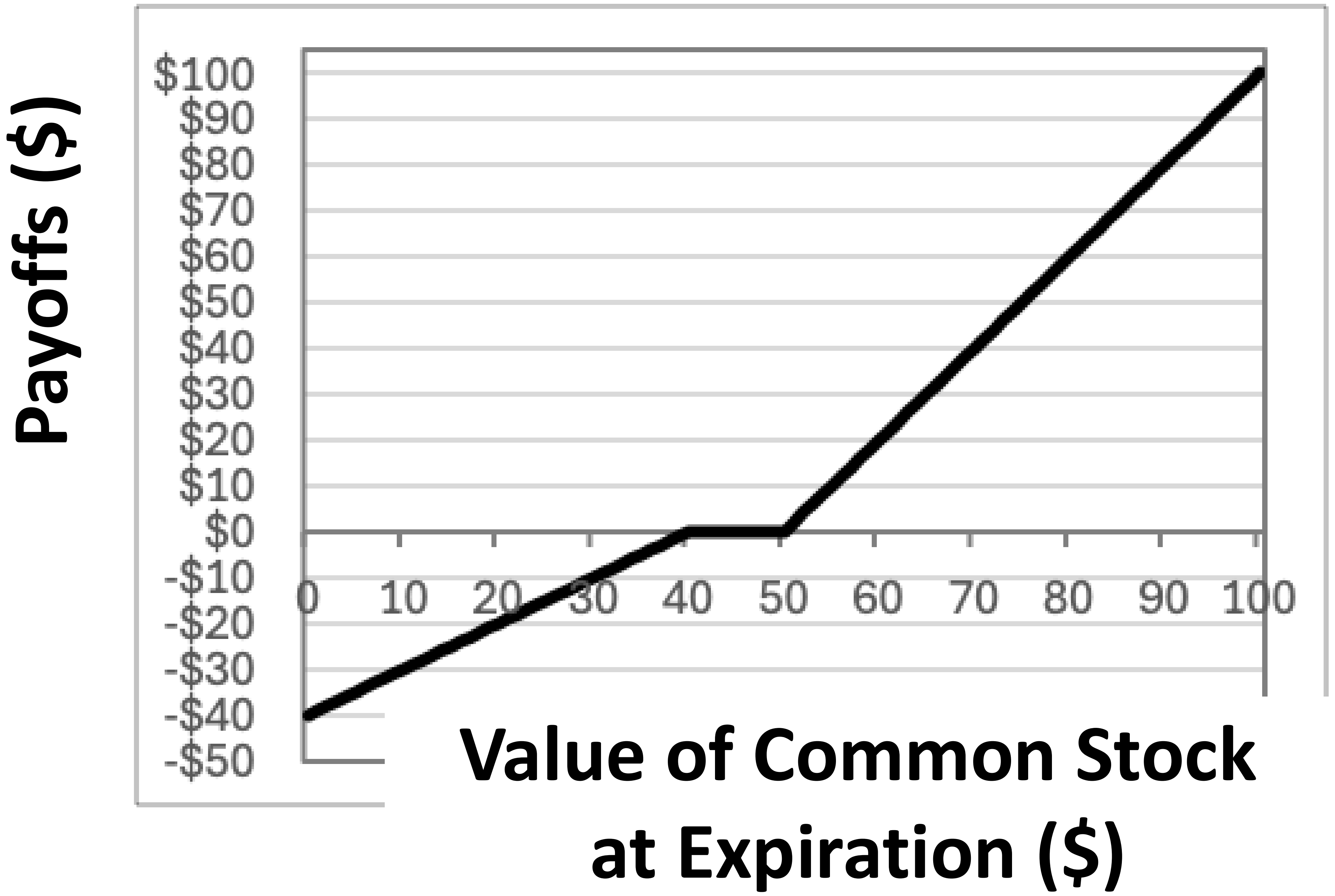
Sell 1 Put Option with Strike Price $40.
Buy 1 Call Option with Strike Price $50

Example 2

Steps

1. Find kink points
2. Assess slopes

Find strategy that recreates the payoff below:



Example 2

Steps

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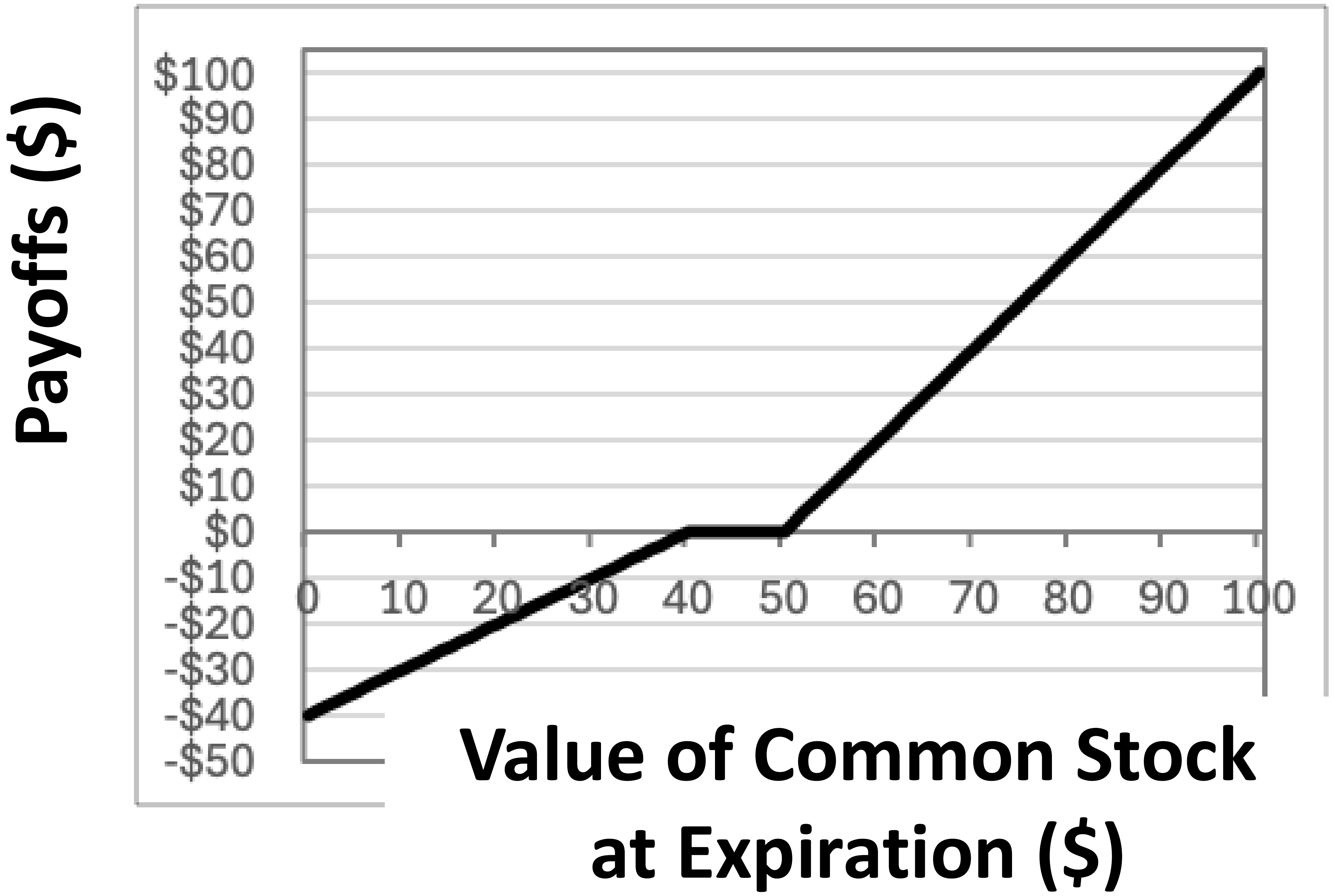


Step 1:
Kink Point 1 at $40
Kink Point 2 at $50

Step 2:
Slope is +1 from $0 to $40
Slope is 0 from $40 to $50
Slope is +2 from $50 onwards

Sell 1 Put Option with Strike Price $40.
Buy 2 Call Options with Strike Price $50

Great Recession

The Queen Asks Economists in November 2009

'It's awful - Why did nobody see it coming?': The Queen gives her verdict on global credit crunch

The Queen Asks Economists in November 2009

'It's awful - Why did nobody see it coming?': The Queen gives her verdict on global credit crunch

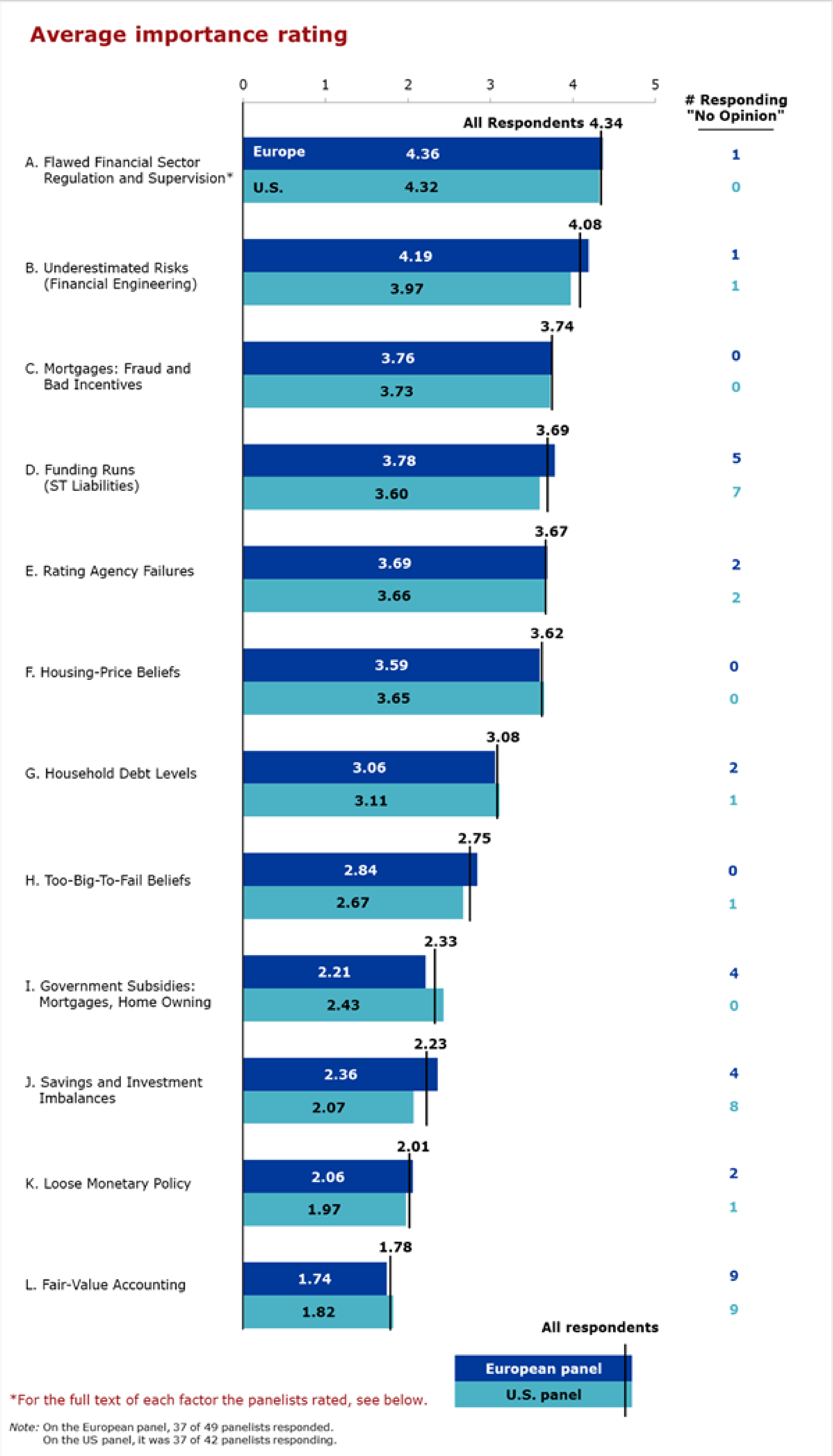
Indeed.

Well To Be Fair A Few Did…But Were Ignored

- Nouriel Roubini (“Dr. Doom”)
- Raghuram Rajan (IMF Head of Research)
- Robert Shiller (Nobel-Prize Winner)
- Steve Keen
- Dean Baker
- Michael Burry (founder of Scion Capital Hedge Fund)

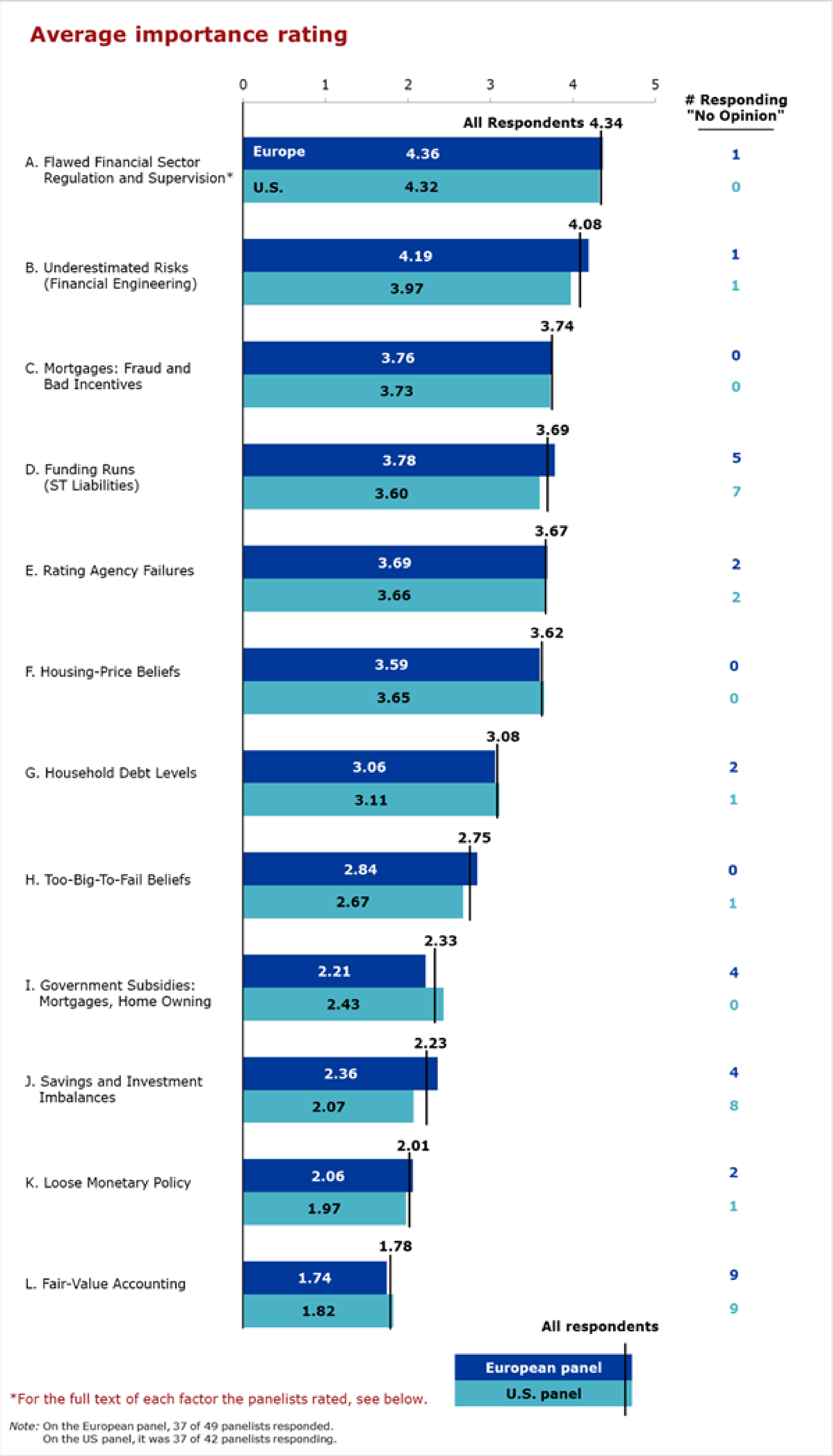
Many Views on Causes of Great Recession

- Not one cause but many
- Broader debates over role of wealth inequality, macroeconomic policy, global imbalances, regulation



Many Views on Causes of Great Recession

- Not one cause but many
- Broader debates over role of wealth inequality, macroeconomic policy, global imbalances, regulation
- Today, I'll focus on on the more financial topics
  - Housing Market
  - Mortgage Lending
  - Securitization
  - Leverage
- While Recession was Global, I'll focus on US

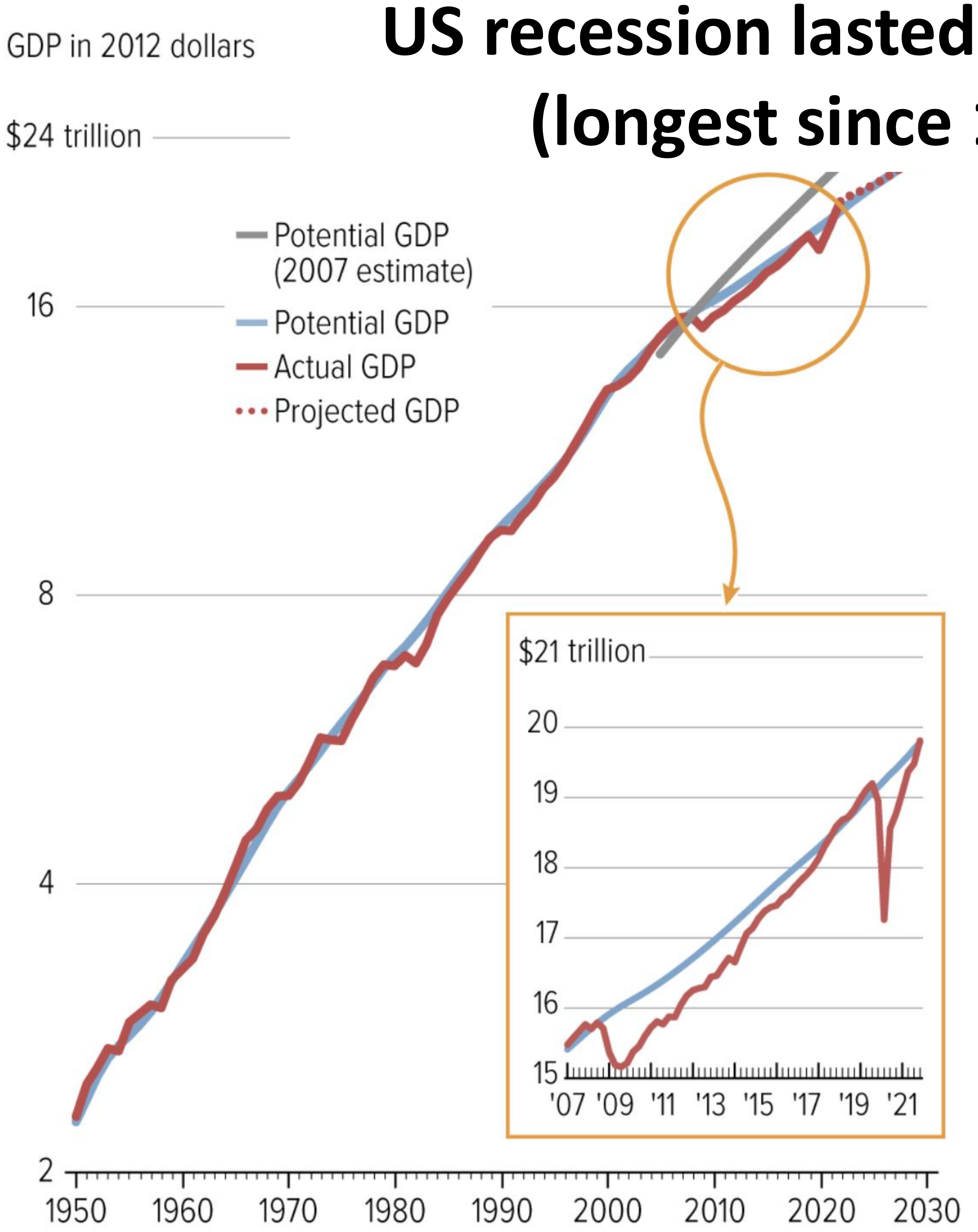


Big Picture

Global Economies Never Fully Recovered from Great Recession

US recession lasted 18 months (longest since 1950s)

–$70,000 in Lifetime Income for average American (Present Value)



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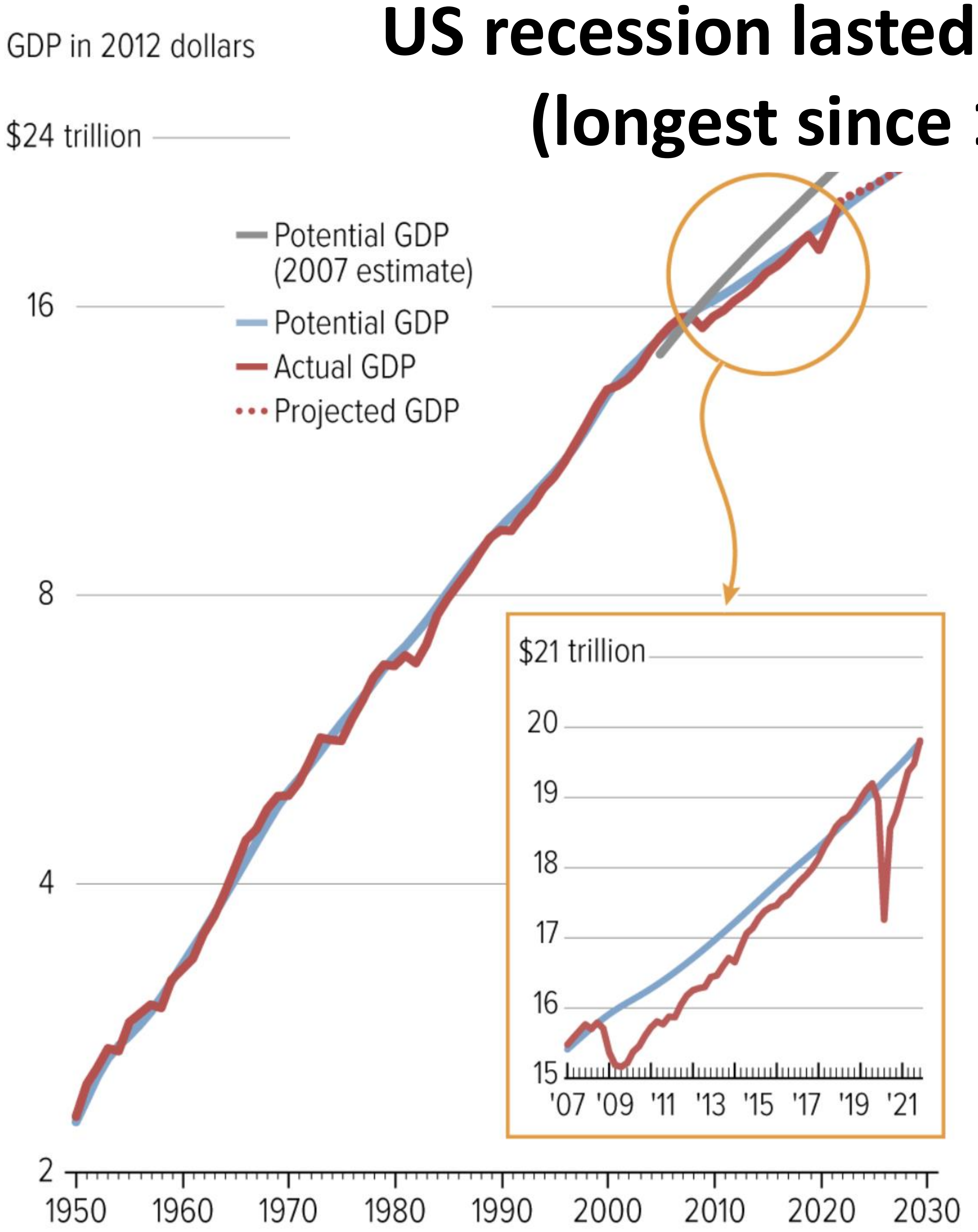
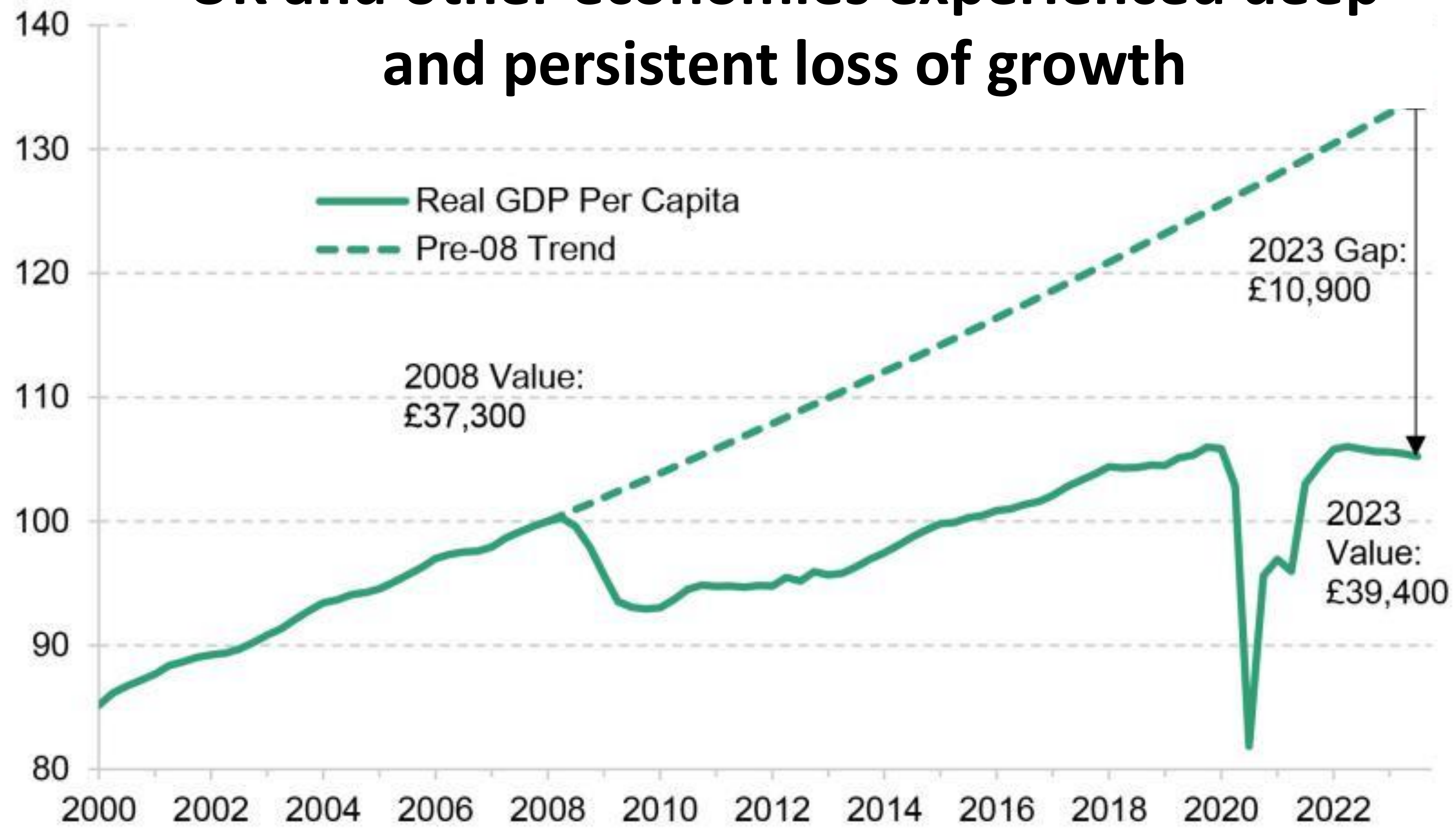
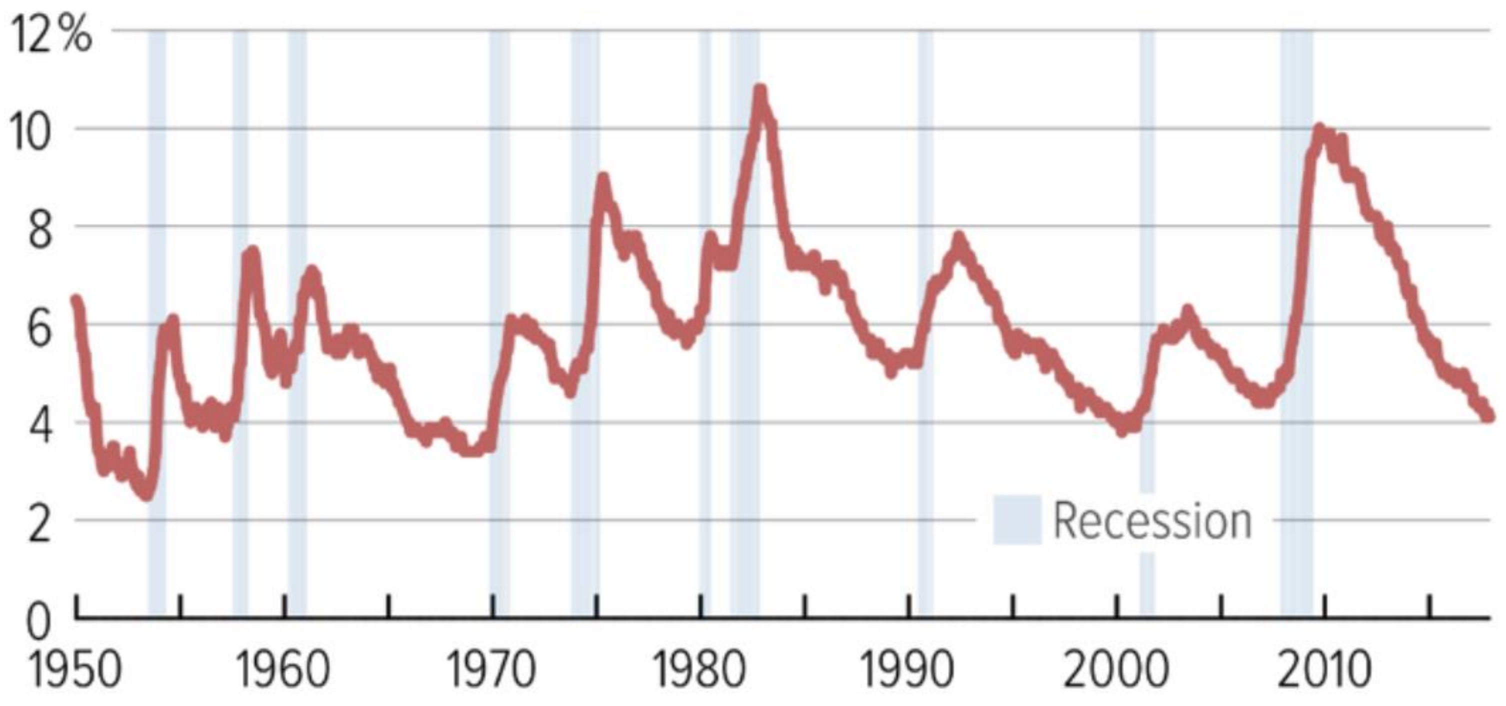


UK and other economies experienced deep and persistent loss of growth

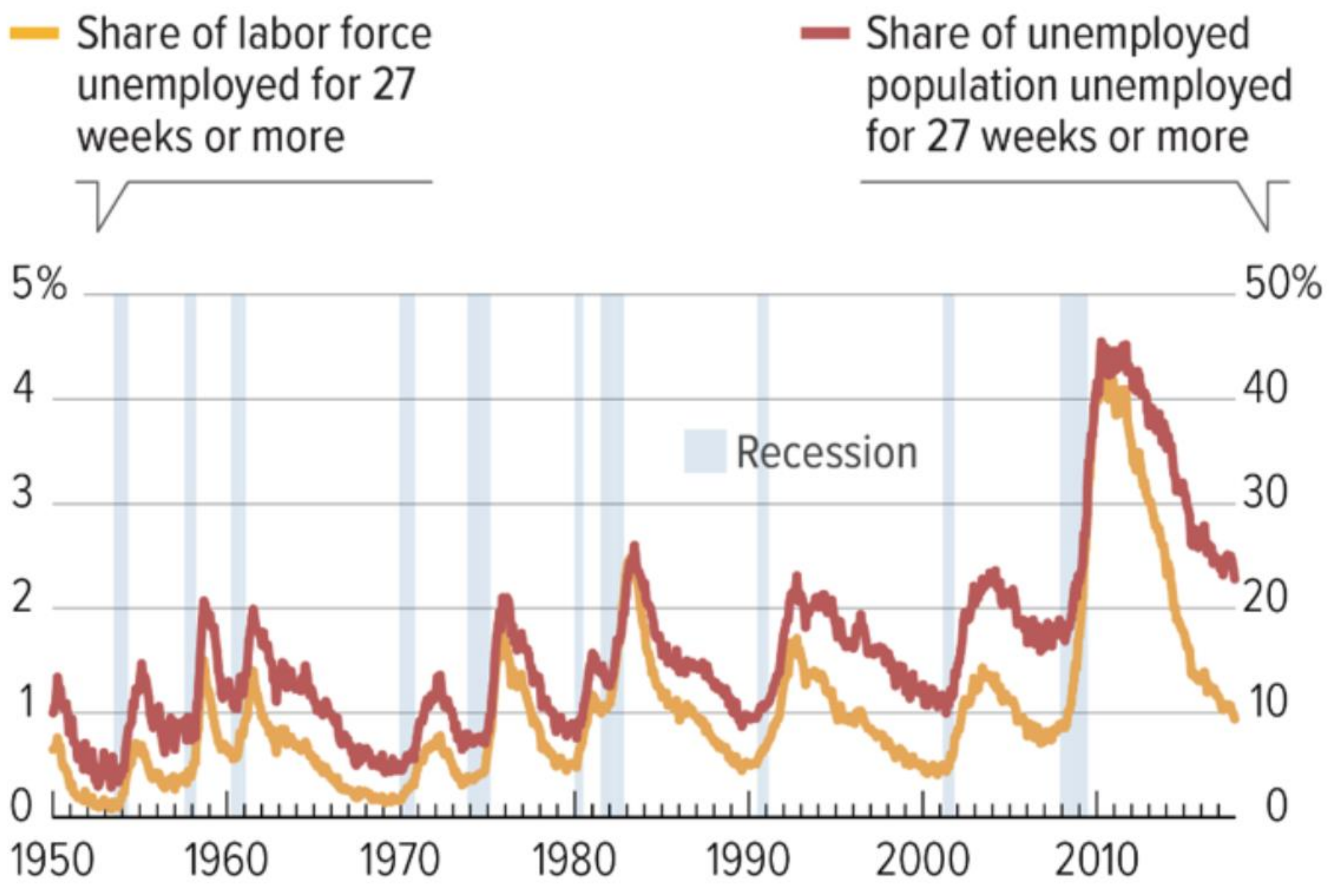


Huge Increases in Unemployment

Unemployment Rate Rose to Near Its Postwar High



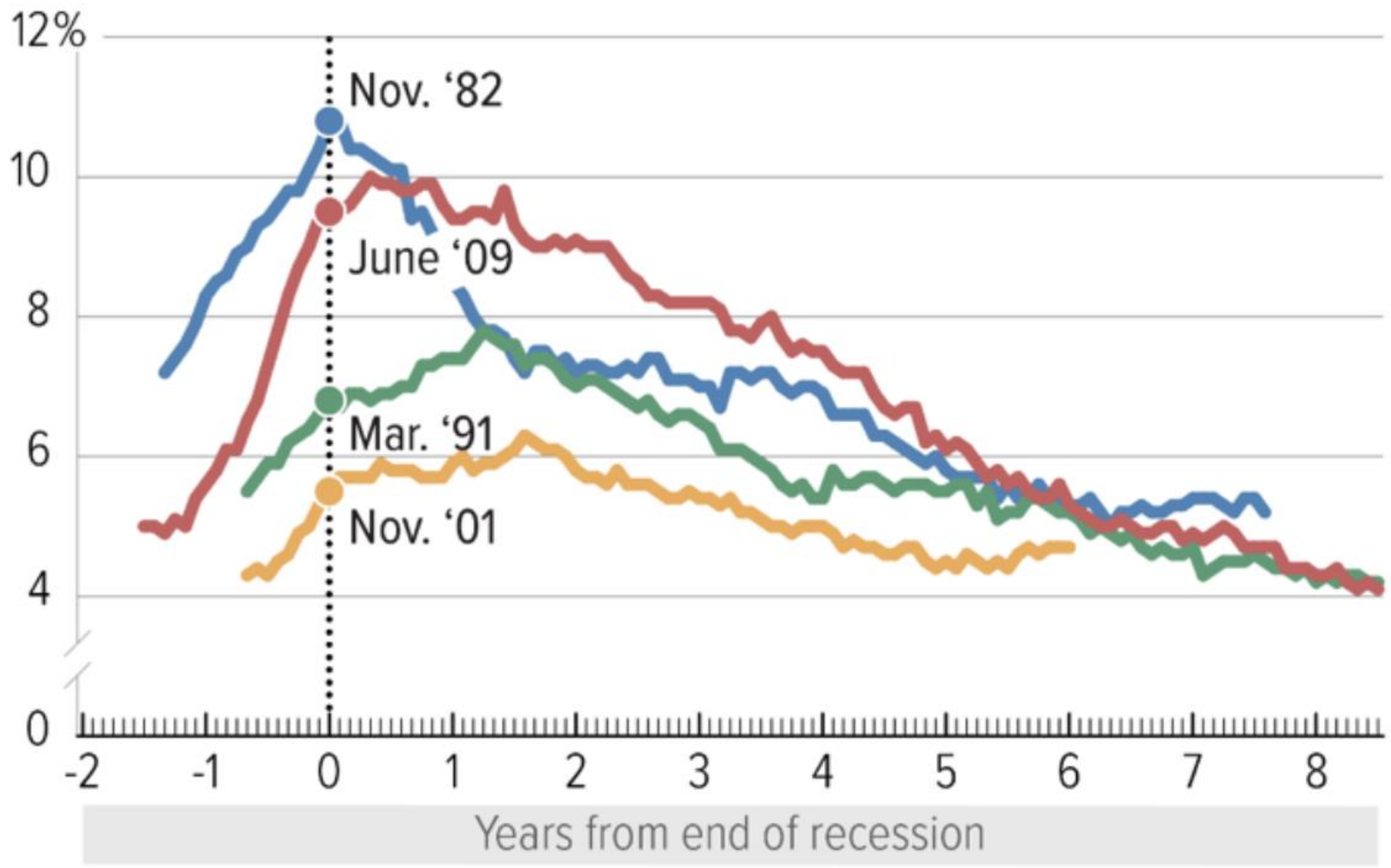
Long-Term Unemployment Rose to Historic Highs



Sources: Bureau of Labor Statistics and National Bureau of Economic Research

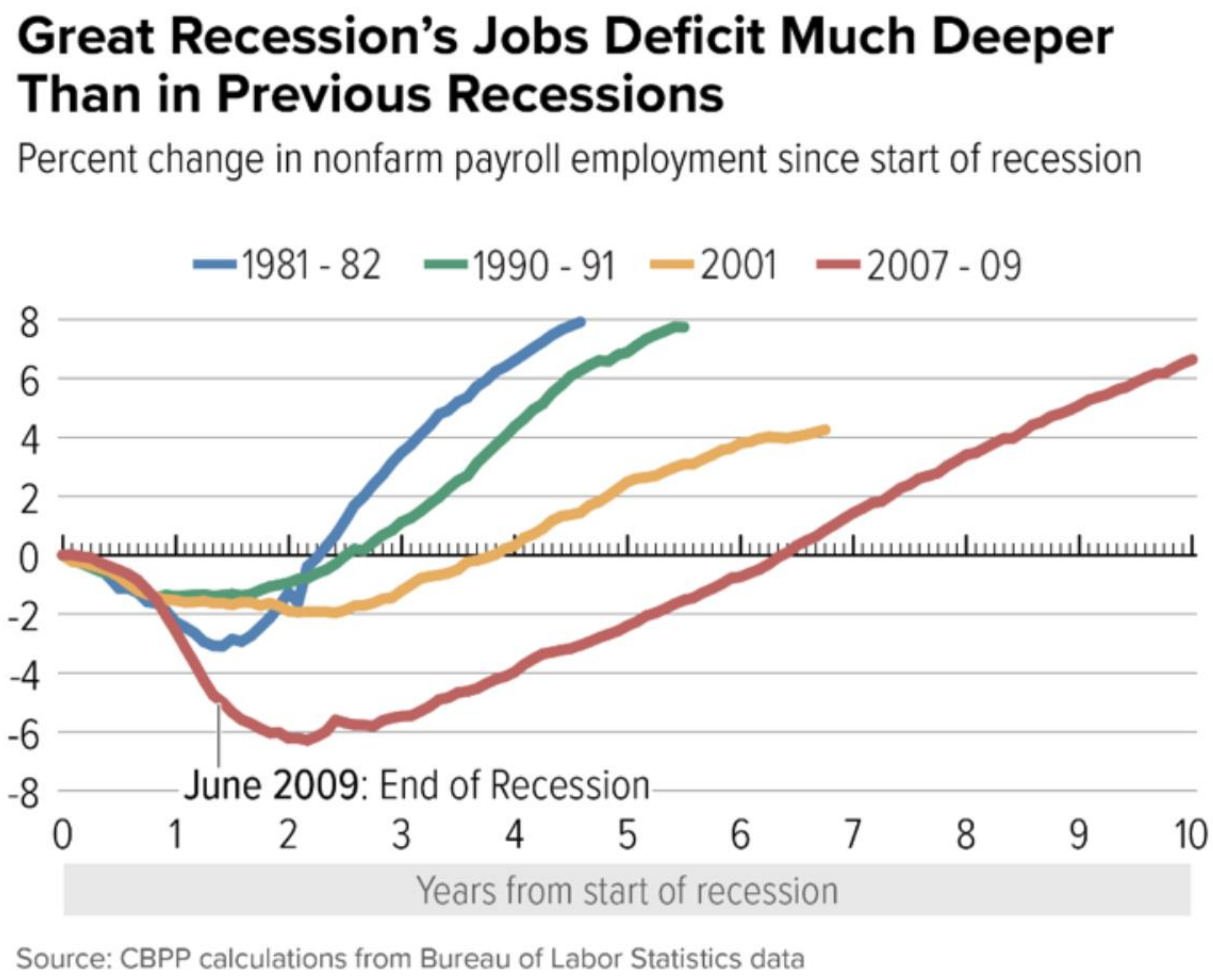
Unemployment Rate Stayed High Long After Great Recession's End

Unemployment rates during recessions and recoveries



Source: CBPP calculations from Bureau of Labor Statistics data

Slow Recovery in Jobs



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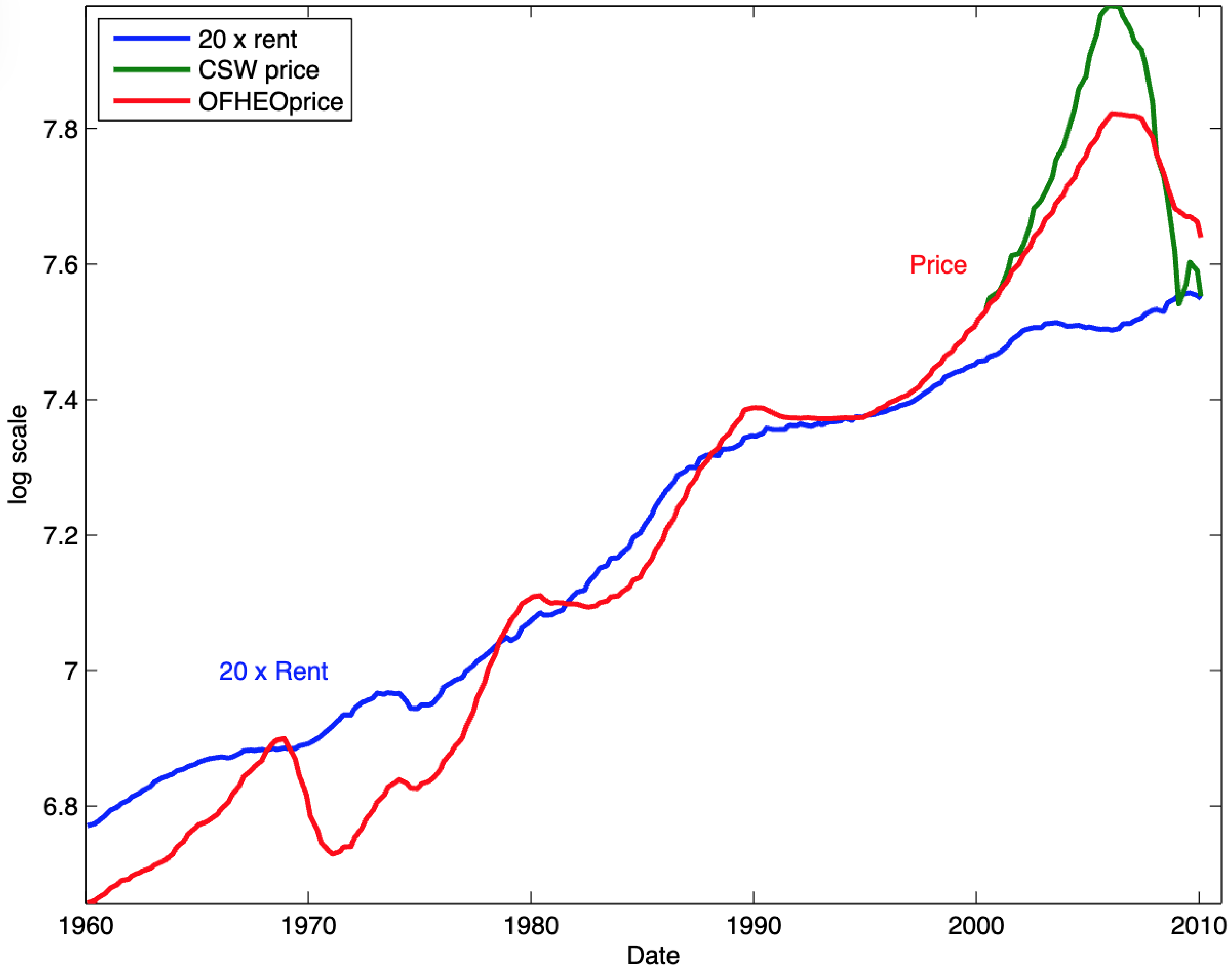
U.S. Housing Market

The Rise and Fall of U.S. House Prices



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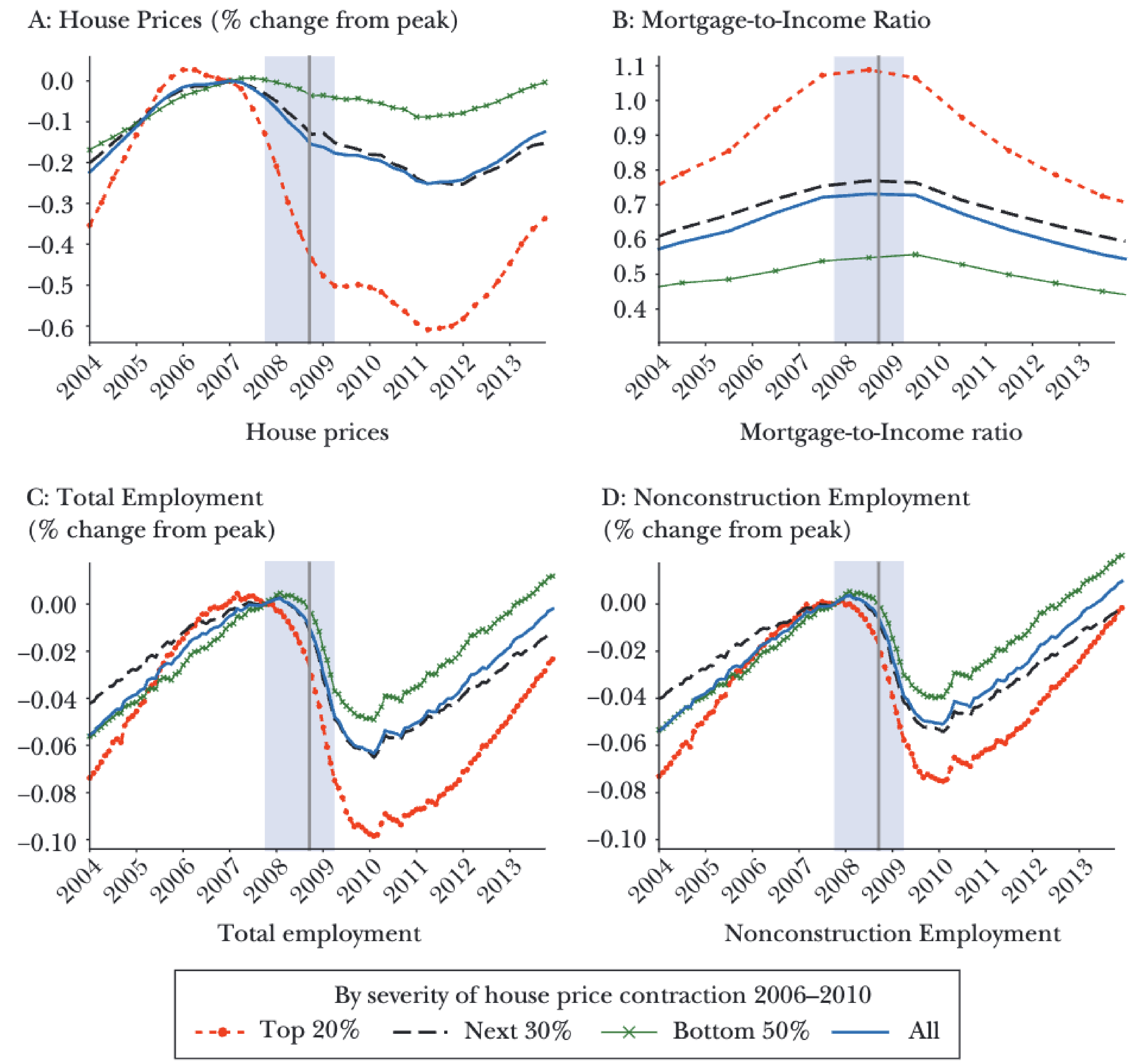
Regions with Biggest Booms in House Prices, Had Larger Falls in House Prices & Employment

State-Level House Prices, Mortgage Debt, and Employment



Note: Figure 6 portrays cross-sectional and time series variation of four variables: house prices, the mortgage-to-income ratio, employment, and nonconstruction employment. The data is quarterly and covers the period from 2004 to 2015. For each variable, we group states into three categories based on the severity of the house price contraction from 2006 to 2010. The first category experienced the largest house price drop and accounts for 20 percent of the population, the middle group contains 30 percent of the population, and the bottom group the remaining 50 percent. The solid line shows aggregate behavior for each variable. The house price and the mortgage/income ratio are population-weighted, while the employment measures are simple aggregates.

Panel (c): Years to recovery and precrisis house price growth in a region

	Delinquencies	Foreclosures	House prices	Auto sales	Unemployment
	(1)	(2)	(3)	(4)	(5)
Precrisis House Price Growth	−0.139 (0.016)	−0.124 (0.013)	−0.719 (0.016)	−0.0144 (0.016)	−0.644 (0.055)
Observations	13,188	13,188	13,188	13,188	1344
Adjusted R-squared	0.005	0.006	0.126	0.00001	0.090

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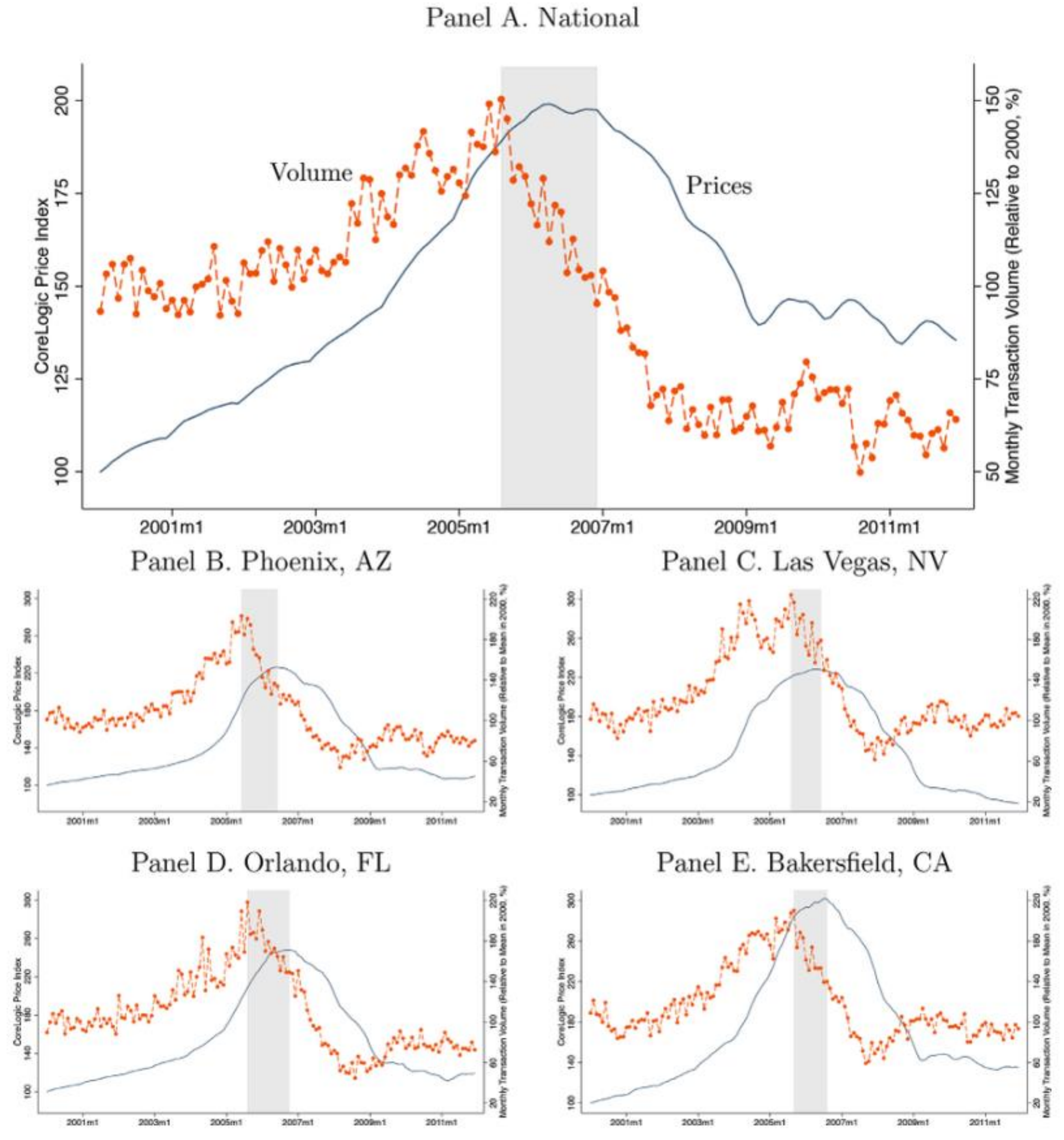
Cities with Larger Price Booms Had Experienced Increases in Volume



ig. 1. The dynamics of prices and volume. This figure displays the dynamic relation between prices and volume in the U.S. housing market between 2000 nd 2011. Panel A shows monthly prices and sales volume at the aggregate level. Panels B–E show analogous series for a set of cities that represent regions vith the largest boom–bust cycles during this time: Phoenix, AZ; Las Vegas, NV; Orlando, FL; and Bakersfield, CA. Monthly price index information comes rom CoreLogic and monthly sales volume is based on aggregated transaction data from CoreLogic for 115 MSAs representing 48% of the U.S. housing stock. We apply a calendar-month seasonal adjustment for volume. Shaded regions denote the quiet, defined as the period between the peak of volume and the ast peak of prices before their pronounced decline.

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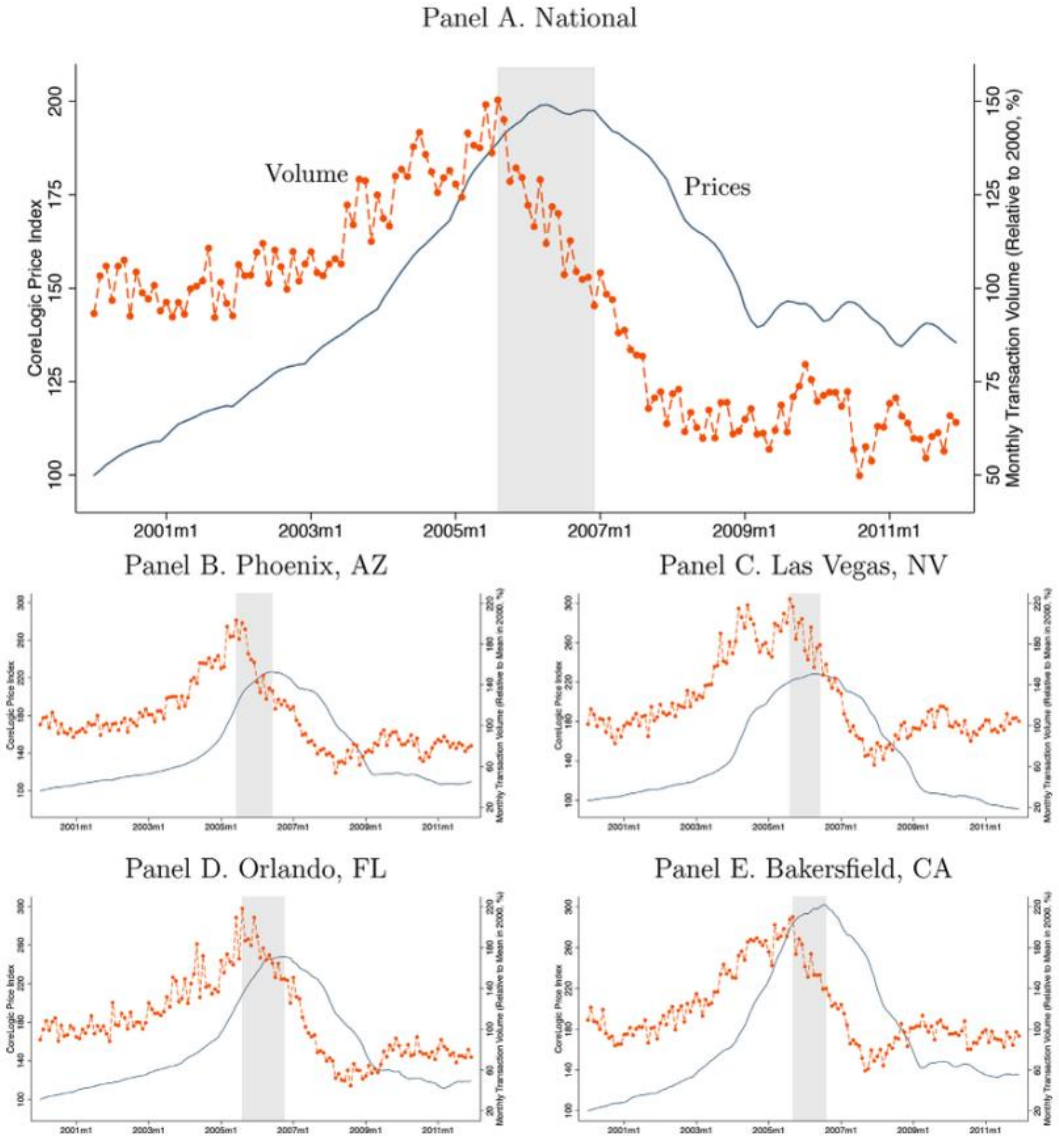
Cities with Larger Price Booms Had Experienced Increases in Construction



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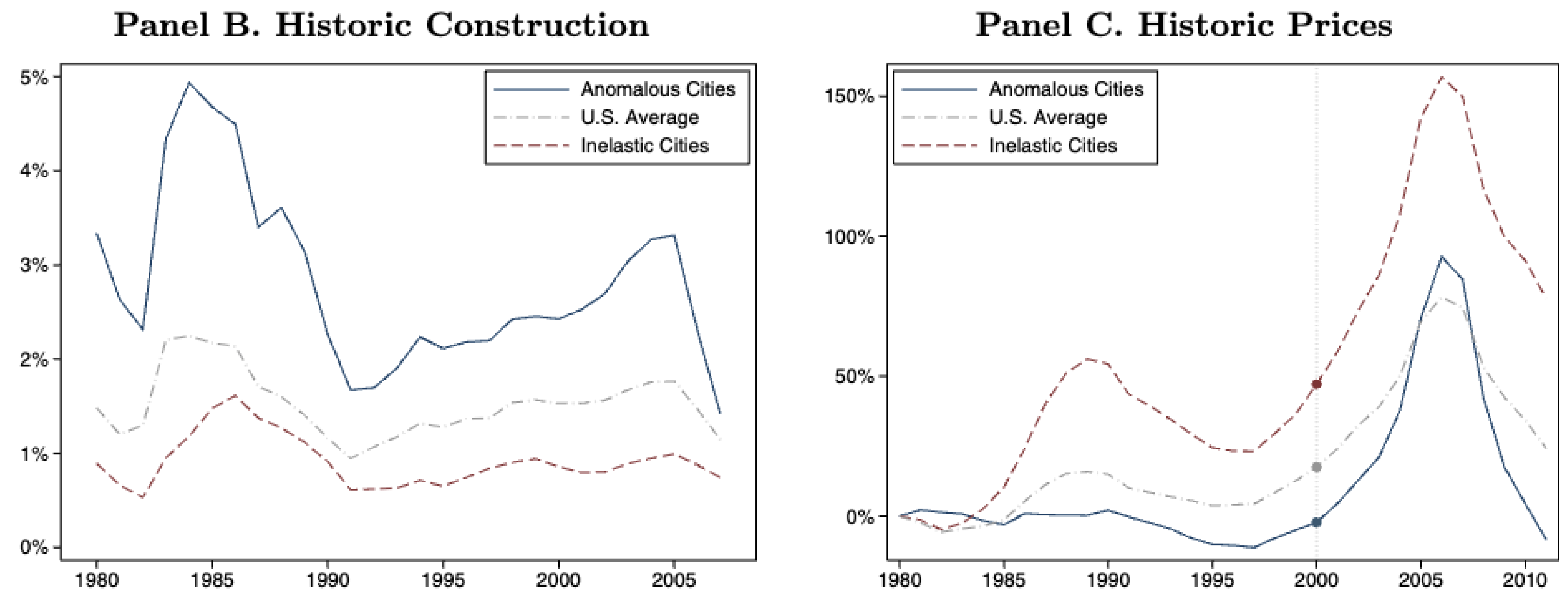
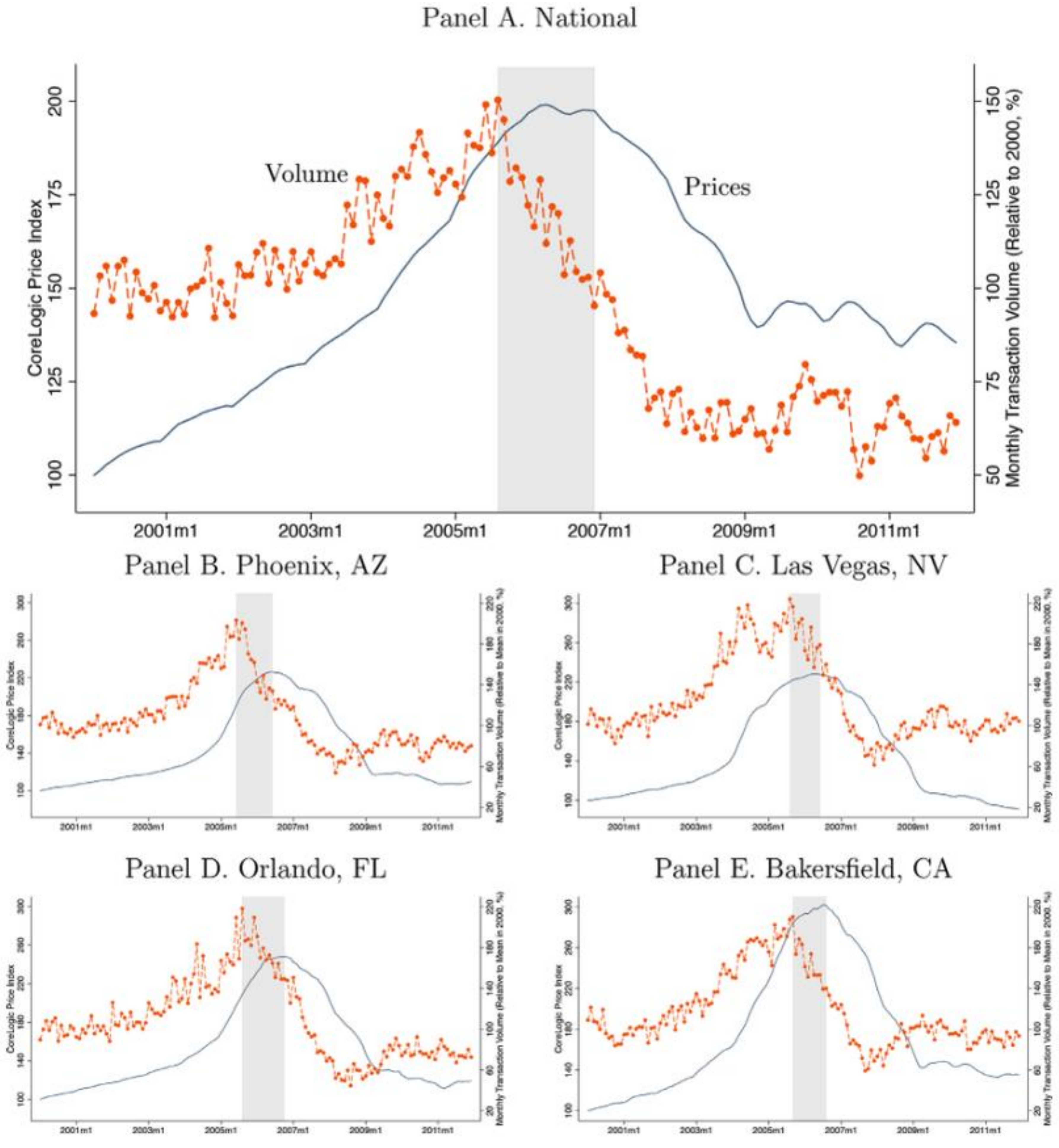


Figure 5. The U.S. housing boom and bust across cities. Anomalous Cities include those in Arizona, Nevada, Florida, and inland California. Inelastic Cities are Boston, Providence, New York, Philadelphia, and all cities on the west coast of the United States We measure the housing stock in each city at an annual frequency by interpolating the U.S. Census's decadal housing stock estimates with its annual housing permit figures. House price data come from the second-quarter FHFA house price index deflated by the CPI-U. The figure includes all metropolitan areas with populations over 500,000 in 2000 for which we have data. Panel A: The cumulative price increase is the ratio of the house price in 2006 to the house price in 2000. Annual housing stock growth is given as the log difference in the housing stock in 2006 and 2000 divided by six. Panels B and C: Each series is an average over cities in a group weighted by the city's housing stock in 2000. Construction is annual permitting as a fraction of the housing stock. Prices represent cumulative returns as of 1980 on the housing in each group. (Color figure can be viewed at wileyonlinelibrary.com)

Cities with Larger Price Booms Had Experienced Increases in Construction and Short-Term Speculation



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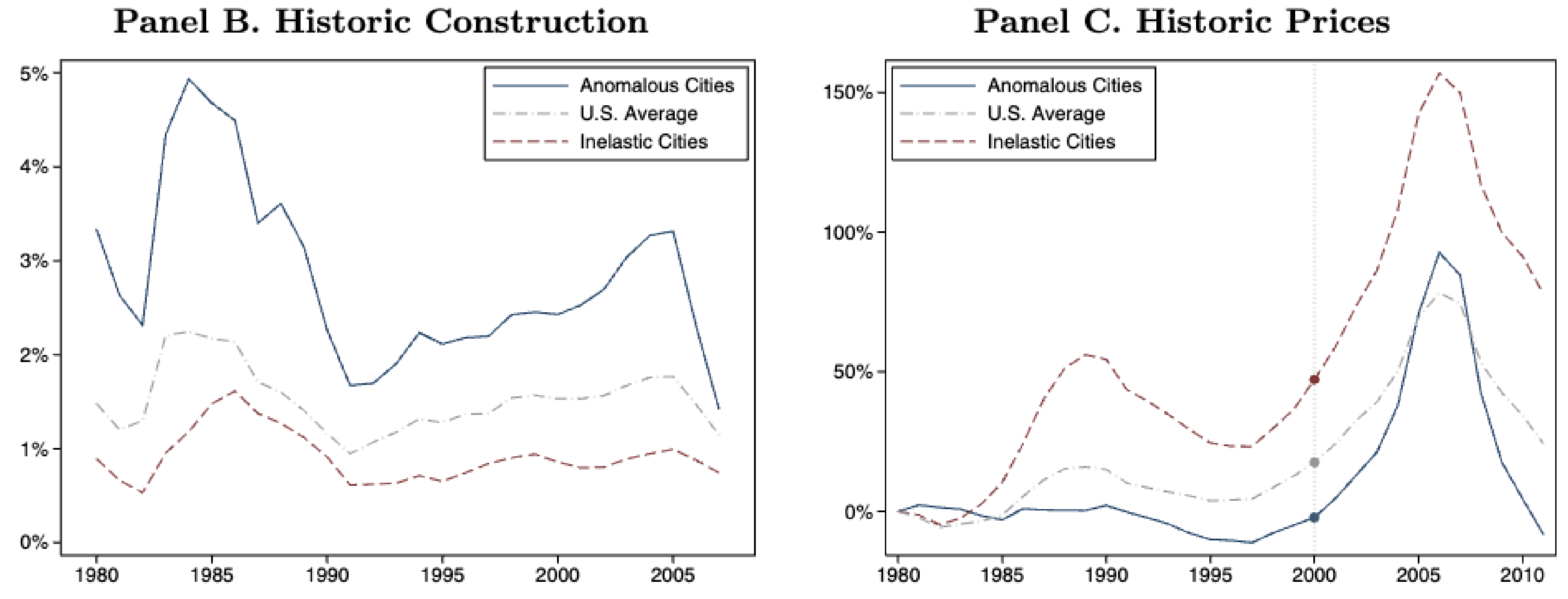

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Fig. 6. The flow of listings for short-holding-period buyers. In this figure, we illustrate the time variation in propensities to list among recent buyers versus all buyers between 2000 and 2011 in the U.S. We link listings micro data to transaction data at the property level to identify short-holding-period listings. We plot monthly aggregate time series for total listings (blue circles) and short-holding-period listings (red squares), defined as a listing where the previous sale occurred within the past three years. The series include observations for the 57 MSAs in our listings sample. Each series is separately normalized relative to its average value in the year 2003 and seasonally adjusted by removing calendar-month fixed effects. The raw counts of each type of listing in the years 2003, 2007, and 2010 are also reported in the upper right corner of the figure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Long-Term Economic Effects (2017 Data)

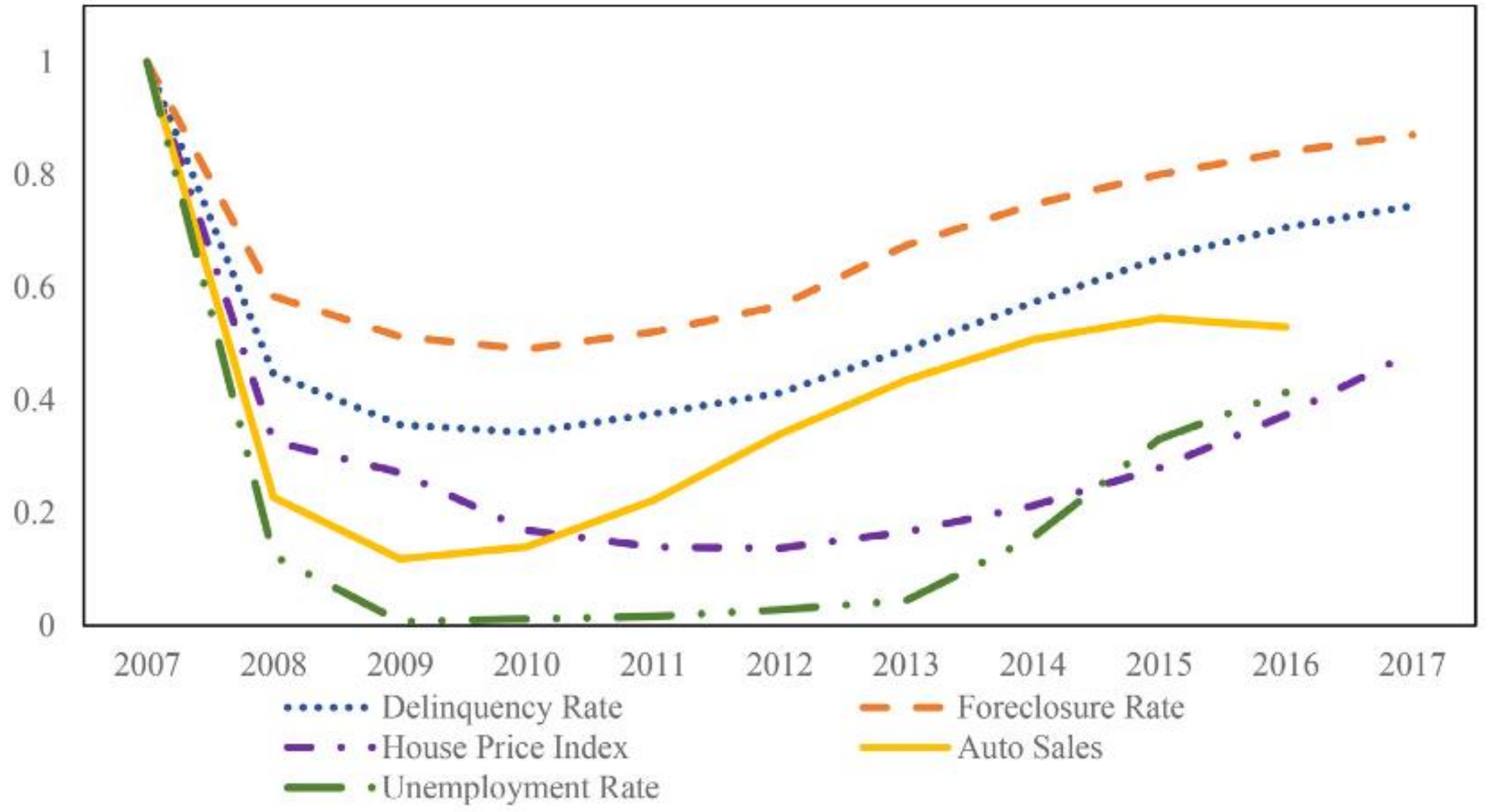


Fig. 3. Slow recovery: Share of regions that recovered to precrisis level over time.
This figure shows the share of regions that recovered to their precrisis levels over time. Precrisis levels are taken as of 2007, so 100% of regions are recovered in 2007 by definition. Delinquency rates, foreclosure rates, house prices, and auto sales are measured at the zip code level, while unemployment rates are measured at the county level. Delinquency rates, foreclosure rates, and house prices are measured from 2007 to 2017, while auto sales and unemployment rates are measured from 2007 to 2016. Data Sources: Delinquency and foreclosure rates come from Equifax, auto sales data from R. L. Polk & Company, house prices from Zillow, and unemployment rates from the U.S. Census Bureau.

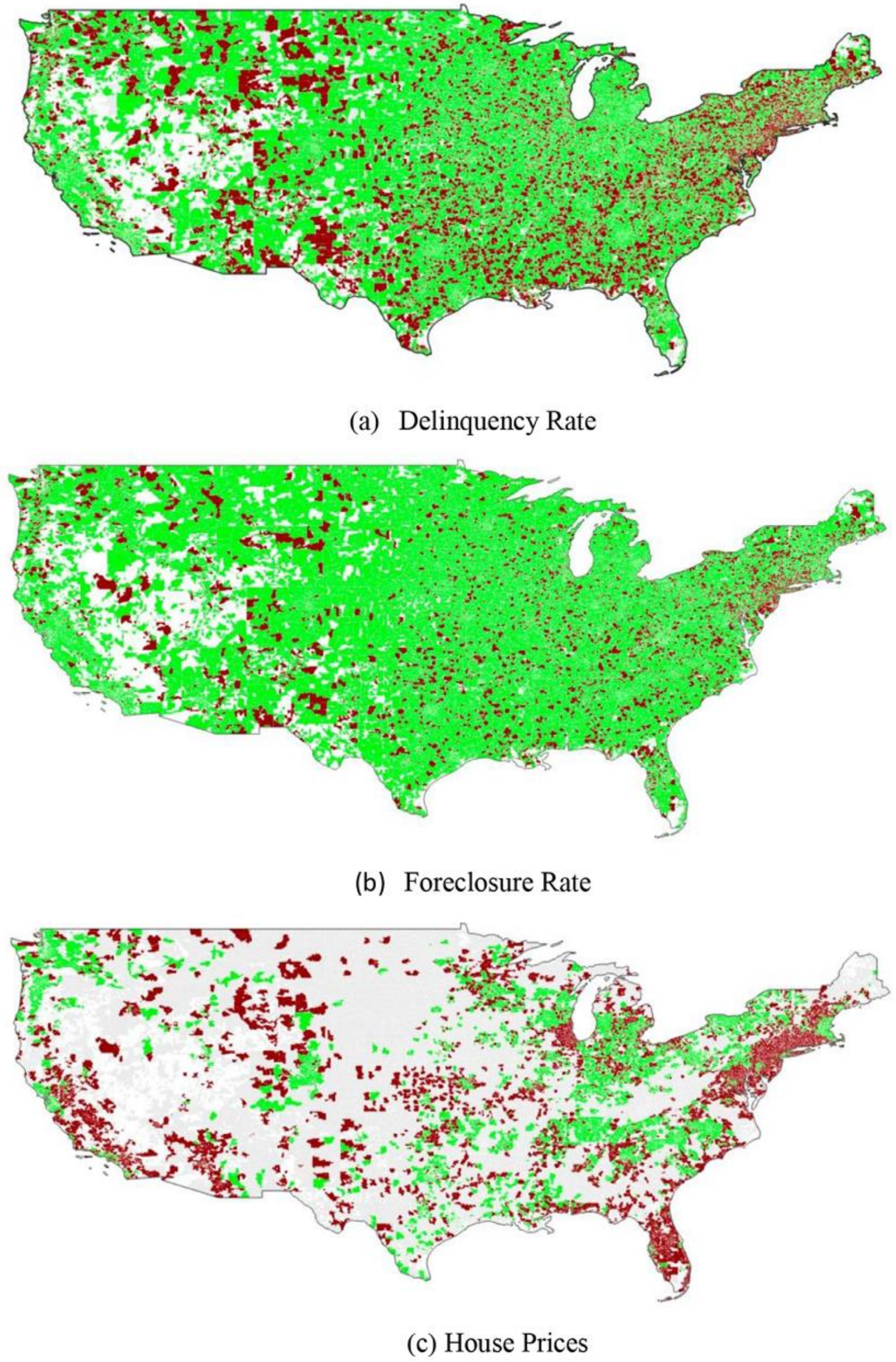


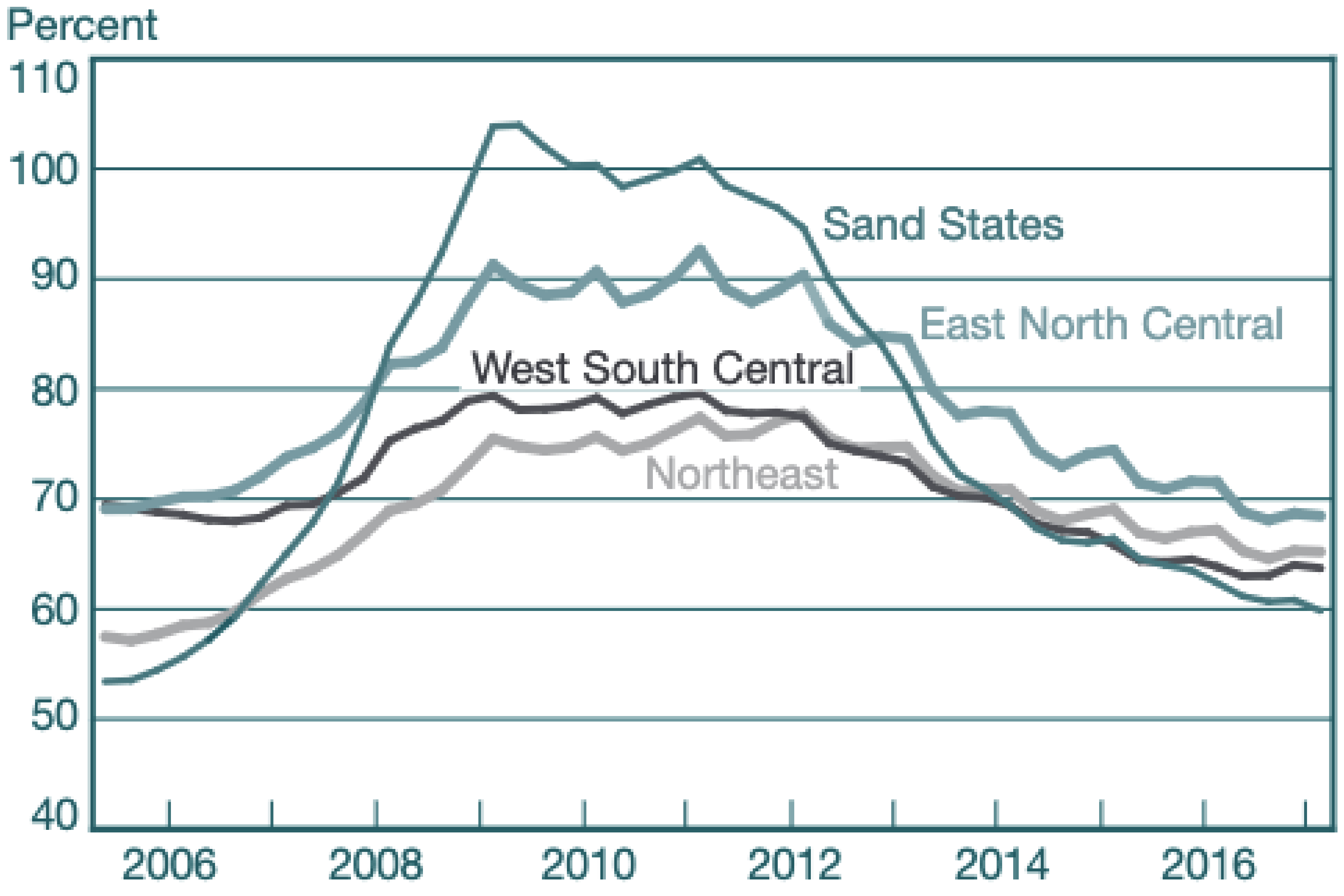
Fig. 4. Uneven recovery: Recovery after the crisis across U.S. regions .
This figure illustrates the regional heterogeneity in the recovery from the crisis with respect to delinquency rates, foreclosure rates, house price levels, auto sales, and unemployment rate. Specifically, we consider whether a region recovered to its precrisis level. The dark color indicates that the region was below its precrisis (2007) level and the light color indicates that the region recovered to its precrisis level. Delinquency rates, foreclosure rates, house prices, and auto sales are measured at the zip code level and unemployment rate is measured at the county level. Delinquency rates, foreclosure rates, and house prices are measured as of 2017, while auto sales and unemployment rates are measured as of 2016. Data Sources: Delinquency and foreclosure rates come from Equifax, auto sales data from R. L. Polk & Company, house prices from Zillow, and unemployment rates from the U.S. Census Bureau.

U.S. Mortgage Market

Mortgage Debt Increased Most in Regions with House Price Booms

1. "Sand states": Arizona, California, Florida, Nevada
2. "East North Central" census division: Illinois, Indiana, Michigan, Ohio, Wisconsin
3. "West South Central" census division: Arkansas, Louisiana, Oklahoma, Texas
4. "Northeast" census region: Connecticut, Massachusetts, Maine, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont

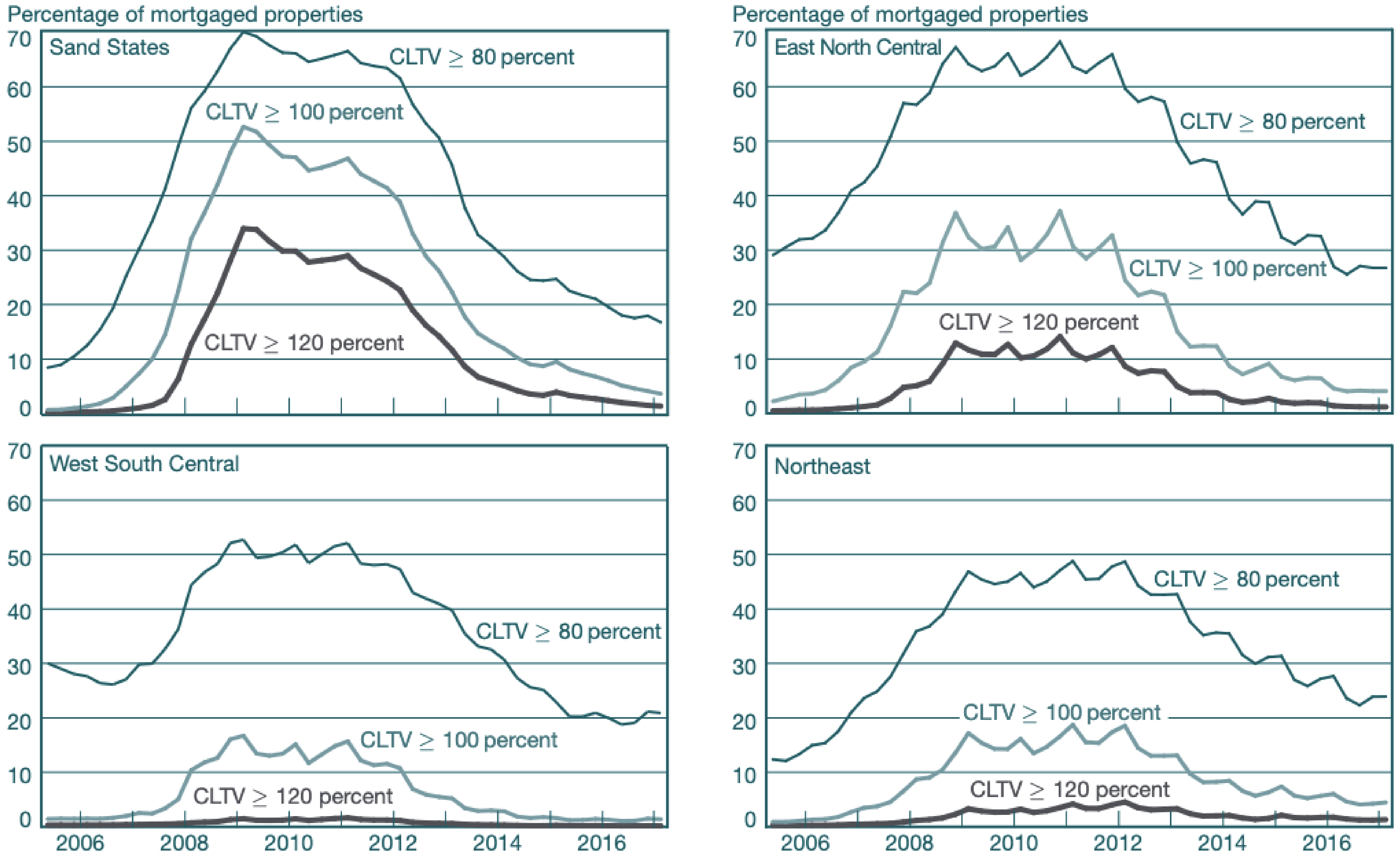
Mean CLTV for Selected Regions, 2005-17



Source: Equifax Credit Risk Insight Servicing McDash (CRISM).

Note: CLTV is combined loan-to-value ratio, as defined in Section 2.1 of this article.

Distribution of CLTVs for Selected Regions, 2005-17



Source: Equifax Credit Risk Insight Servicing McDash (CRISM).

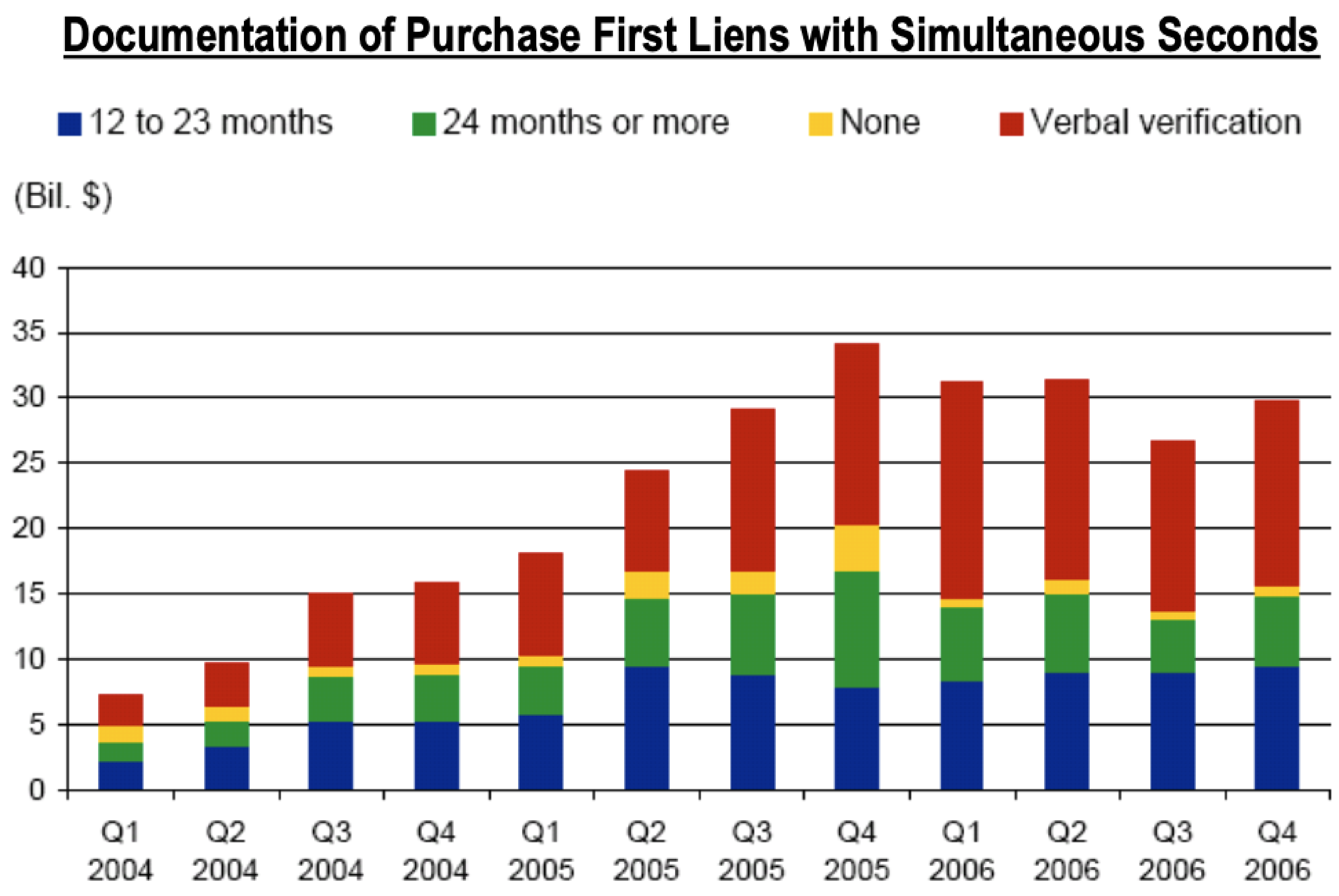
Notes: All distributions are balance-weighted. CLTV is combined loan-to-value ratio, as defined in Section 2.1 of this article.

Not Just Taking Out One Loan But Many Loans

- Speculation: Multiple Homes, Multiple Loans, Flipping Houses
- Often little-to-no quality control on lending

Sub-Prime: Relaxed Lending Standards

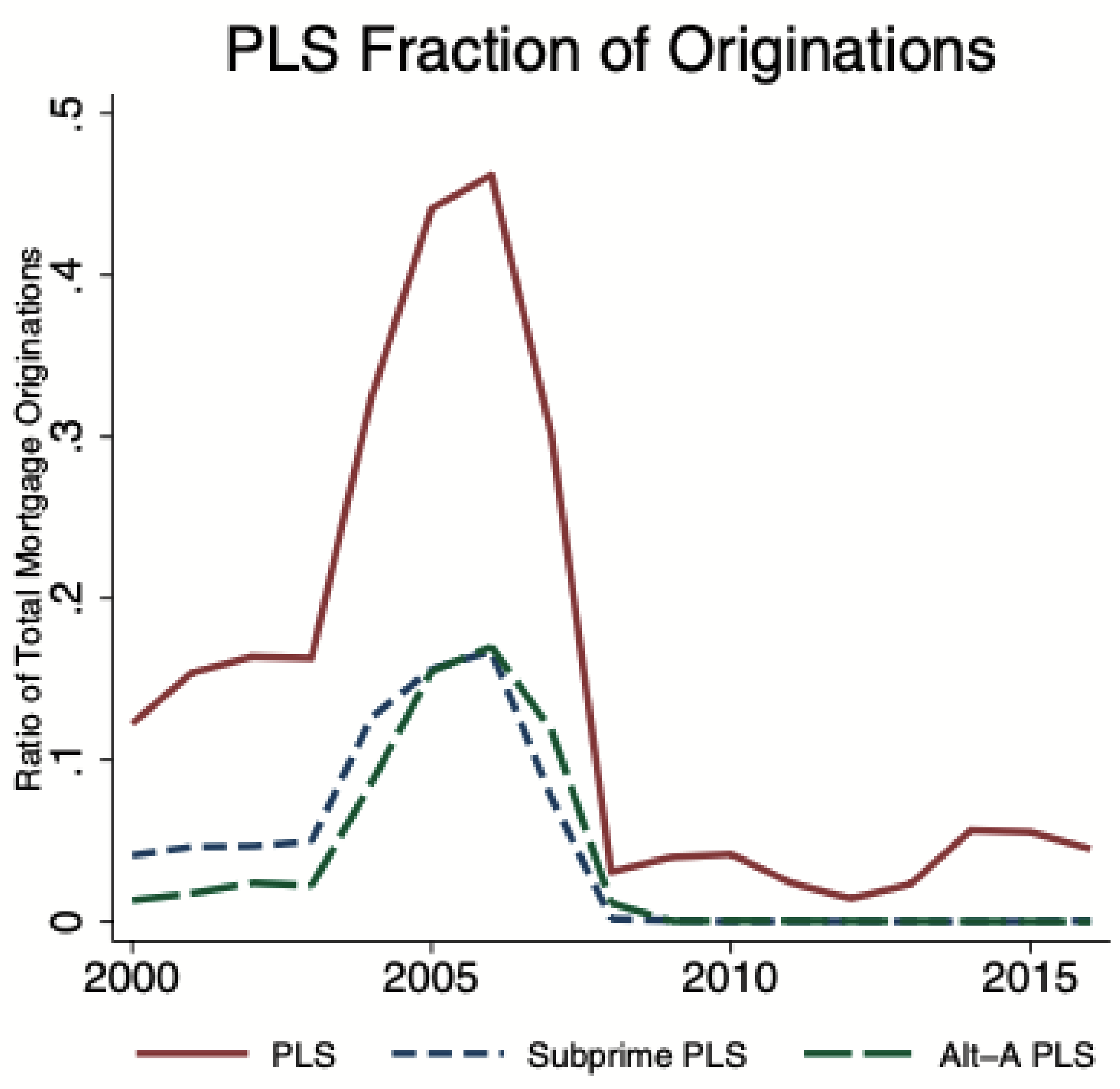
Growth in higher-LTV loans fueled by lower verification standards



Source: Standard & Poor's

Boom in Subprime Lending

- Subprime Loans: Often adjustable-rate mortgages (ARMs), poor credit score, little-or-no downpayment, fraudulent underwriting. Not backed by government.
- NINA ("NINJA") Loans: No Income, No Job, No Assets
- Growth in subprime lending contributed to house prices & construction booms



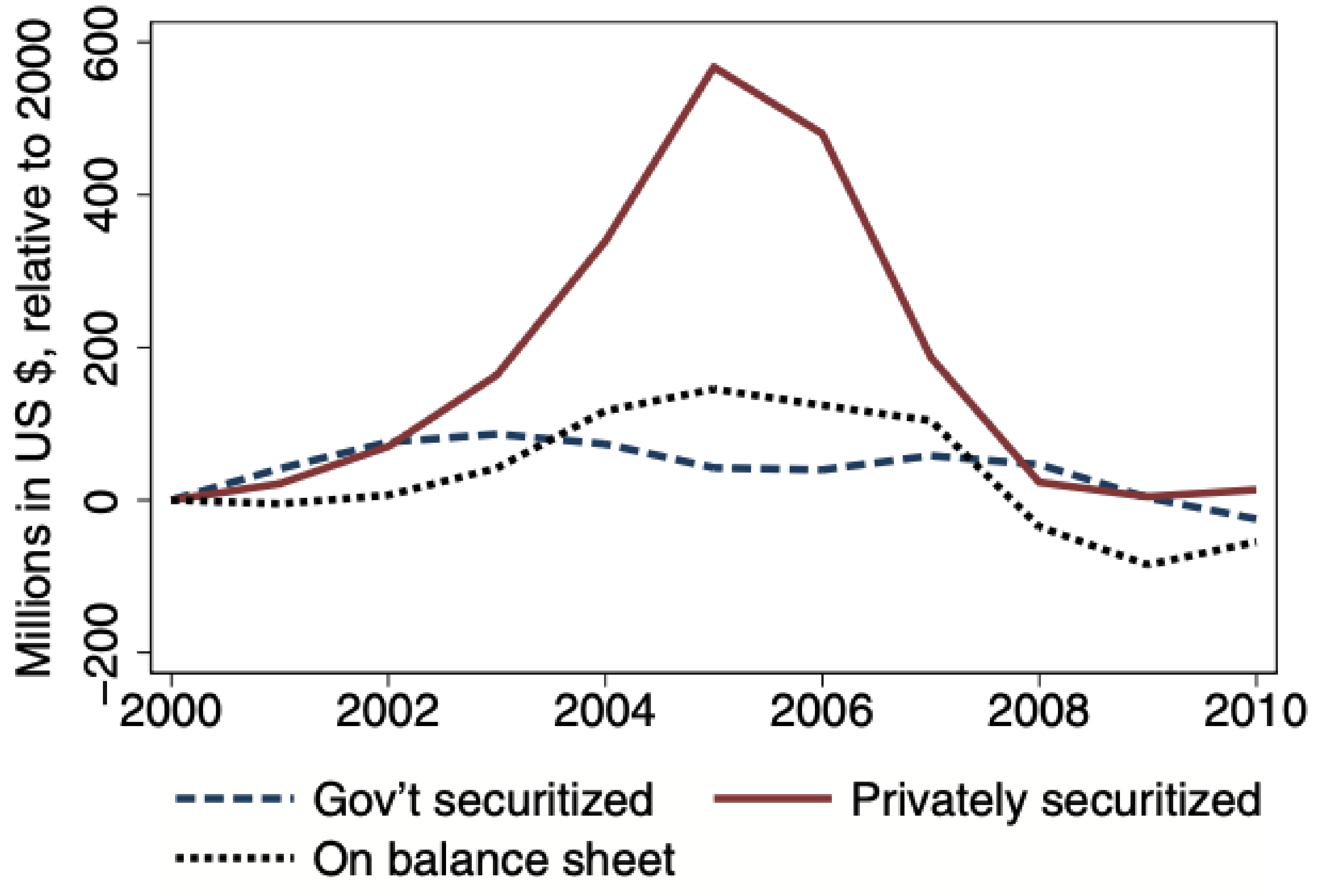


Figure 2
Mortgage originations for home purchase by type
See text for the precise definition of a mortgage sold into the private securitization. *Source*: HMDA.

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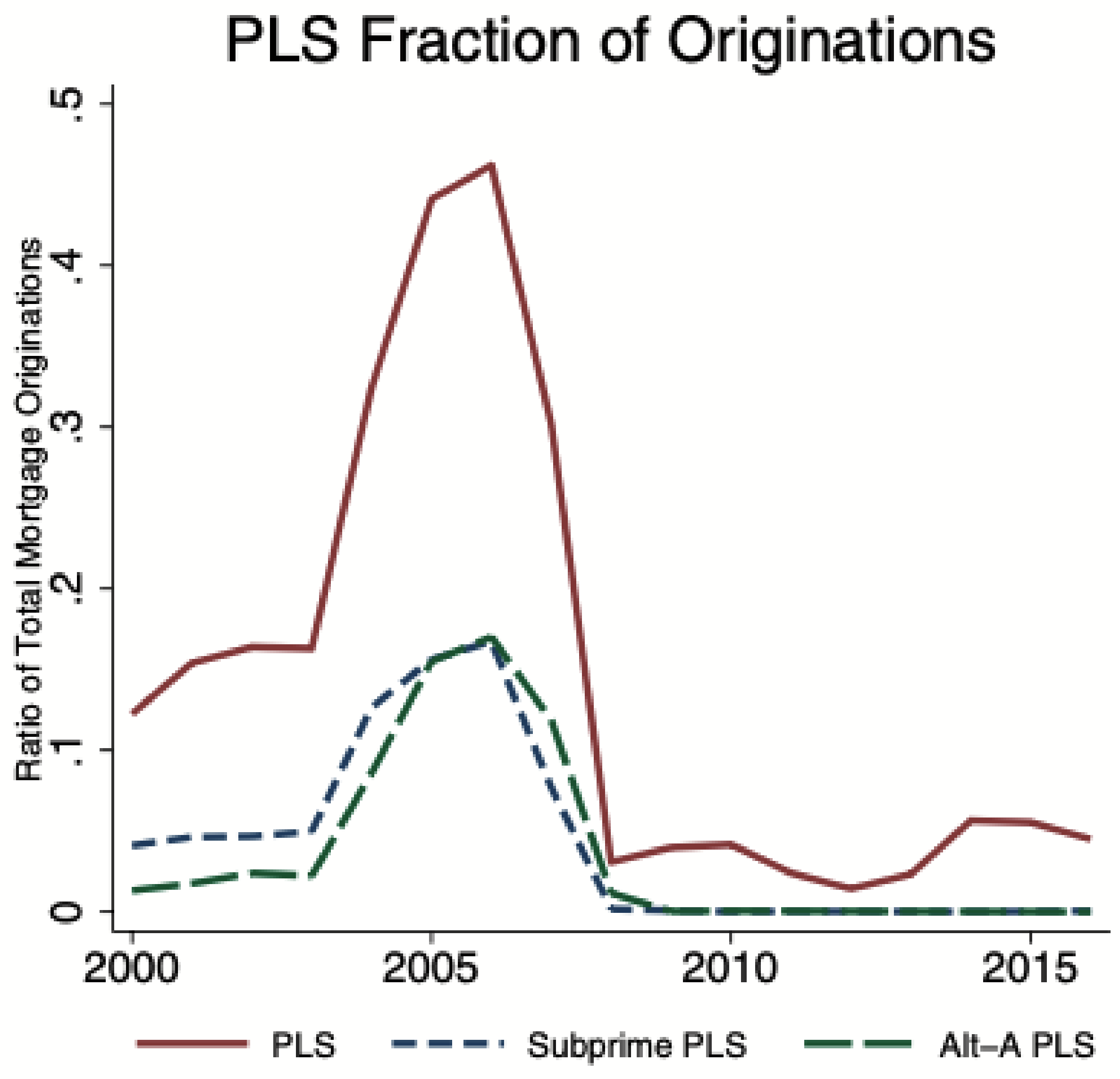
- Growth in subprime lending contributed to house prices & construction booms



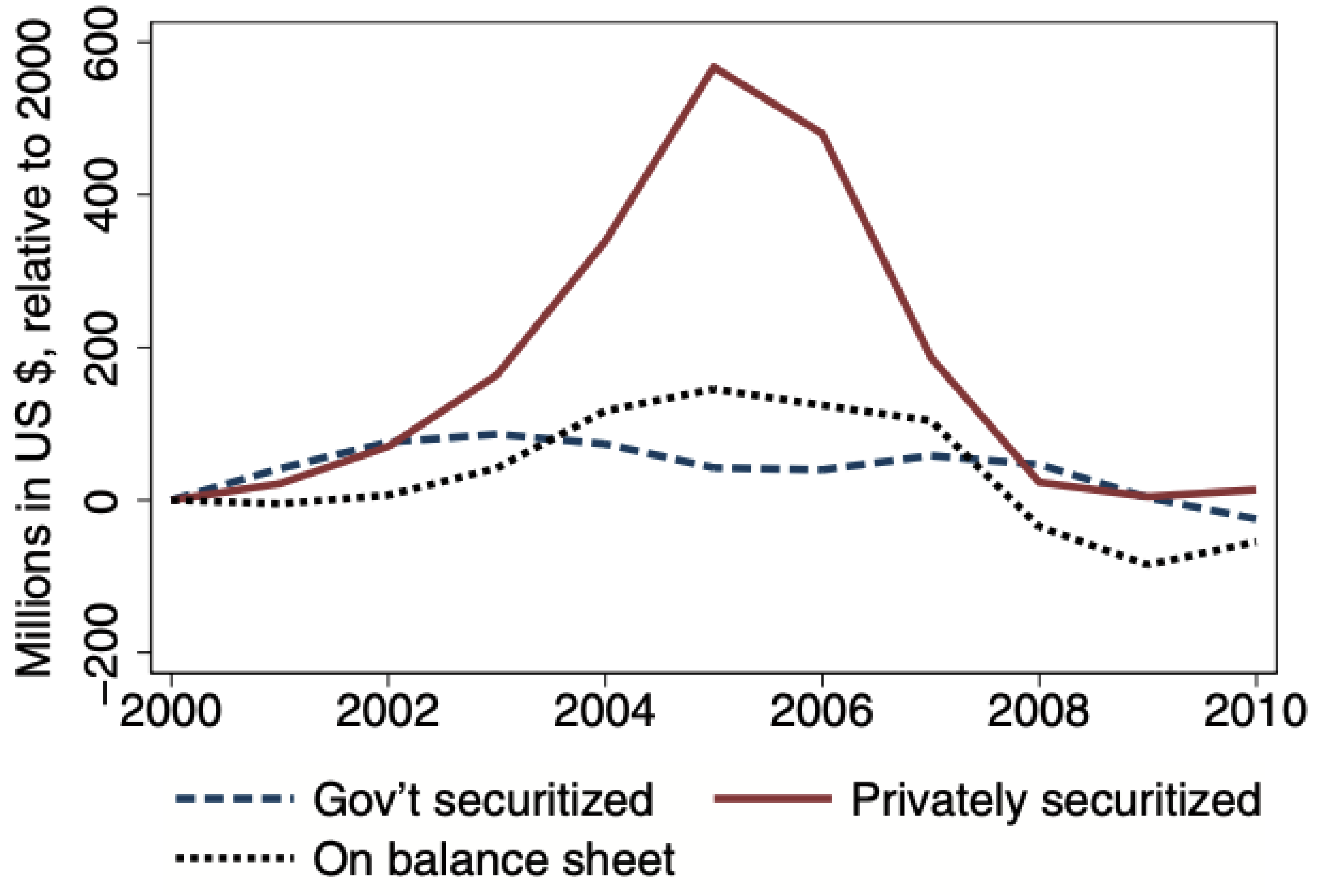
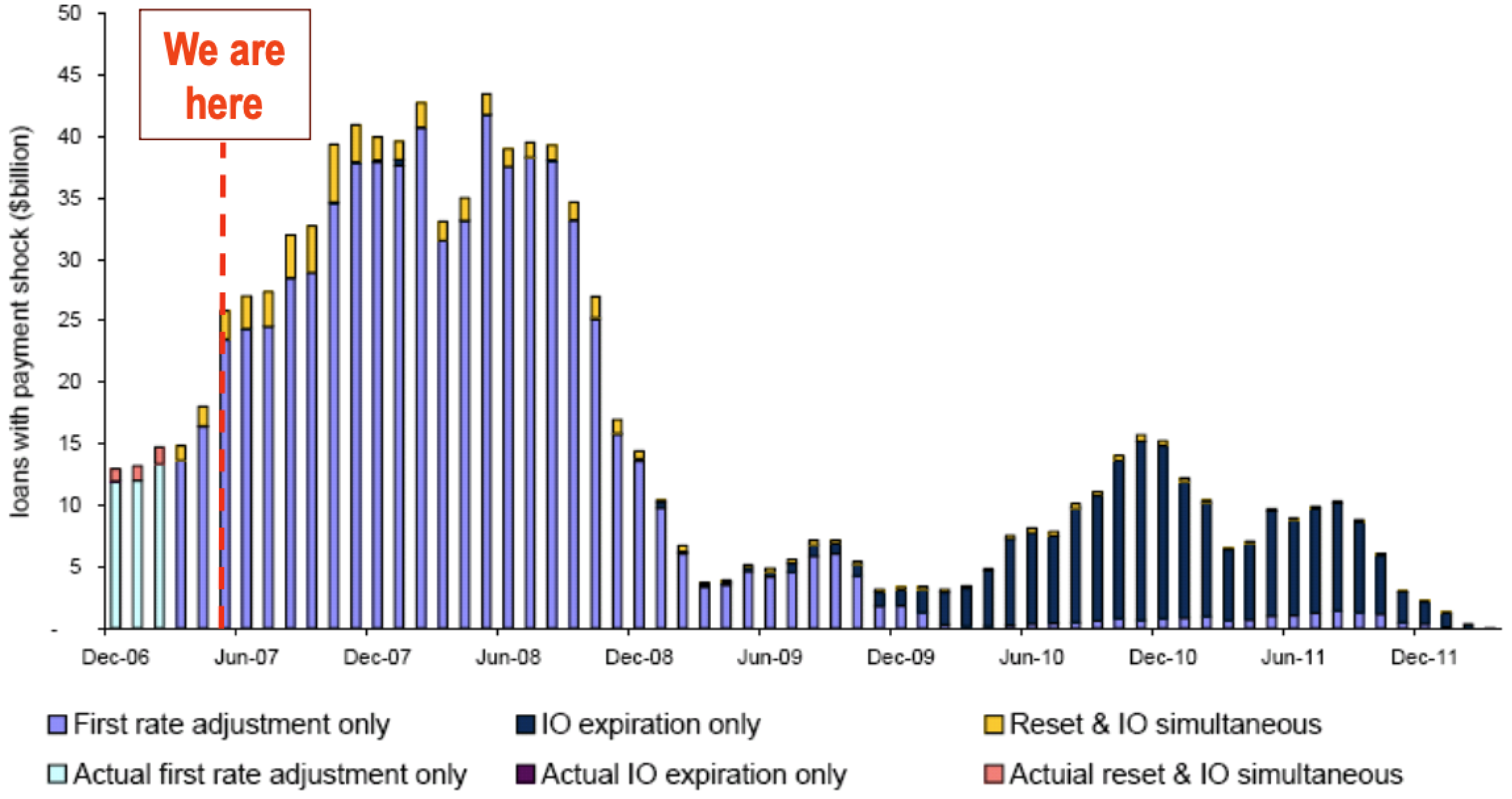


Figure 2
Mortgage originations for home purchase by type
See text for the precise definition of a mortgage sold into the private securitization. *Source*: HMDA.

Figure 11
The effect of PLS credit supply expansion on house prices and construction
The figure plots the coefficients $\{\beta_k\}$ from Equation (5). The left panel uses the natural logarithm of house prices in a ZIP code, and the right panel uses the natural logarithm of new housing permits in an MSA as the dependent variable. The regressions are weighted by the share of total occupied housing units in ZIP code z or MSA m in 2000. Standard errors are clustered at the MSA level.

Subprime Loans Exposed to Higher Interest Rates (ARMs Reset) and Experienced Rising Defaults

~$800 Billion of sub-prime mortgages to reset



Sources: LoanPerformance, Deutsche Bank

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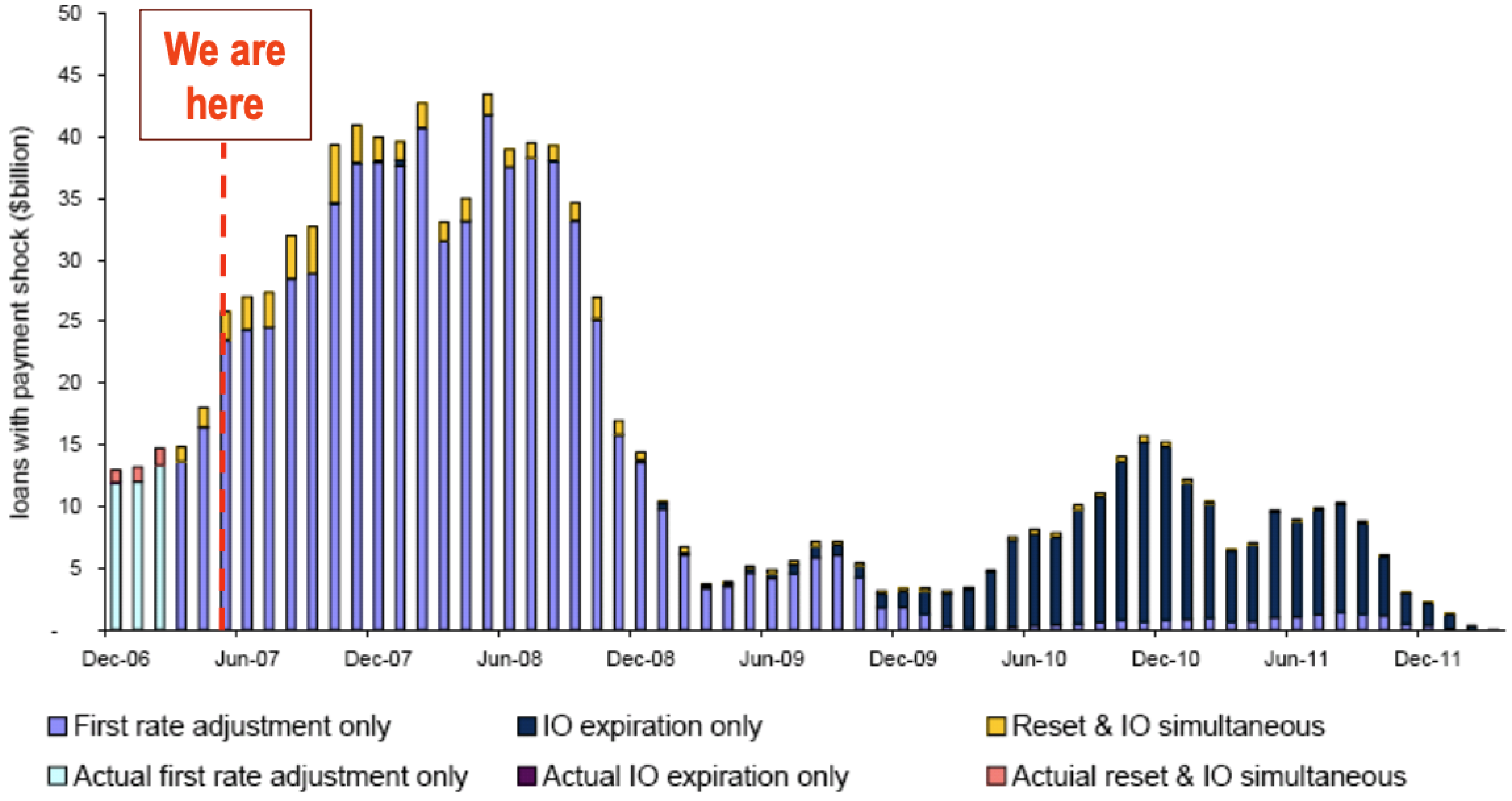
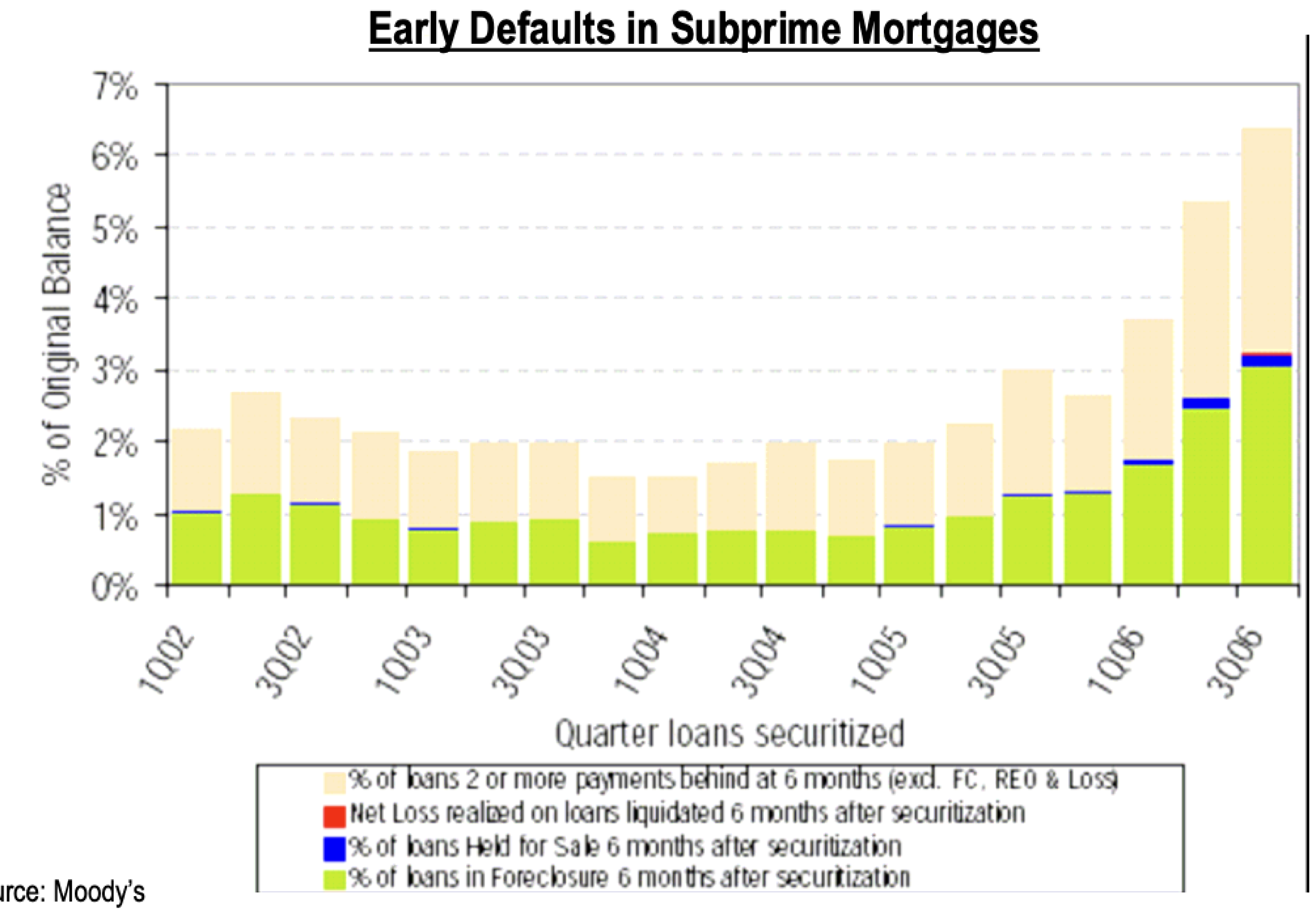


Sources: LoanPerformance, Deutsche Bank

More loans are experiencing early defaults



Source: Moody's

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Mortgage Credit Increased Across Credit Scores
Defaults to Prime Consumers Made Crisis Much Bigger

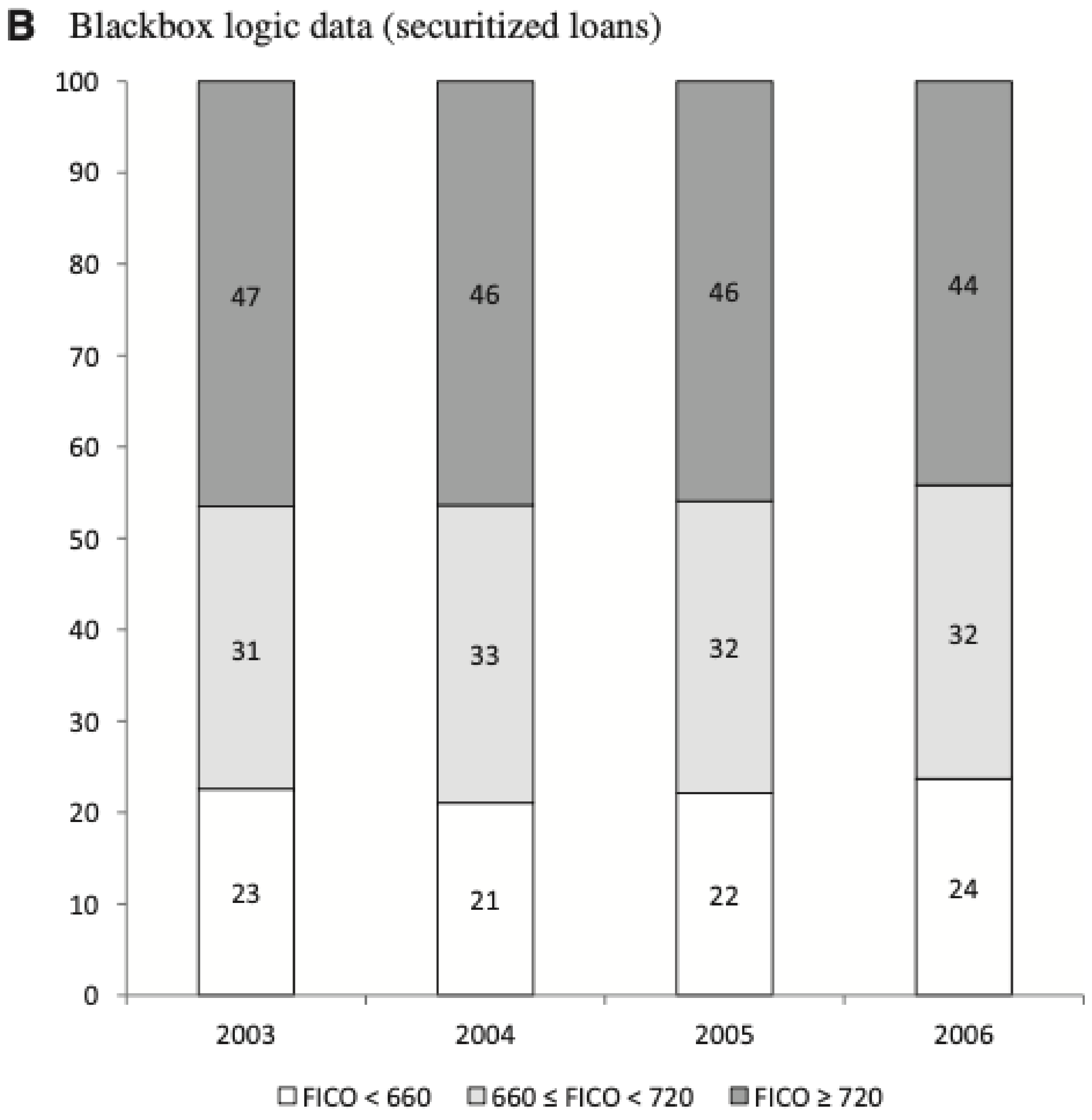

Figure 2
Mortgage origination by credit score
This figure shows the fraction of total dollar volume of purchase mortgages in the LPS data (panel A) and in the Blackbox Logic data (panel B) split by FICO score. A FICO score of 660 corresponds to a widely used cutoff for subprime borrowers, and 720 is near the median FICO score of borrowers in the LPS data (the median is 721 in 2003, 716 in 2004, 718 in 2005, and 715 in 2006). The sample includes ZIP codes with nonmissing house price data from Zillow.

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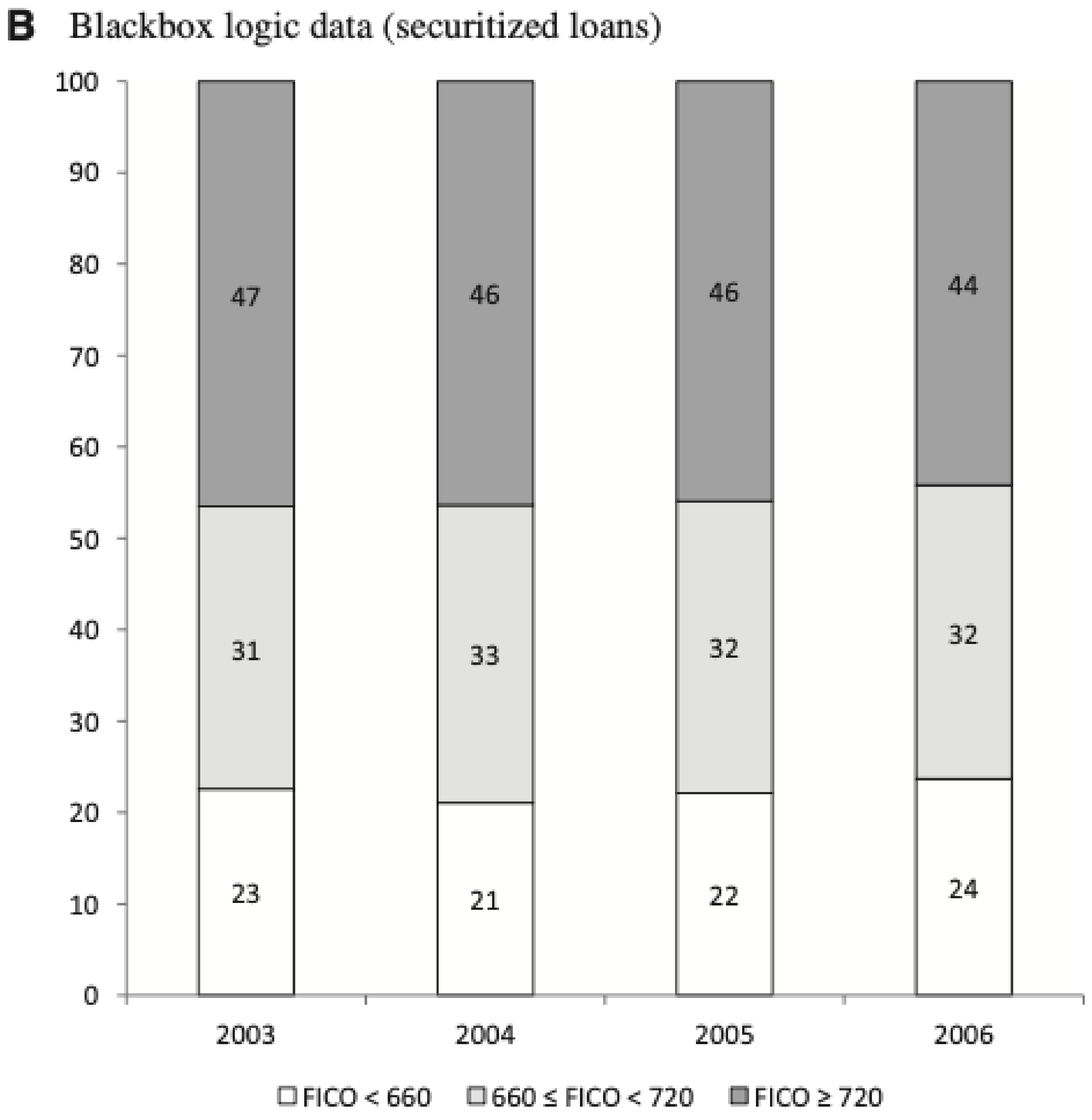

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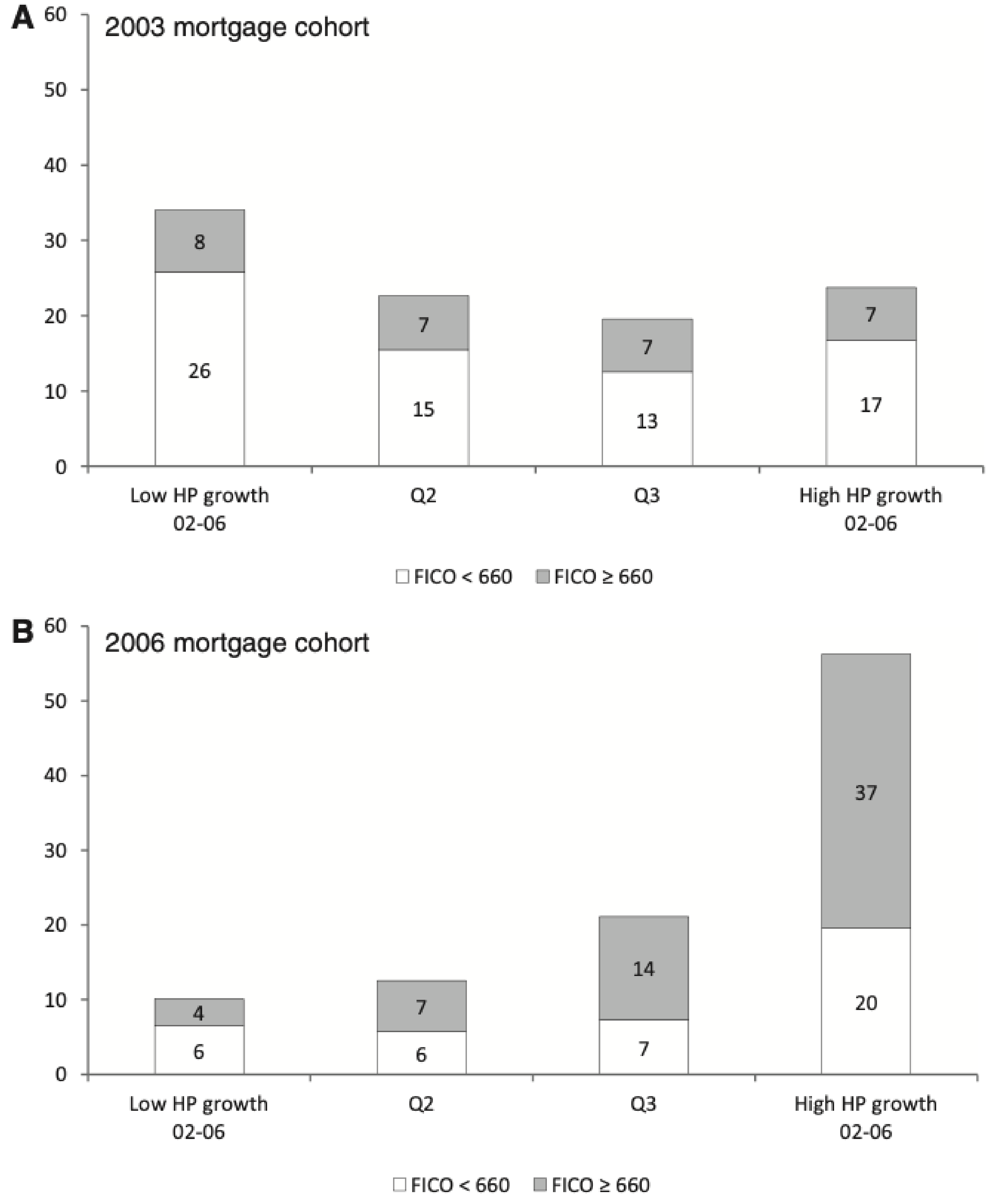
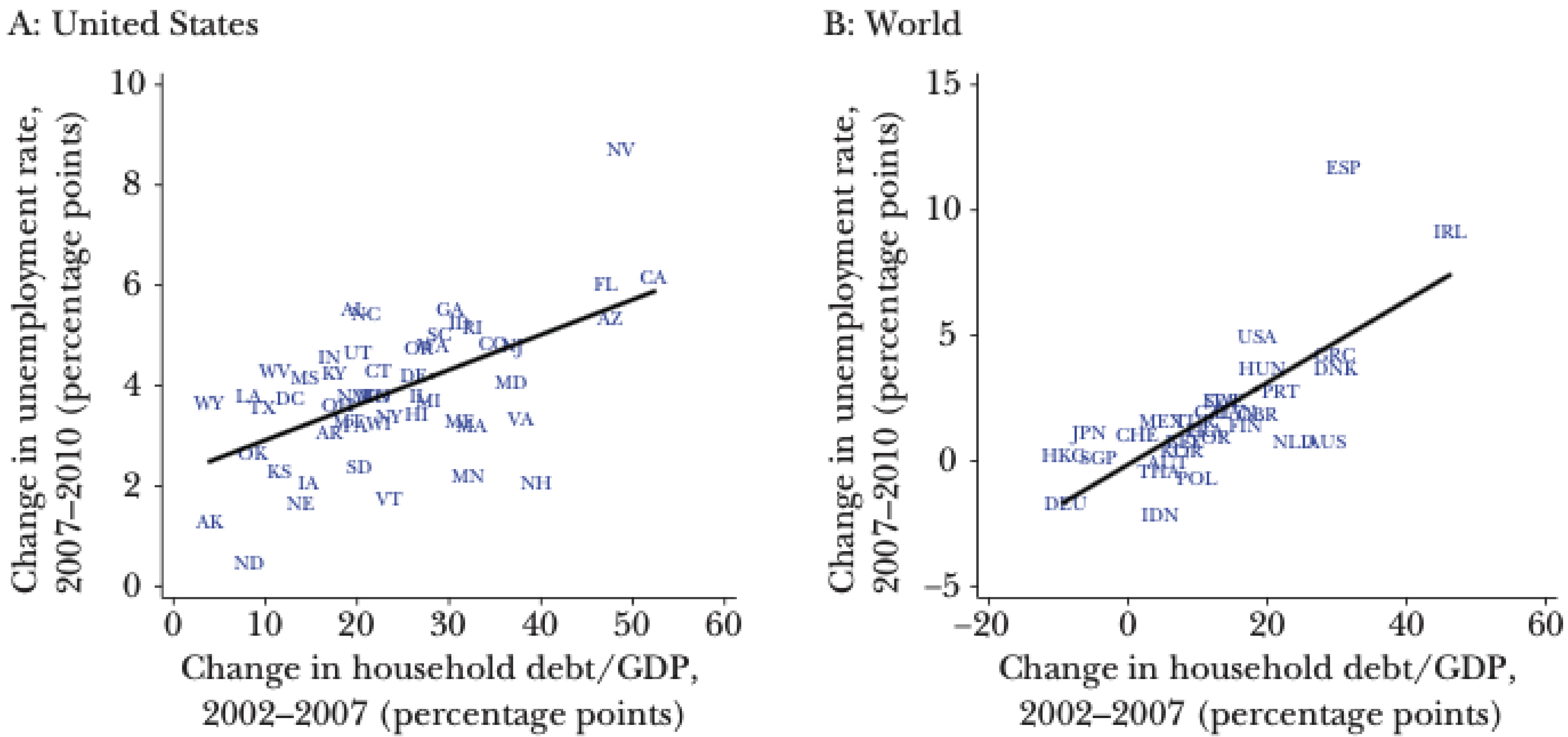


Figure 7
Delinquency by house price growth and credit score
The figure shows the fraction of the dollar volume of purchase mortgages more than ninety days delinquent at any point during the three years after origination for the 2003 and 2006 origination cohorts. Panels show splits by quartiles of house price appreciation that the ZIP code experienced between 2002 and 2006, as well as by whether the borrower is above or below a credit score of 660 (a common FICO cutoff for subprime borrowers). In each panel fractions sum to 100 (the total amount of delinquent mortgages for each cohort), up to rounding error. Data are from the 5% sample of the LPS data set, and the sample includes ZIP codes with nonmissing Zillow house price data.

Areas (in U.S. and Globally) Where Household Debt/GDP Increased the Most 2002–2007, Experienced Larger Increases in Unemployment 2007–2010

Figure 1
Household Debt and Unemployment



Note: Figure 1 shows the relationship between change in household leverage and change in unemployment rate, both across states within the United States and across countries in the world.

Boom Cycle

Bust Cycle

Securitization

Jargon-Busting Securitization

- Create tradeable securities from illiquid assets (e.g., mortgages). Combine loans into a “pool” and sell claims on the whole pool (instead of individual loans).
- Examples Retail Mortgage Backed Securities (RMBS) and Commercial Mortgage Backed Securities (CMBS) are types of Asset Backed Securities (ABS)
- Simplest version of RMBS (no tranching):
  - All investors who buy RMBS are treated equally
  - E.g., 100 loans, each with 10% default rate, and 100 investors owning 1/100 of pool of mortgages.
  - Each investor expects 10% default rate.
- Part of the idea is by pooling you diversify risks.

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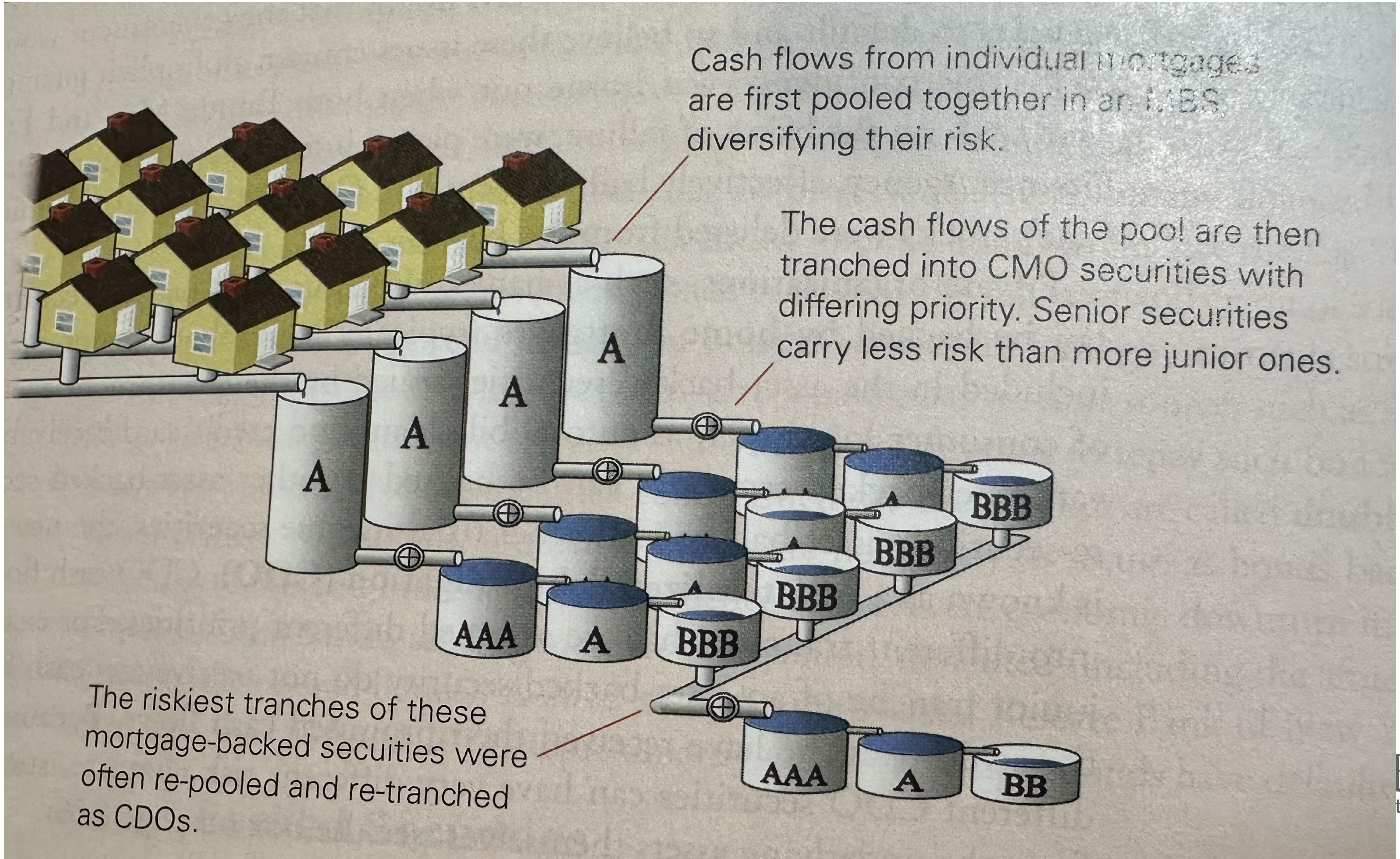
  - Each investor expects 10% default rate.
- Part of the idea is by pooling you diversify risks.
- Mortgages been securitized since 1980s by Government Sponsored Enterprises (GSEs”) Federal National Mortgage Association (“Fannie Mae”) and Federal Home Loan Mortgage Corporation (“Freddie Mac”), Government National Mortgage Association (“Ginnie Mae”). Also, for student loans Student Loan Marketing Association (”Sallie Mae”)
  - Buy mortgages, pool them, guarantee them, and sell them.
  - Guarantee default risk but not risk of early payment (prepayment risk)
  - Limit risk by only taking on low risk loans (good credit score, downpayment, not too large loans etc.)
  - Ginnie Mae backed by U.S. government
  - Fannie Mae, Freddie Mac, and Sallie Mae not explicitly backed by government
- Private-Label Mortgage Backed Security (PLS)
  - 80% fall in in trading by mid-2008. Meant couldn’t value securities. Panic selling.

Securitization



CMO and CDO Tranches

- Collateralized Mortgage Obligations (CMOs) take bunch of MBS.
- Re-securitize CMOs into Collateralized Debt Obligation (CDO). Can also produce Synthetic CDOs (replicate cashflows).
- Re-securitize CDOs into CDO-Squared!

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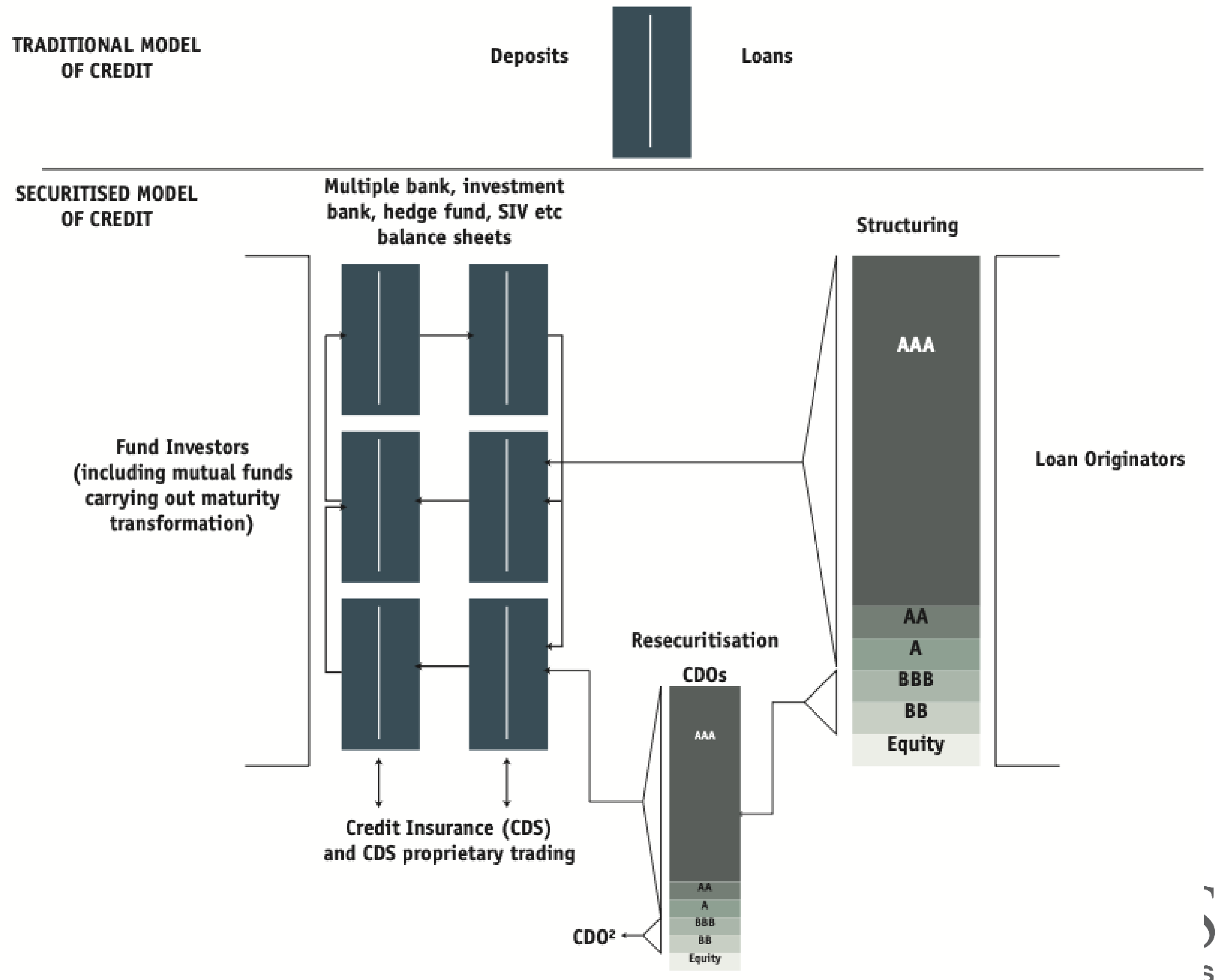
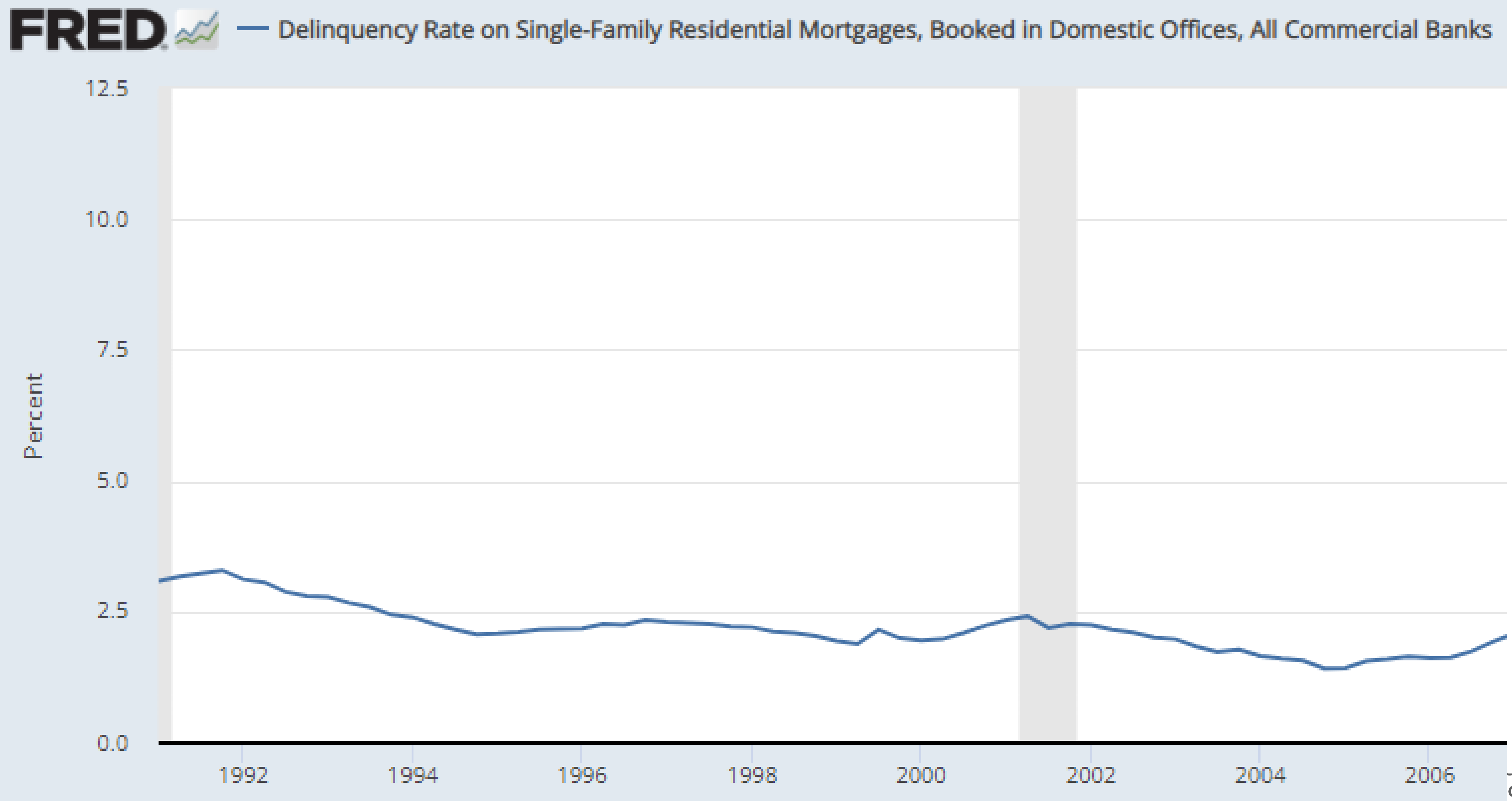
Tranches:

- **Senior Tranches:** Highest priority (only experience losses if all loans in mezzanine and equity defaulted)
- **Mezzanine:** Middle priority
- **Equity Tranches:** Lowest priority (any defaults go here first)



Historically, Loan Defaults Low Pre-2007
Defaults Appeared Independent Pre-2007



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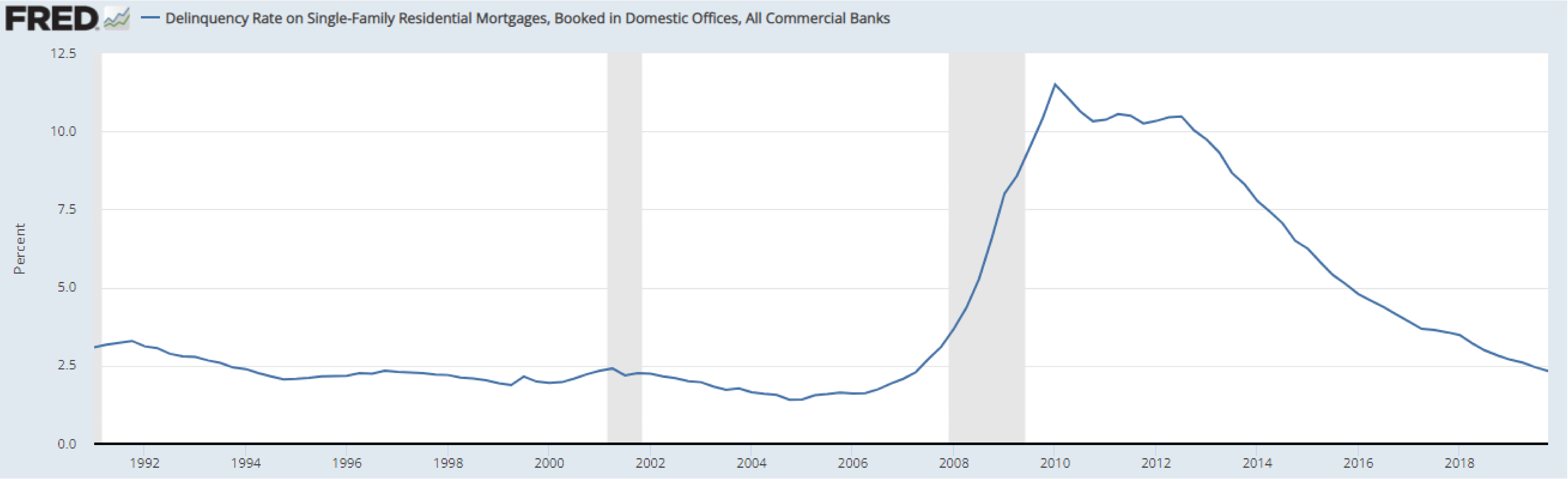
Loan Defaults Spiked 2007-2010
Defaults Appeared Correlated



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Credit Ratings Agencies (CRAs)

- Ratings agencies (S&P, Fitch, Moody's) supposed to independently evaluate risk.
- However, they are paid by the firm issuing the security.
- This crates conflict of interest.
- If they didn't give a high rating, there was threat that customer would go to their competitors.
- Hard to evaluate risks of many products as they were bespoke and traded over-the-counter.

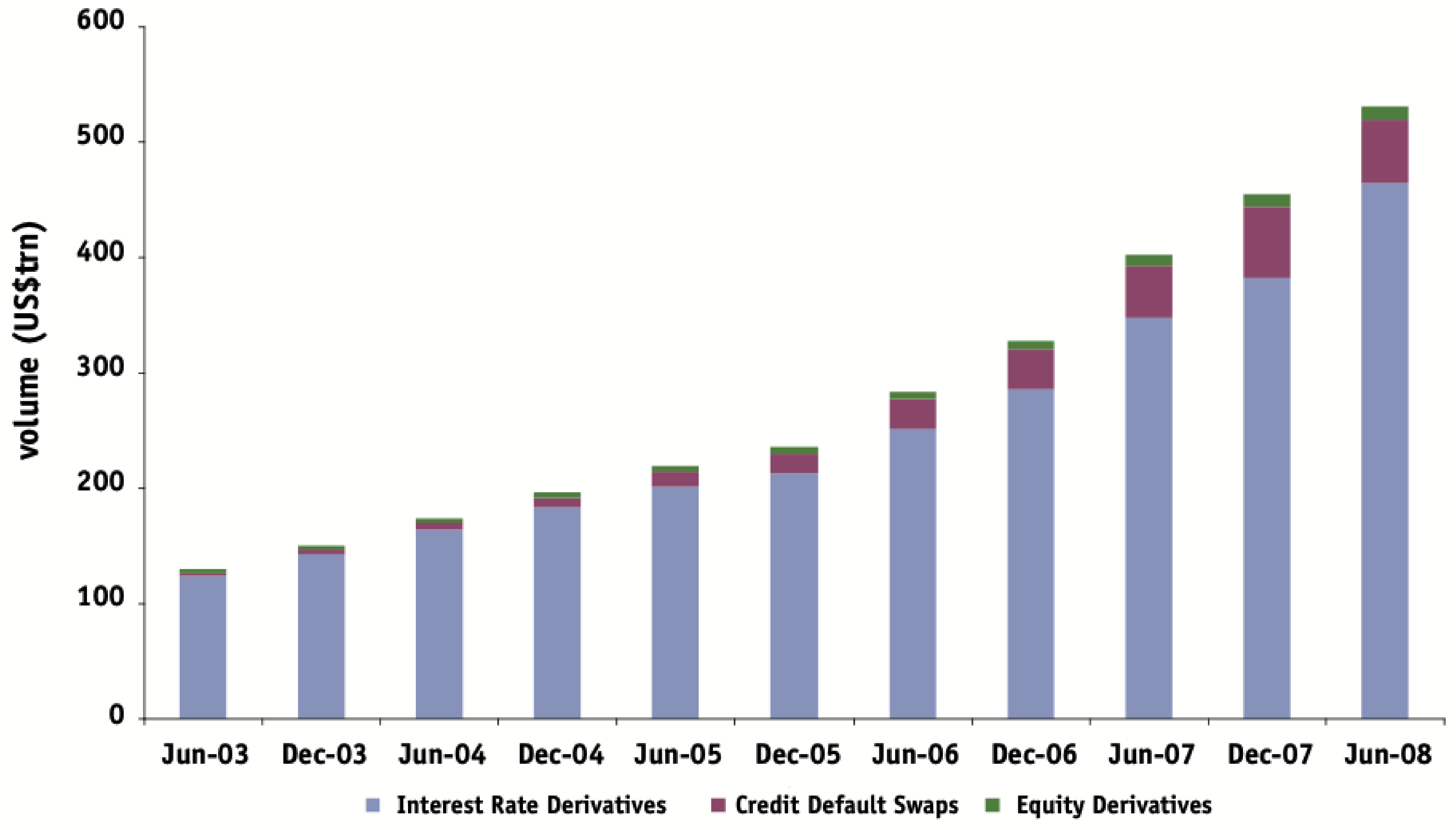
Leverage

Banks Highly Leveraged in 2007

	Trading Book	Year-on-Year Increase	Total Assets	Gross Leverage
Bear	**122.5**	**13.0**	**395.4**	**33.5**
Lehman	**313.1**	**86.5**	**691.1**	**30.7**
Citigroup	**539.0**	**145.1**	**2,187.6**	**19.3**
JPMorgan	**491.4**	**125.7**	**1,562.1**	**12.7**
Merrill	**234.7**	**30.9**	**1,020.1**	**32.0**
Goldman	**406.4**	**118.0**	**1,119.8**	**26.2**
Totals	**2,107.2**	**519.2**	**6,976.1**	**20.2**

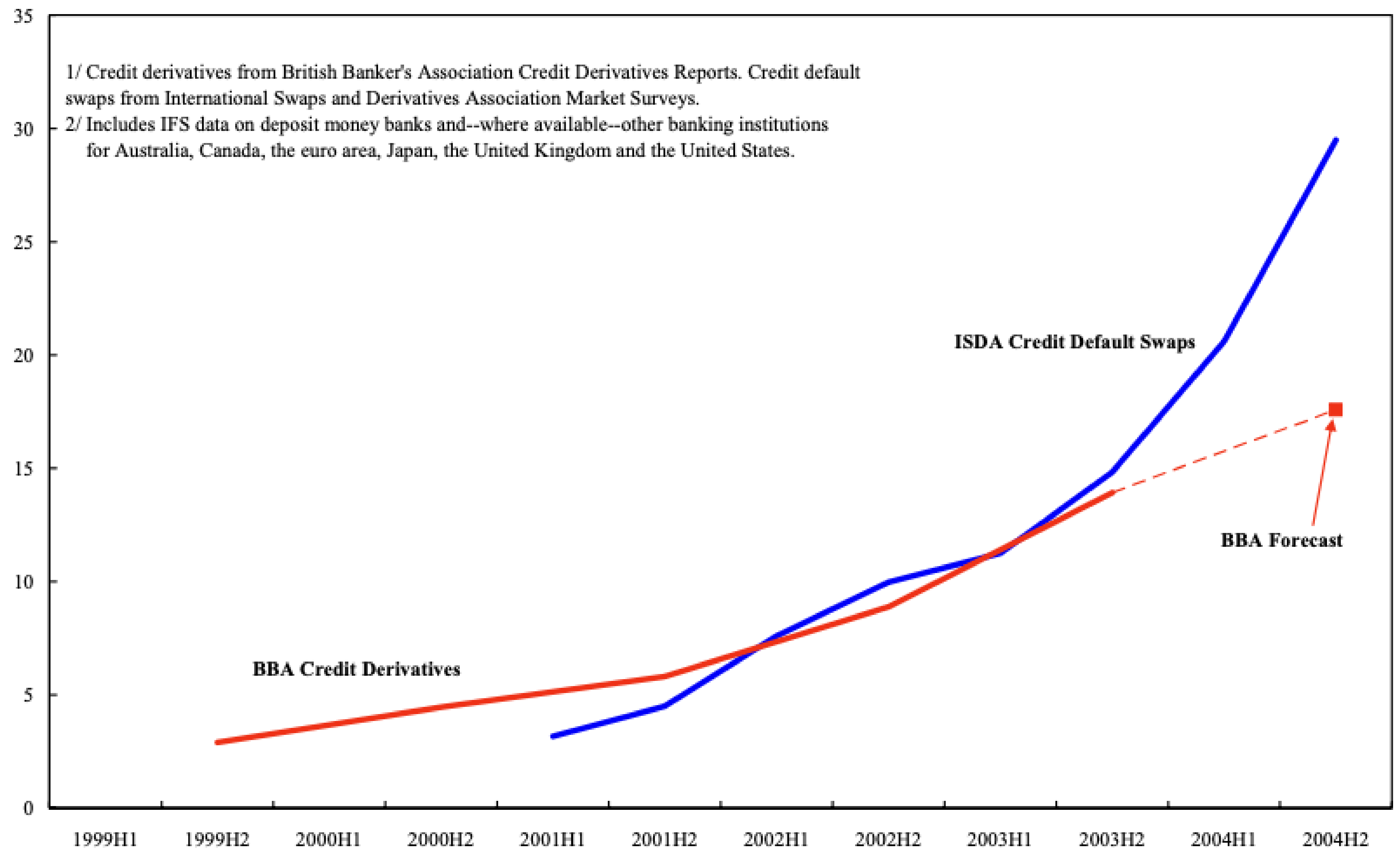
Growth in OTC Derivatives - especially Credit Default Swaps (CDS) Meant That Defaults Propagated Through Financial System

Exhibit 2.8: OTC derivative volume by product type



Source: ISDA

Figure 1: Credit Derivatives and Credit Default Swaps 1/
(In Percent of Private Sector Bank Credit 2/)



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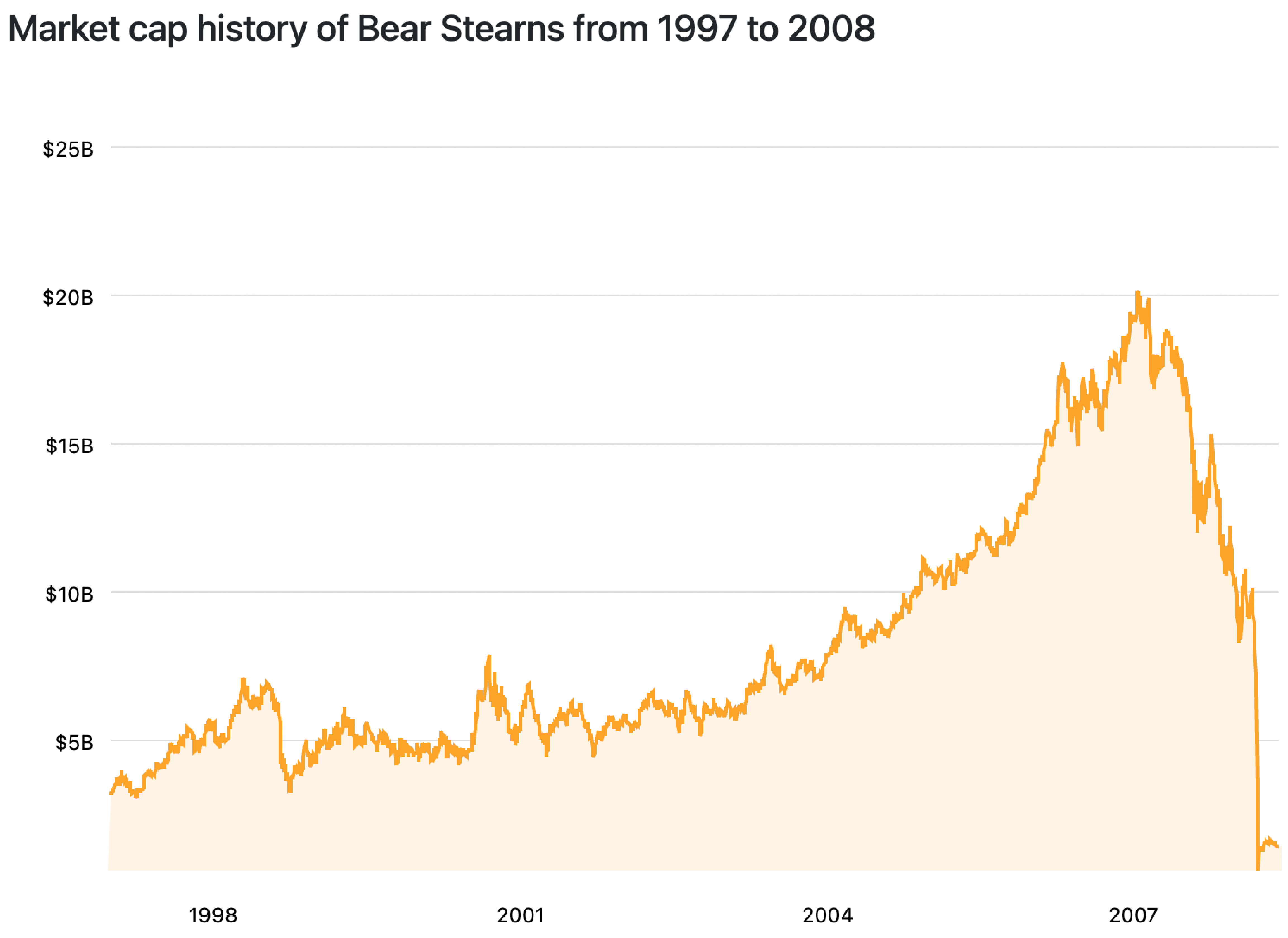
Bear Sterns Failed in March 2008: Federal Reserve Bailout Loan & Sold to JP Morgan Chase (Approved in May)

- Bought for $1.2 bn ($10 a share, previously traded $170+ in 2006/2007)



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Lehman Brothers Filed for Bankruptcy on Monday 15 September 2008

Market cap history of Lehman Brothers from 1997 to 2008

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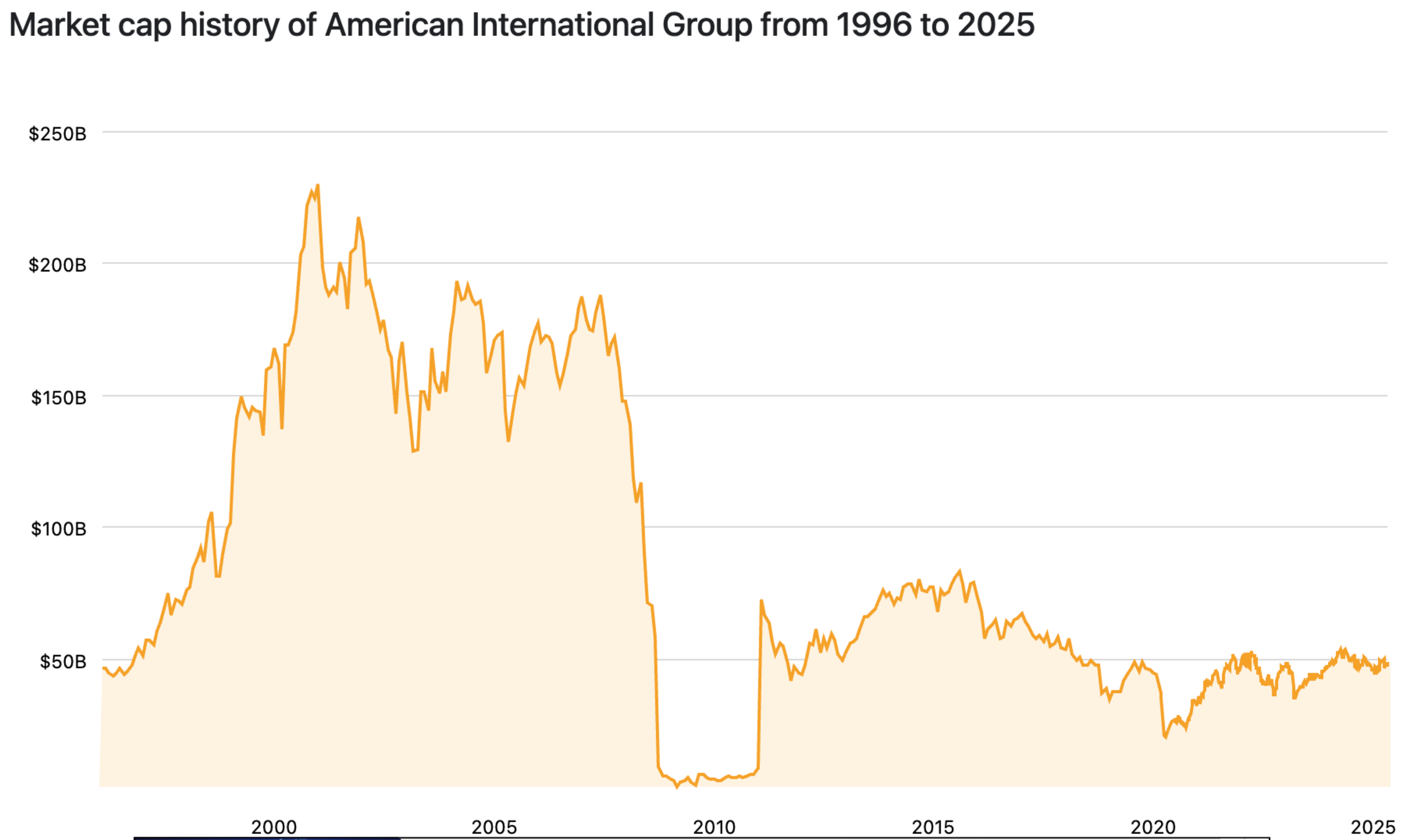
AIG Failed 16 September 2008. Bailed out by government at cost of ~$85bn initially but cost grew to ~$180bn

- AIG had written ~$500 bn in Credit Default Swaps (CDS)
- But ~$80bn CDS on CDOs were the real problem.
- Much were done without collateral or hedging.
- Downgrade in AIG meant it had to post more collateral it didn't have ($12 bn shortfall).



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Noone Wanted to Buy (Liquidity Dried Up)
Assets Sold at Large Discounts (Haircuts)
With Limited Trading, Hard to Value Assets

Exhibit 2.7: Largest CRAs global structured finance 1 year transition rates

%	2007			2008		
	Downgrade	Upgrade	Stable	Downgrade	Upgrade	Stable
AAA	1.0	n/a	99.0	23.4	0.0	76.6
AA	4.4	3.5	92.1	34.9	1.8	63.3
A	11.3	4.2	83.9	36.9	2.1	61.0
BBB	20.2	2.9	77.0	40.2	1.0	58.1
BB	21.0	2.3	76.6	44.8	1.7	53.9
B	11.1	1.8	87.1	55.5	1.3	43.5
CCC	34.9	0.7	32.7	78.5	1.0	21.4

CCC figures for 2007 may not add to 100% because transitions to default are excluded.

Source: 1991-2006 & 2007 figures are average from Fitch Ratings, Moody's Investors Services and S&P, while the 2008 figures are for S&P only.

it 2.6: S&P's corporate finance cumulative default rates 1981-2008 (%)

%	Years after the issue of security		
	5	10	15
AAA	0.27	0.55	0.65
AA	0.34	0.83	1.20
A	0.72	1.94	2.91
BBB	2.43	5.16	7.70
BB	9.07	16.02	19.33
B	20.58	28.41	33.14
CCC/C	44.93	50.44	52.93

ate finance refers to long-term local currency credit ratings. It includes industrials, utilities and ial institutions, but excludes public sector ratings and all structured finance vehicles.

Exhibit 1.14

Typical haircut or initial margin
In per cent

	April 2007	August 2008
US Treasuries	0.25	3
Investment grade bonds	0–3	8–12
High-yield bonds	10–15	25–40
Investment grade corporate CDS	1	5
Senior leveraged loans	10–12	15–20
Mezzanine leveraged loans	18–25	35+
ABS CDOs		
AAA	2–4	95[1]
AA	4–7	95[1]
A	8–15	95[1]
BBB	10–20	95[1]
Equity	50	100[1]
AAA CLO	4	10–20
Prime MBS	2–4	10–20
ABS	3–5	50–60

ABS = asset-backed security; CDO = collateralised debt obligation; CDS = credit default swap; CLO = collateralised loan obligation; MBS = mortgage-backed security; RMBS = residential mortgage-backed security. [1] Theoretical haircuts as CDOs are no longer accepted as collateral.

Source: IMF

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Source: IMF

- Government bailout (Troubled Asset Relief Program, TARP) injecting liquidity into system by allowing Treasury to buy up to ~$700bn assets
- Was used less than that and actually turned a slight profit for the Treasury in the end.

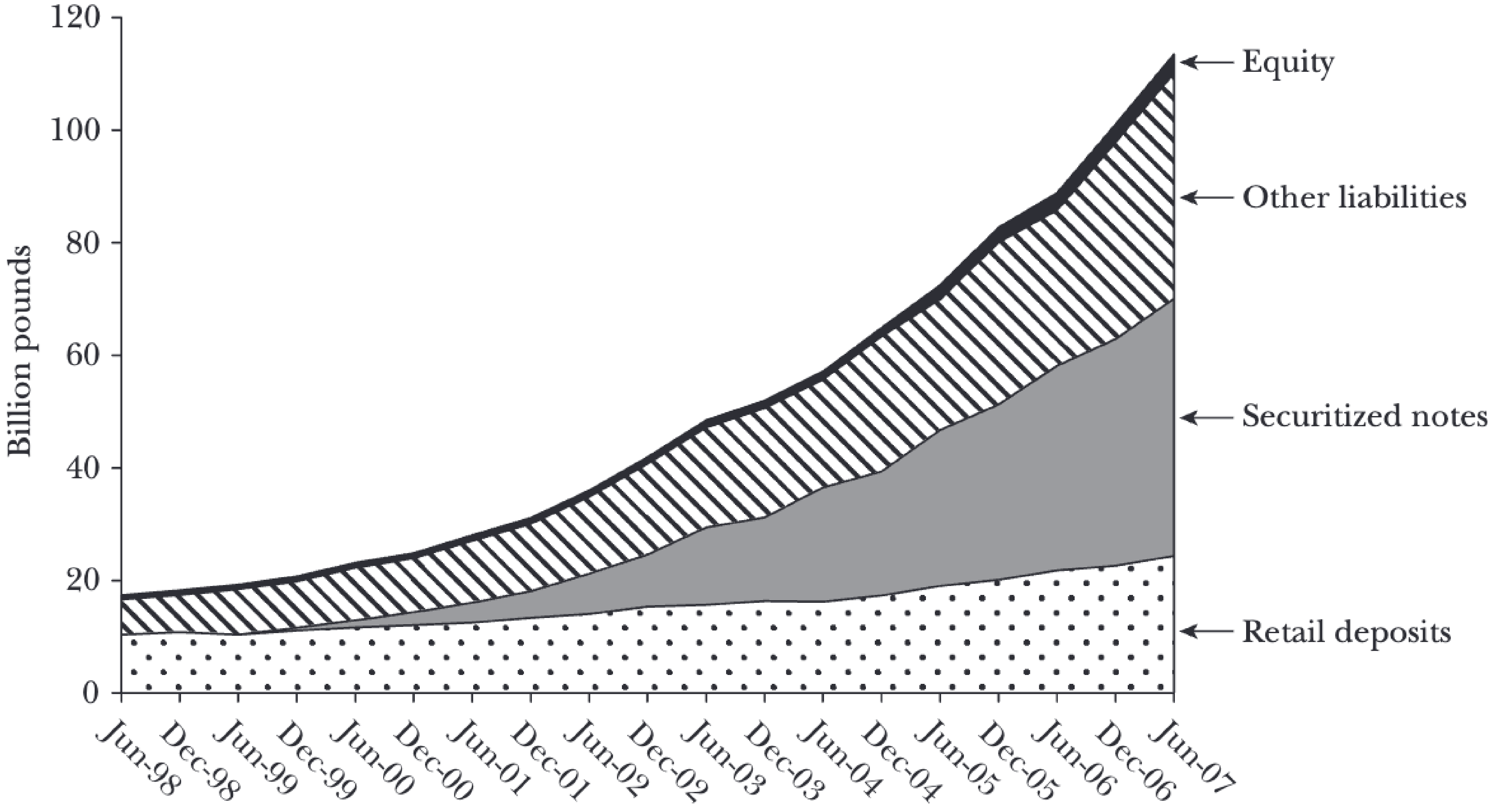
Northern Rock

Northern Rock Bank Run

Police called to break-up Northern Rock panic queues as customers withdraw millions

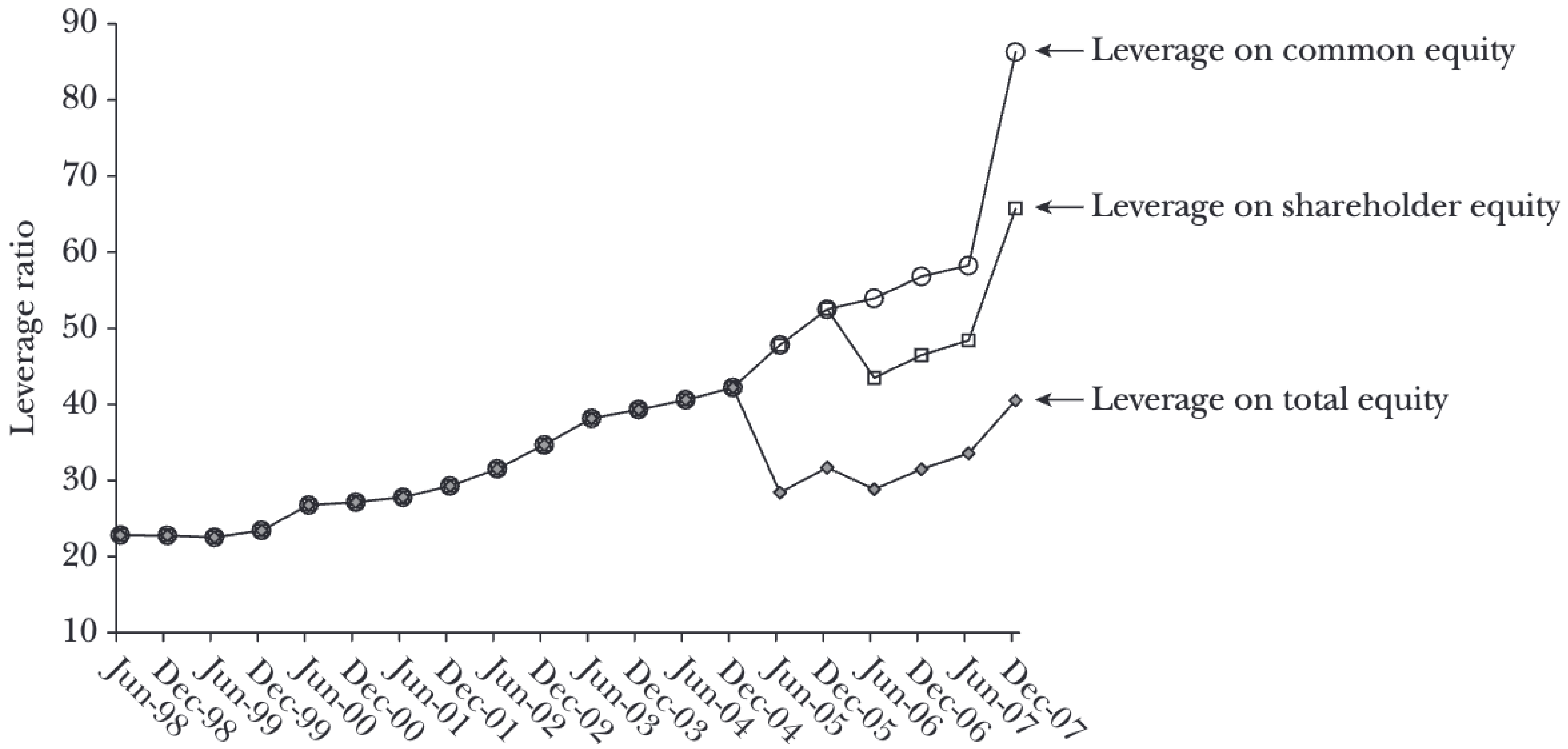
Northern Rock Grew its Leverage

Composition of Northern Rock's Liabilities, June 1998–June 2007



Source: Northern Rock, annual and interim reports, 1998–2007.

Northern Rock's Leverage, June 1998–December 2007



Source: Northern Rock, annual and interim reports, 1998–2007.
Note: The leverage ratio is the ratio of total assets to equity.

Northern Rock

- Business model based on financing from money markets as a cheaper source of finance than depositors.
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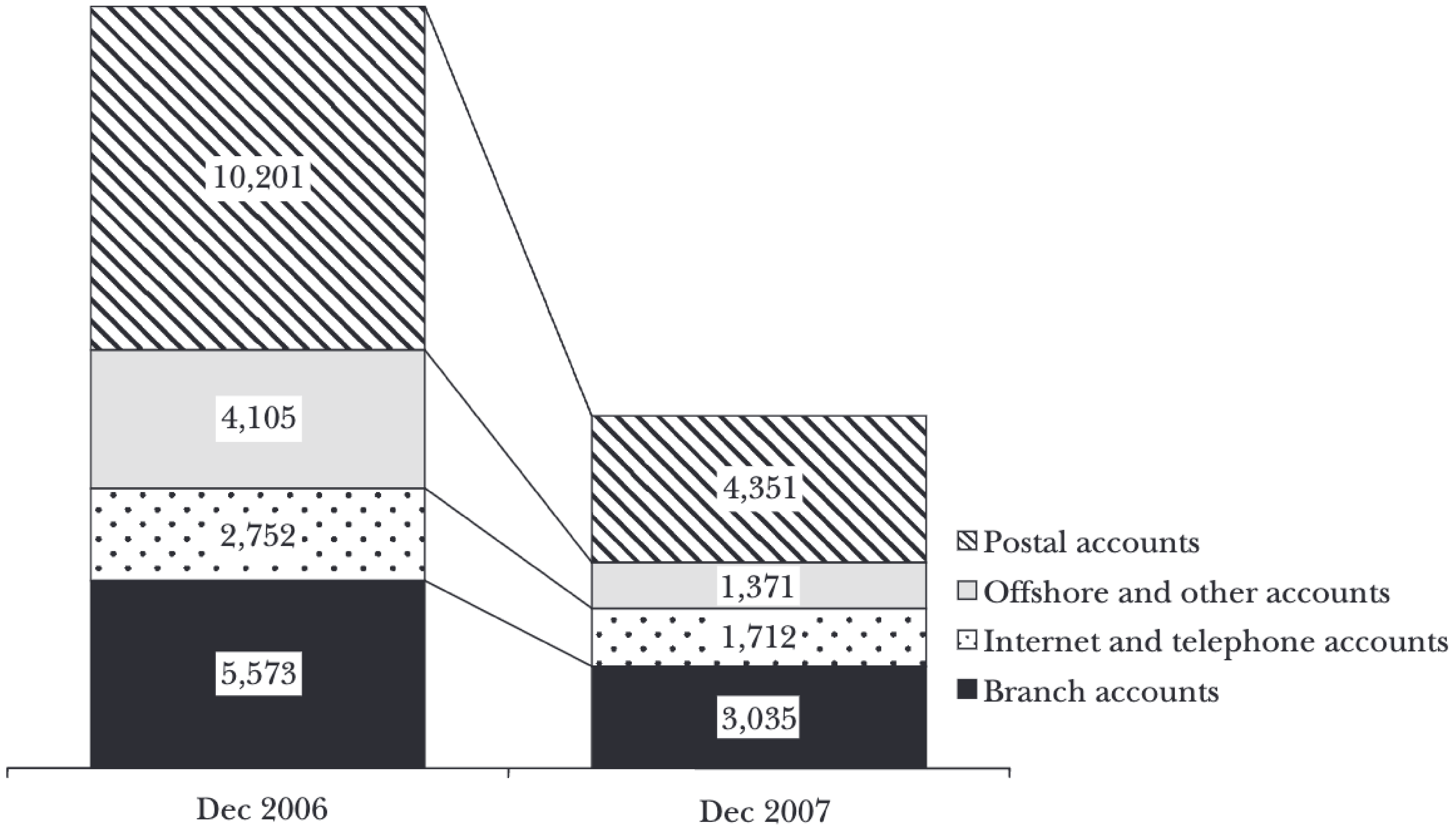
- Bank of England as 'lender of last resort' (viewed firm as solvent but illiquid)
- Savers worried and pulled ~£1bn deposits in one day, another ~£2bn, worsening Northern Rock's situation. 40% fall in share price.
- Government guaranteed savers' deposits.
- Bank nationalized in February 2008 following failed private bids. Sold to Virgin Money in 2011 for £747 m.

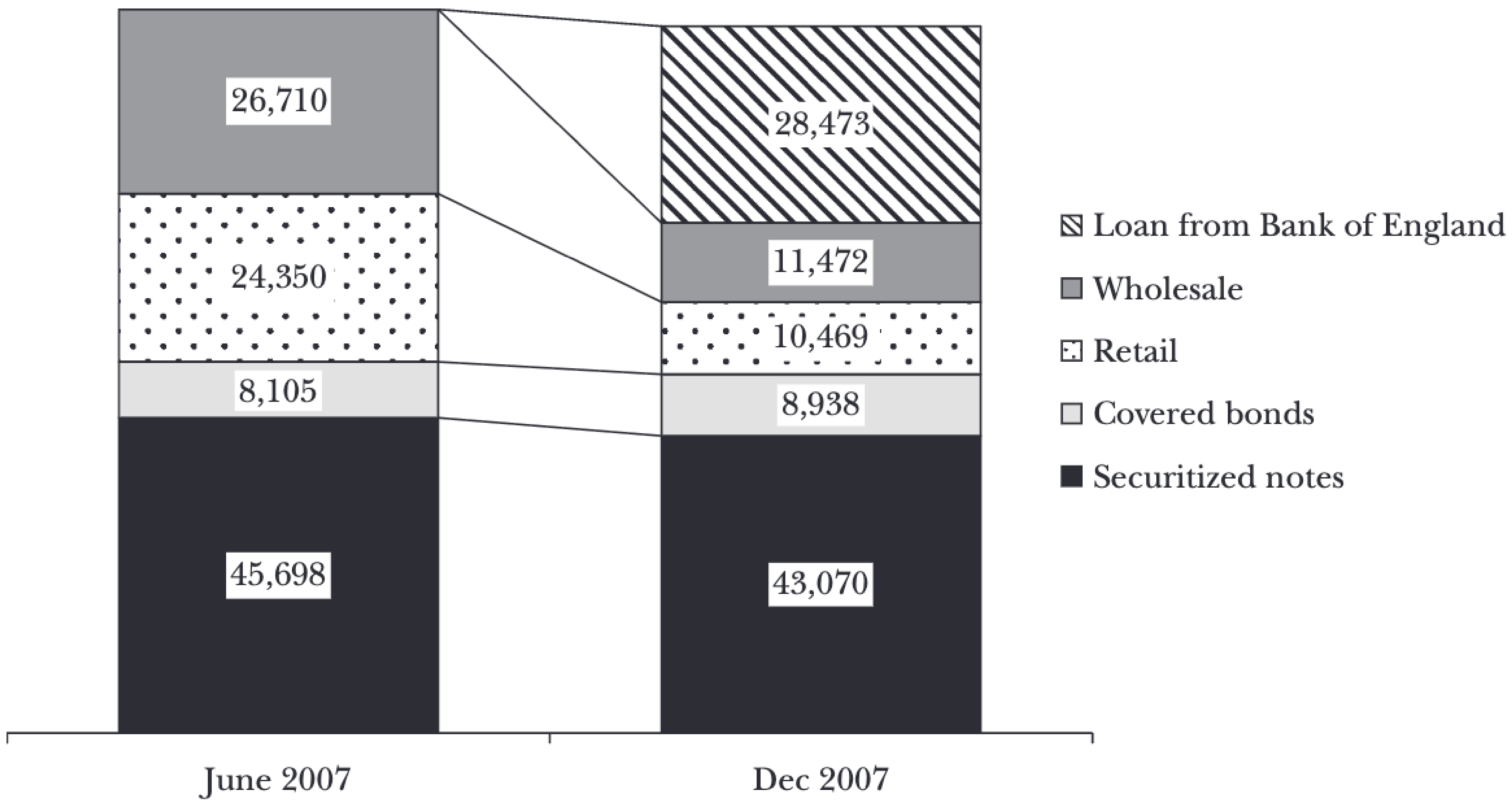
Northern Rock Collapse

Composition of Retail Deposits of Northern Rock Before and After Run
(million pounds)



Source: Northern Rock, annual report for 2007.

Composition of Northern Rock's Liabilities Before and After the Run
(millions of pounds)



Source: Northern Rock annual report for 2007.

Great Recession Recap

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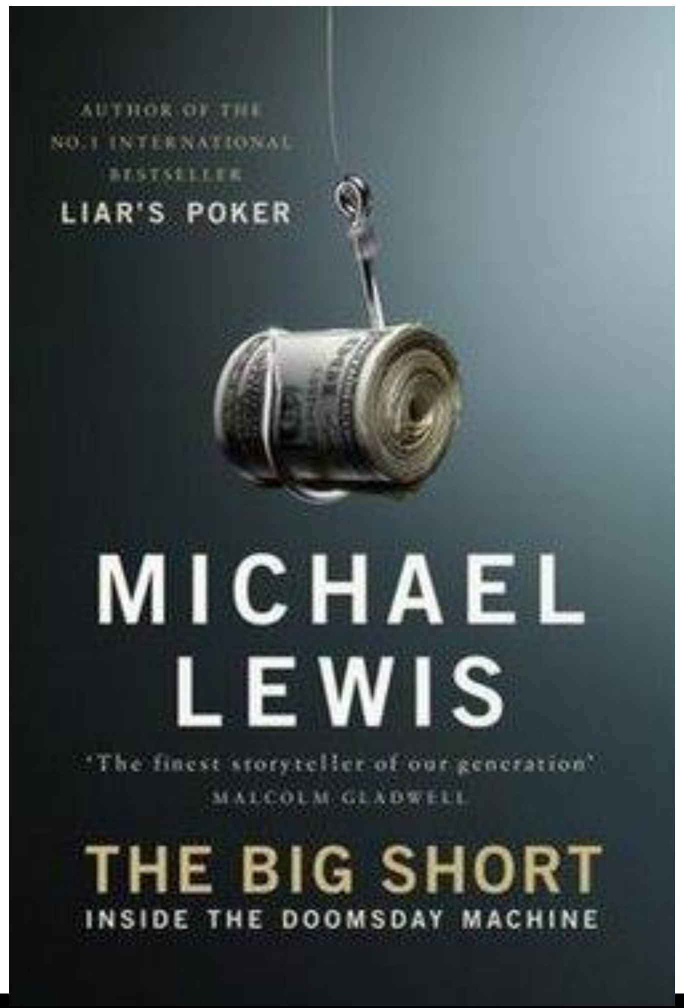
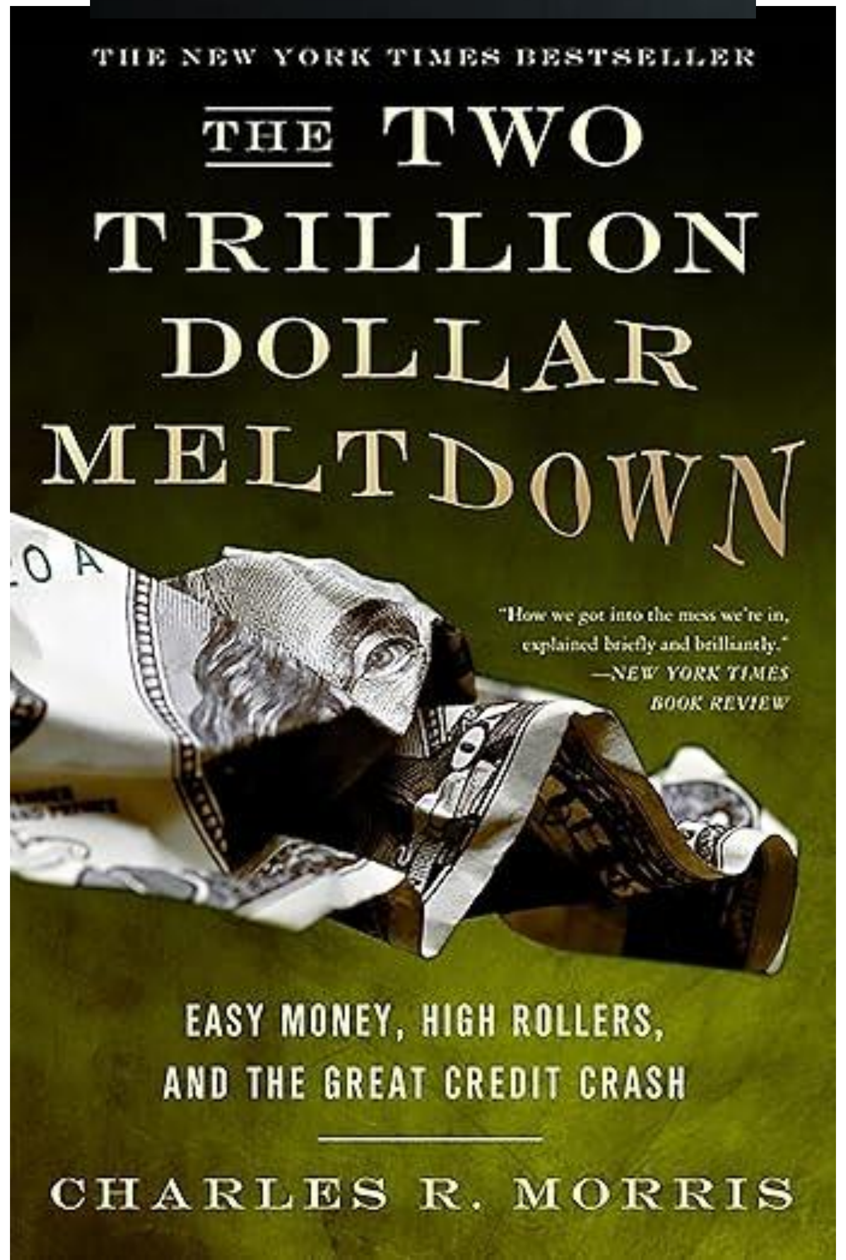
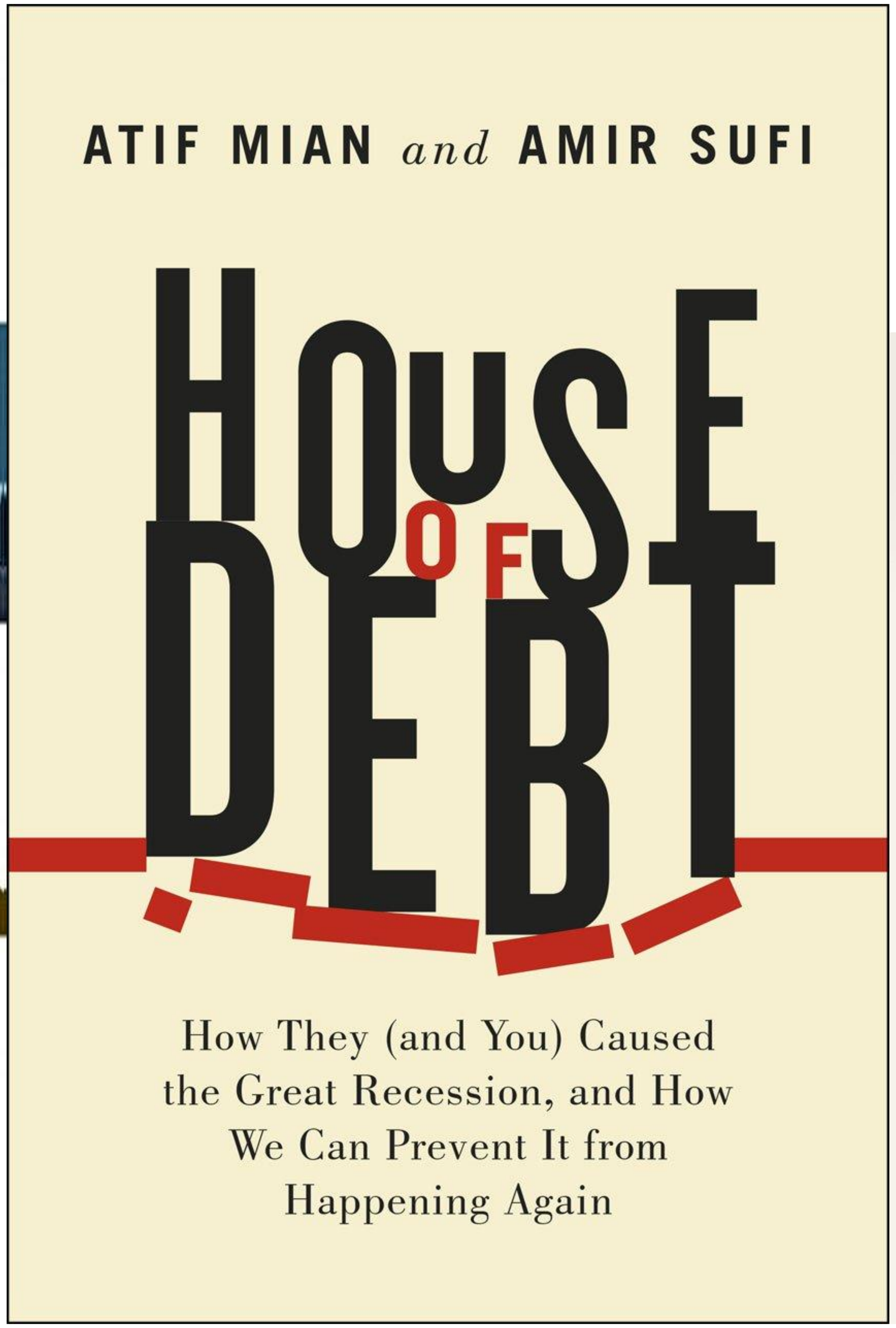
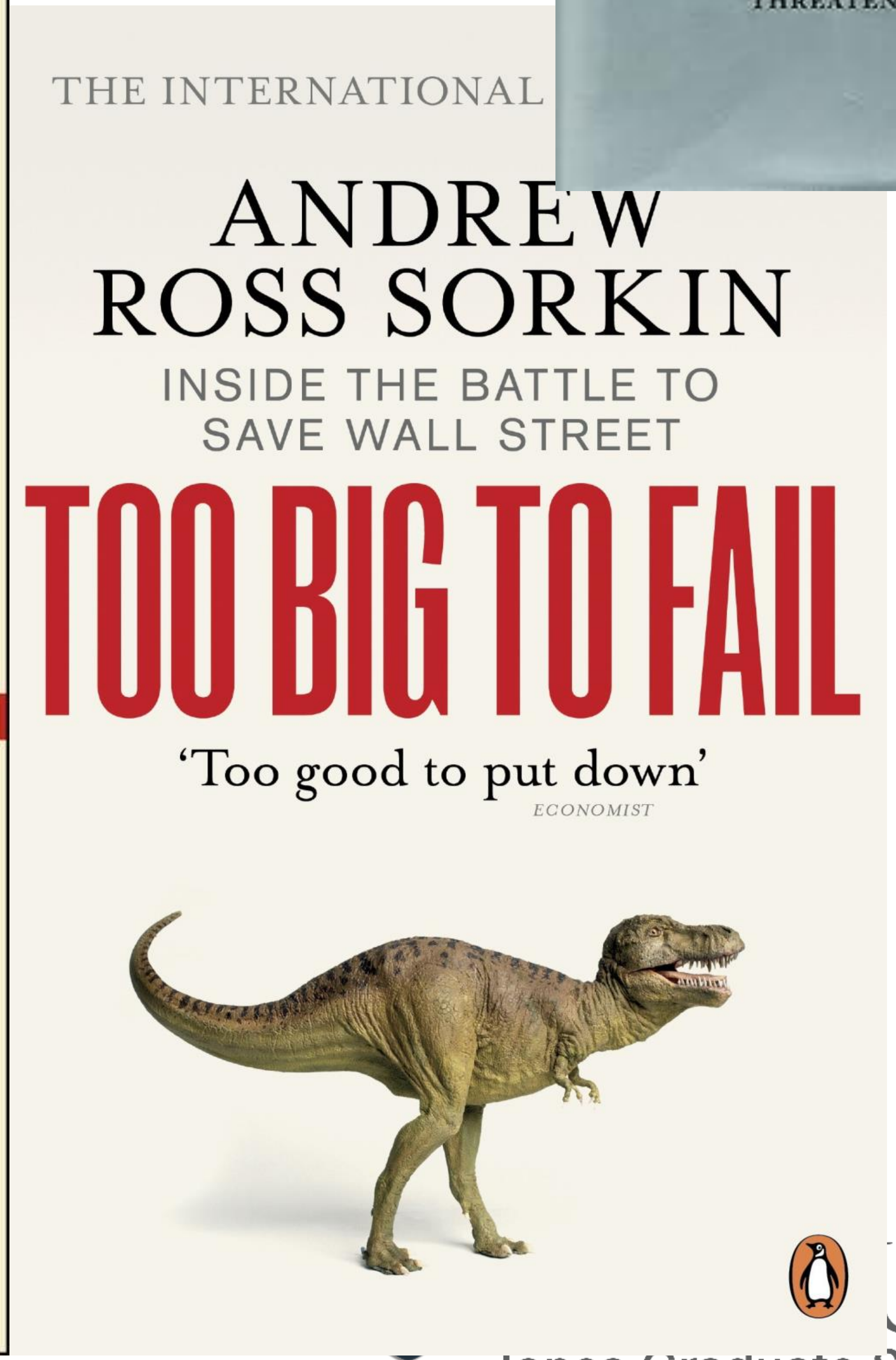
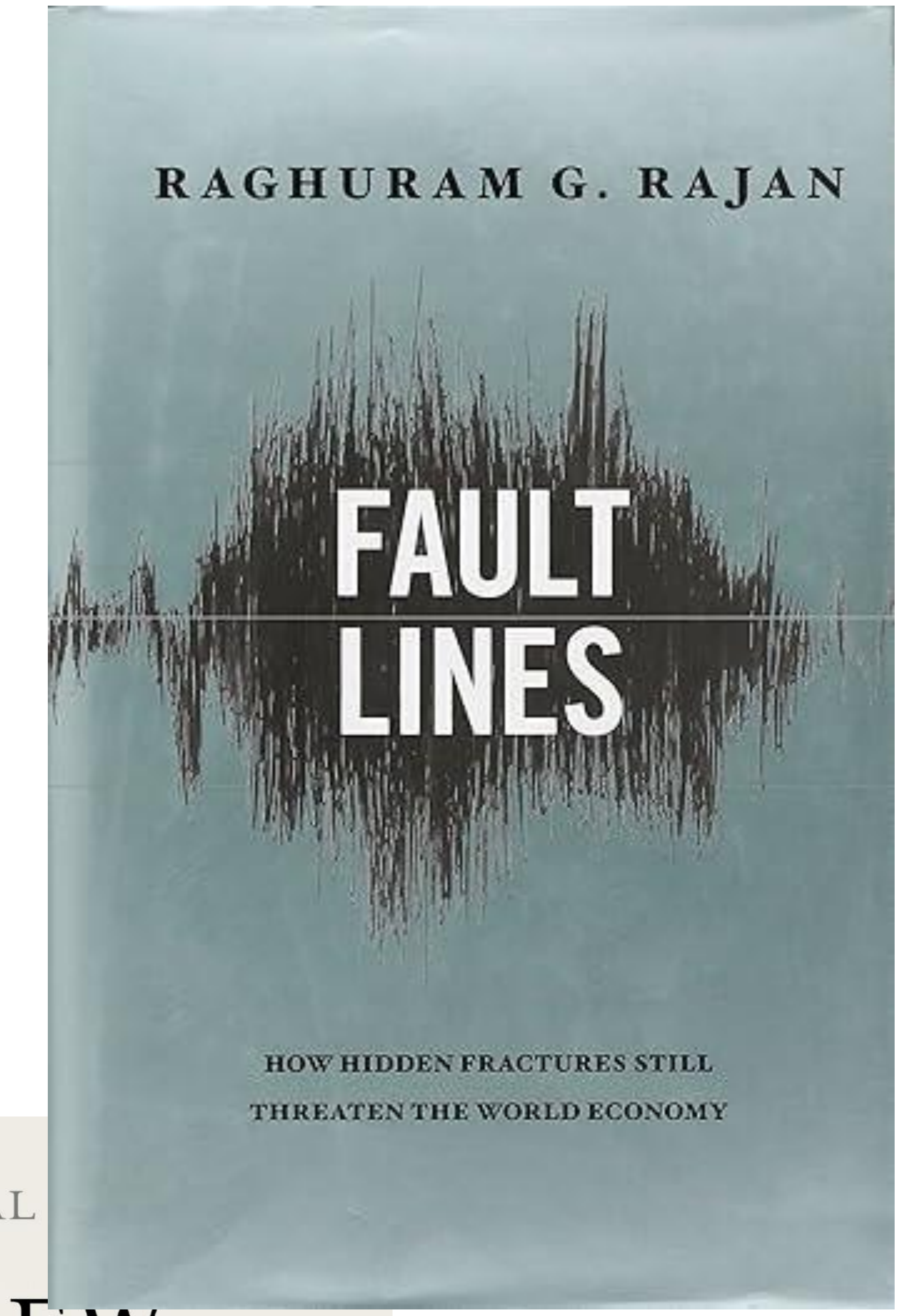
Further Watching & Reading!











Great Recession Led to Bailouts, Higher Government Debt, & More Regulation

- Housing or Mortgage Market Alone Couldn't Cause Crisis
- Needed Propagation Mechanism of Leverage, CDOs, CDSs (Banking Crisis)
- Government actions prevented a Great Depression 2 but…problems remain of banks being "Too Big To Fail" and, more generally, "Crony Capitalism" (Luigi Zingales)
- Many regulations introduced under Dodd-Frank Act to try to prevent happening again:
  - Stress testing
  - Regulated derivatives
  - Credit rating agency oversight
  - Volker rule limiting risky trading by banks of depositors' money
  - Tightened mortgage lending
  - Consumer Financial Protection Bureau (CFPB) created to "ensure that markets for financial products are fair, transparent, and competitive…and to protect consumers from unfair, deceptive, or abusive acts and practices"

Consumer Finance

U.S. Household Assets

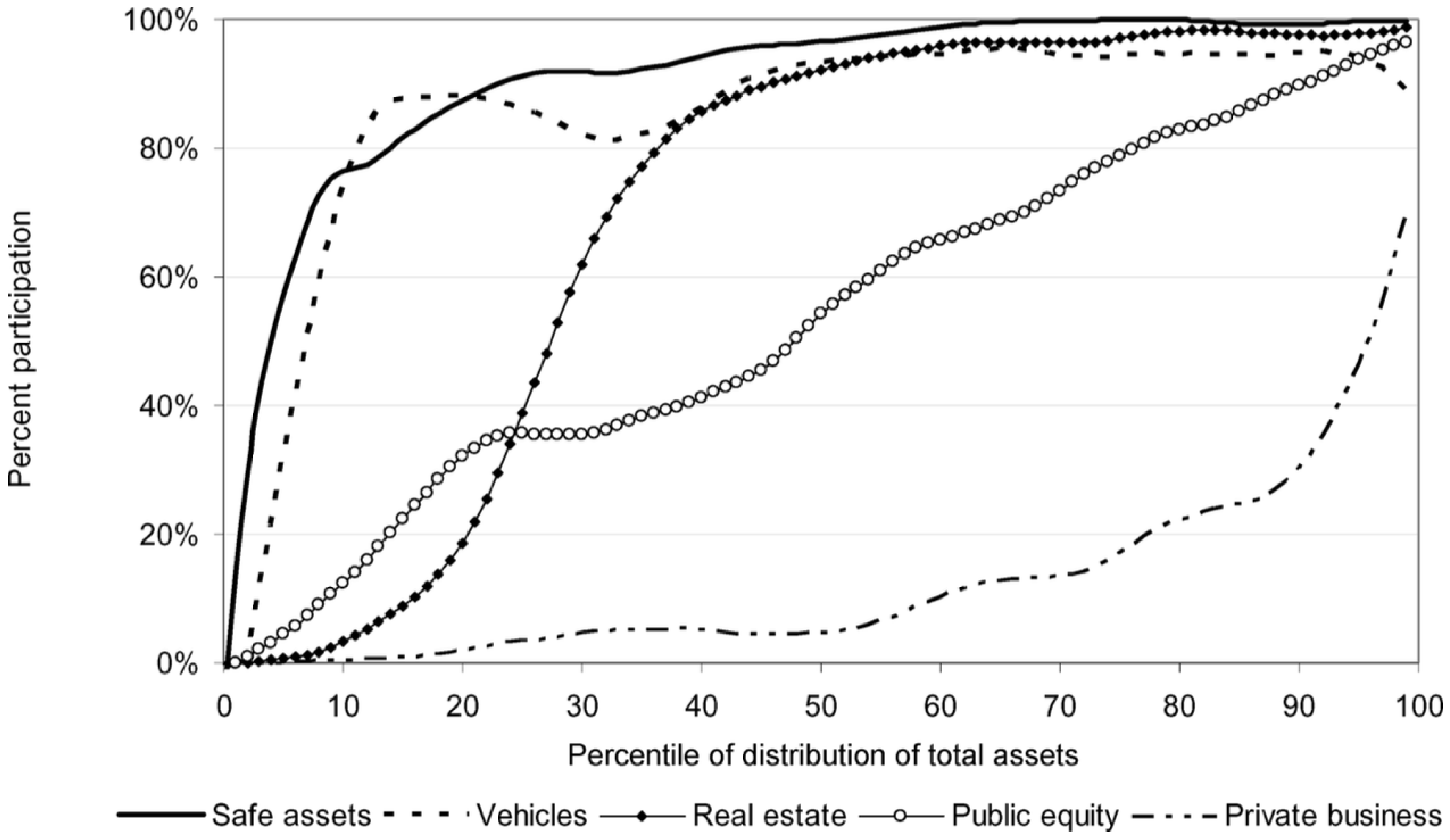
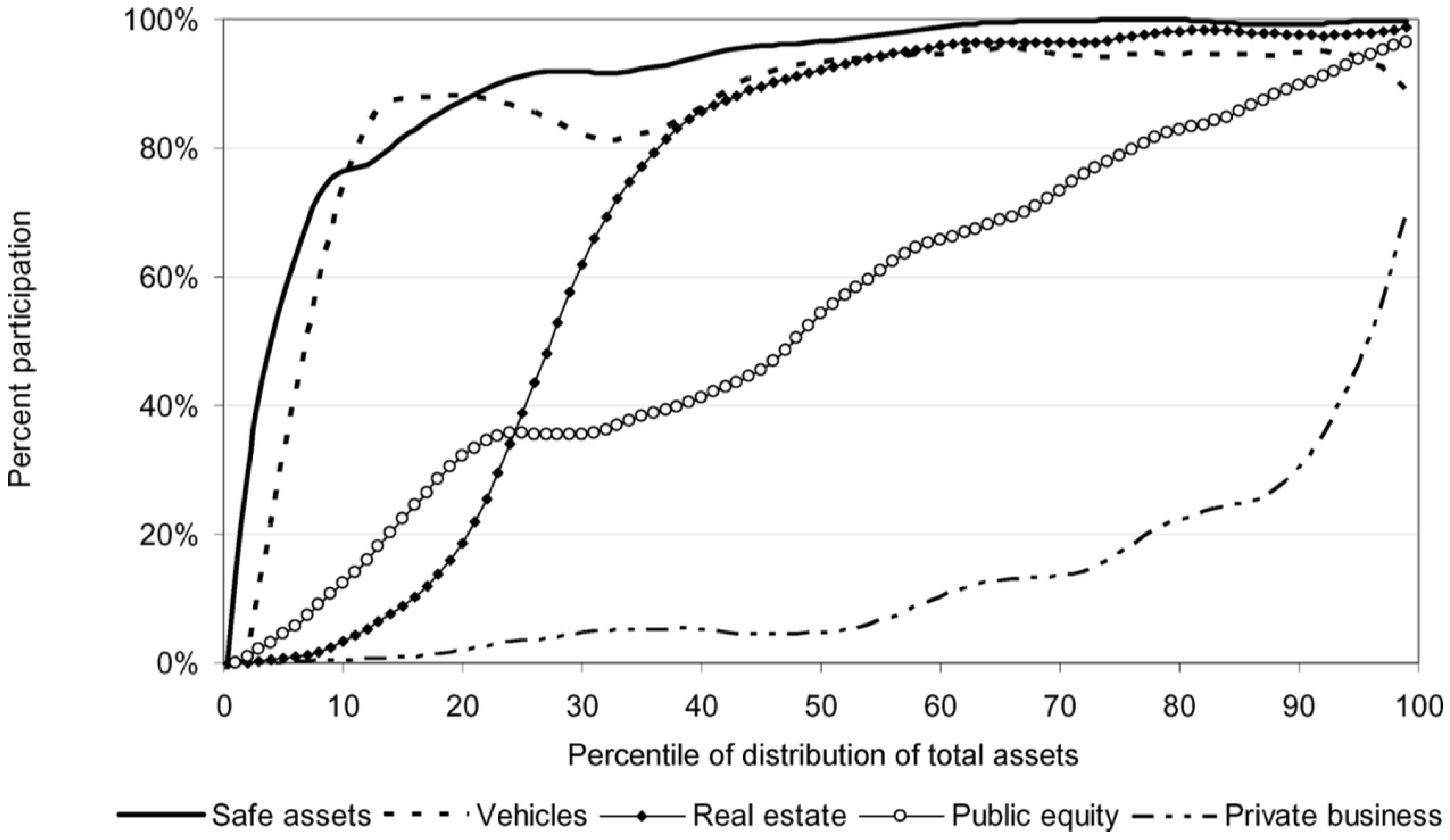


Figure 2. Participation rates by asset class. The cross-sectional distribution of asset class participation rates in the 2001 Survey of Consumer Finances.

U.S. Household Assets



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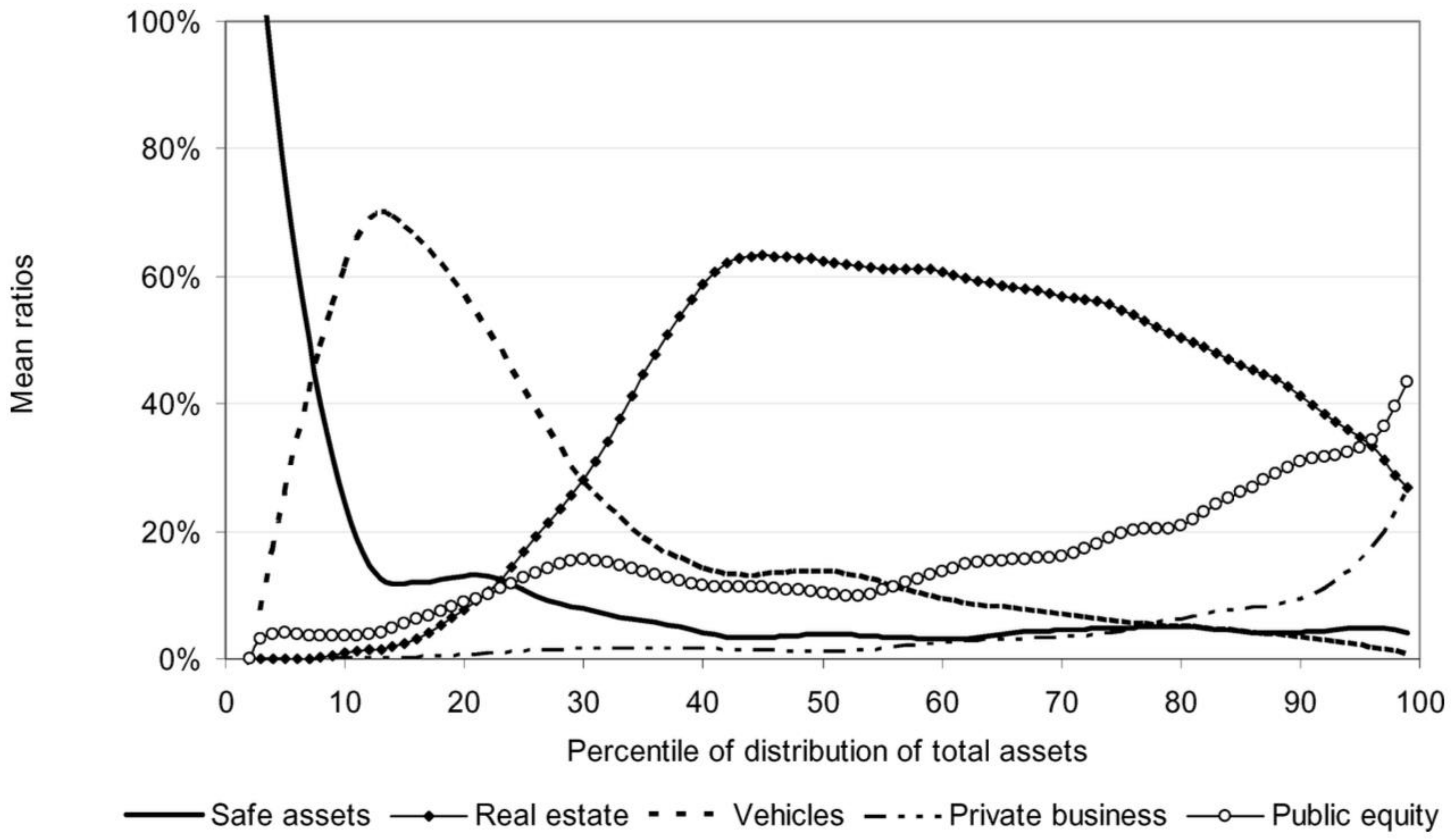
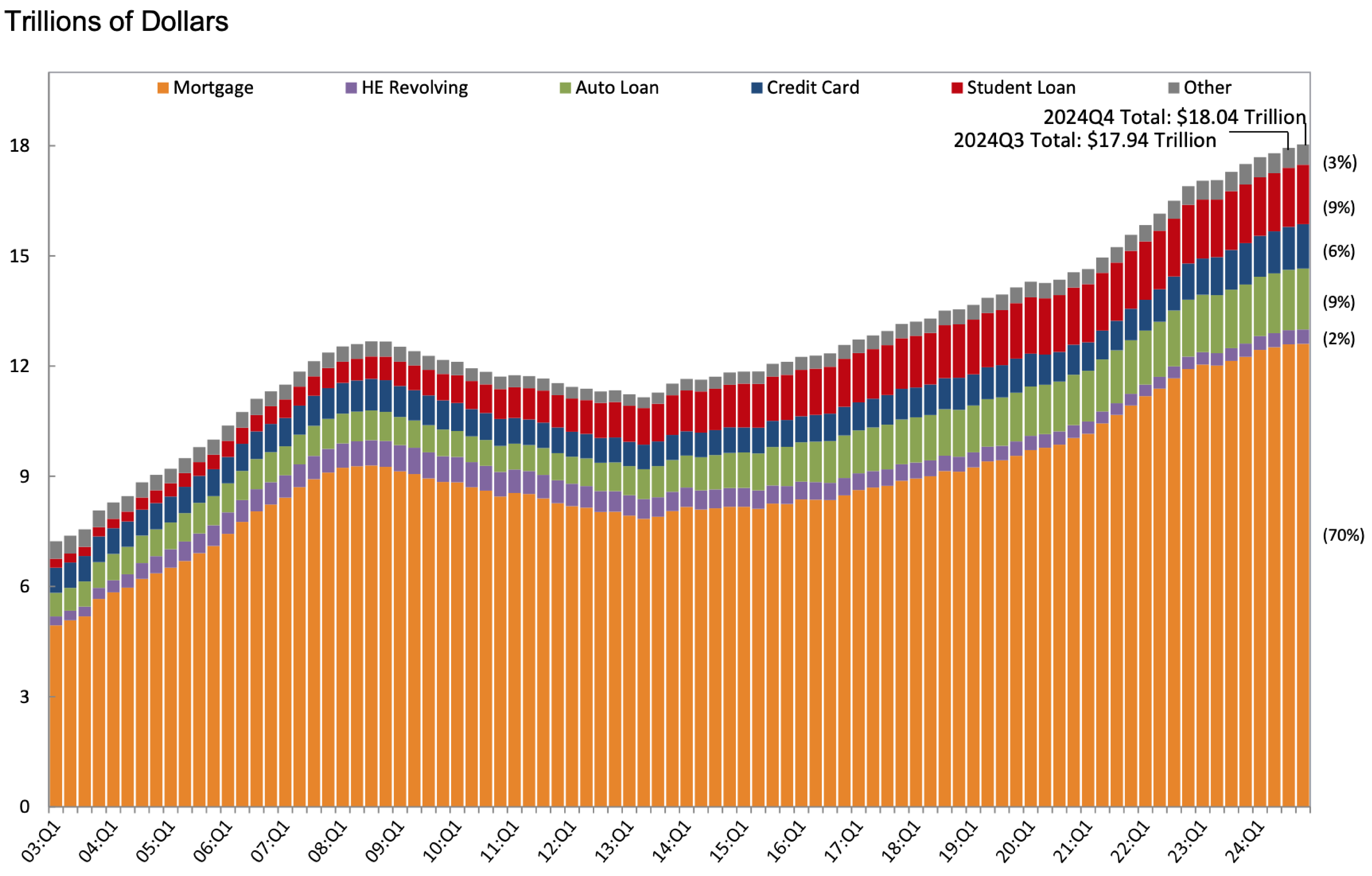


Figure 3. Asset class shares in household portfolios. The share of each asset class in the aggregate portfolio of households at each point in the wealth distribution, in the 2001 Survey of Consumer Finances.

U.S. Household Debt



Source: New York Fed Consumer Credit Panel/Equifax

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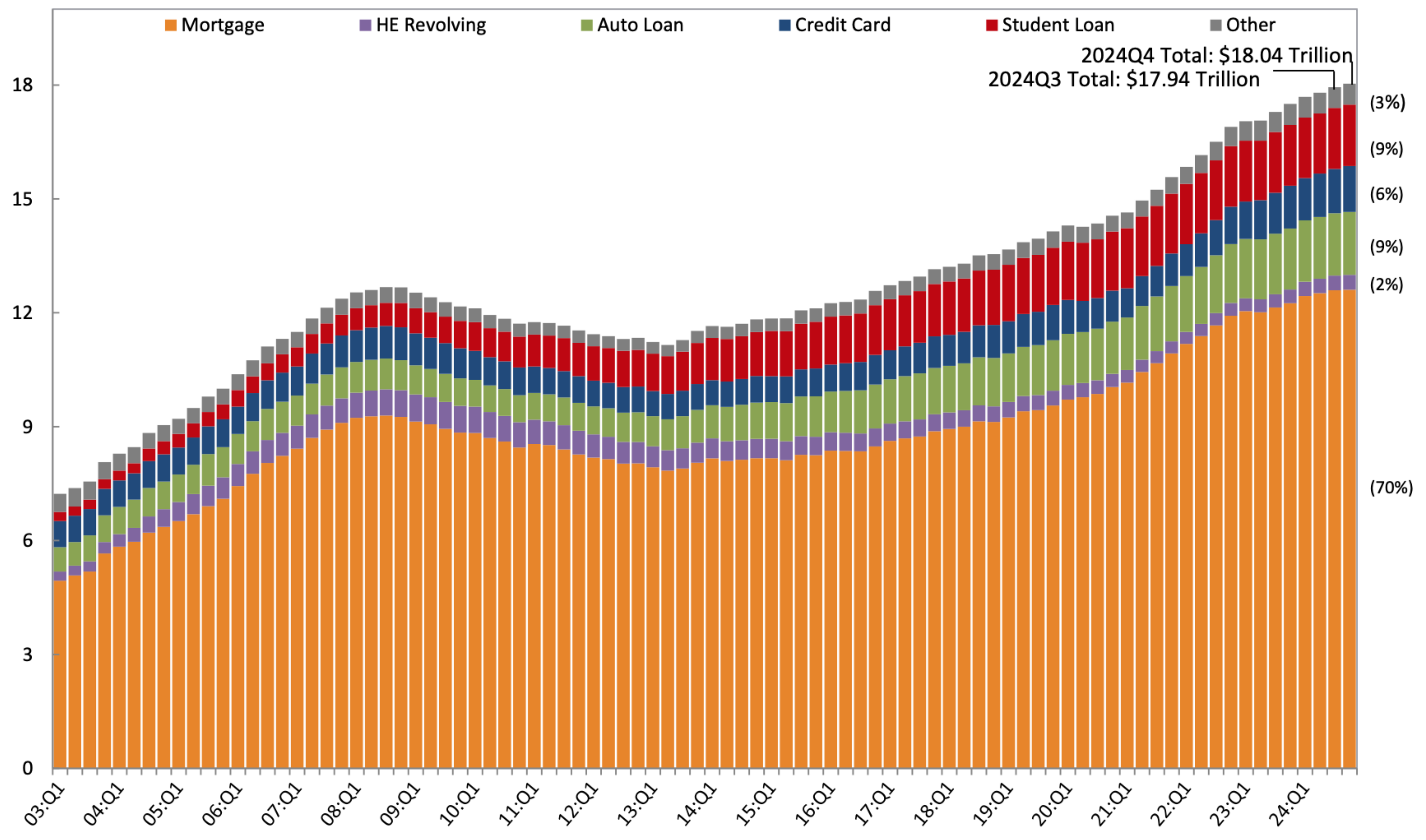
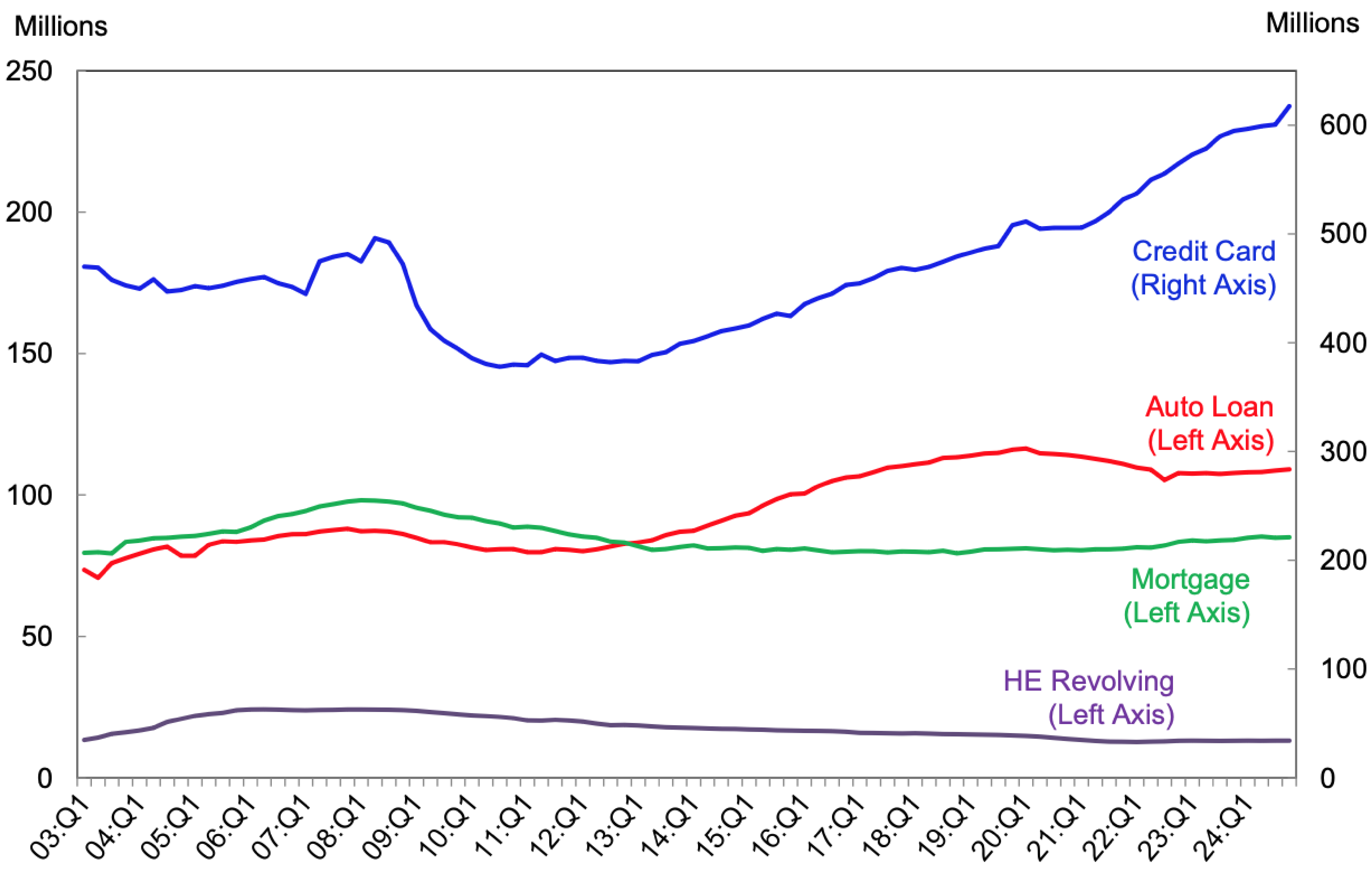
U.S. Household Debt



Source: New York Fed Consumer Credit Panel/Equifax



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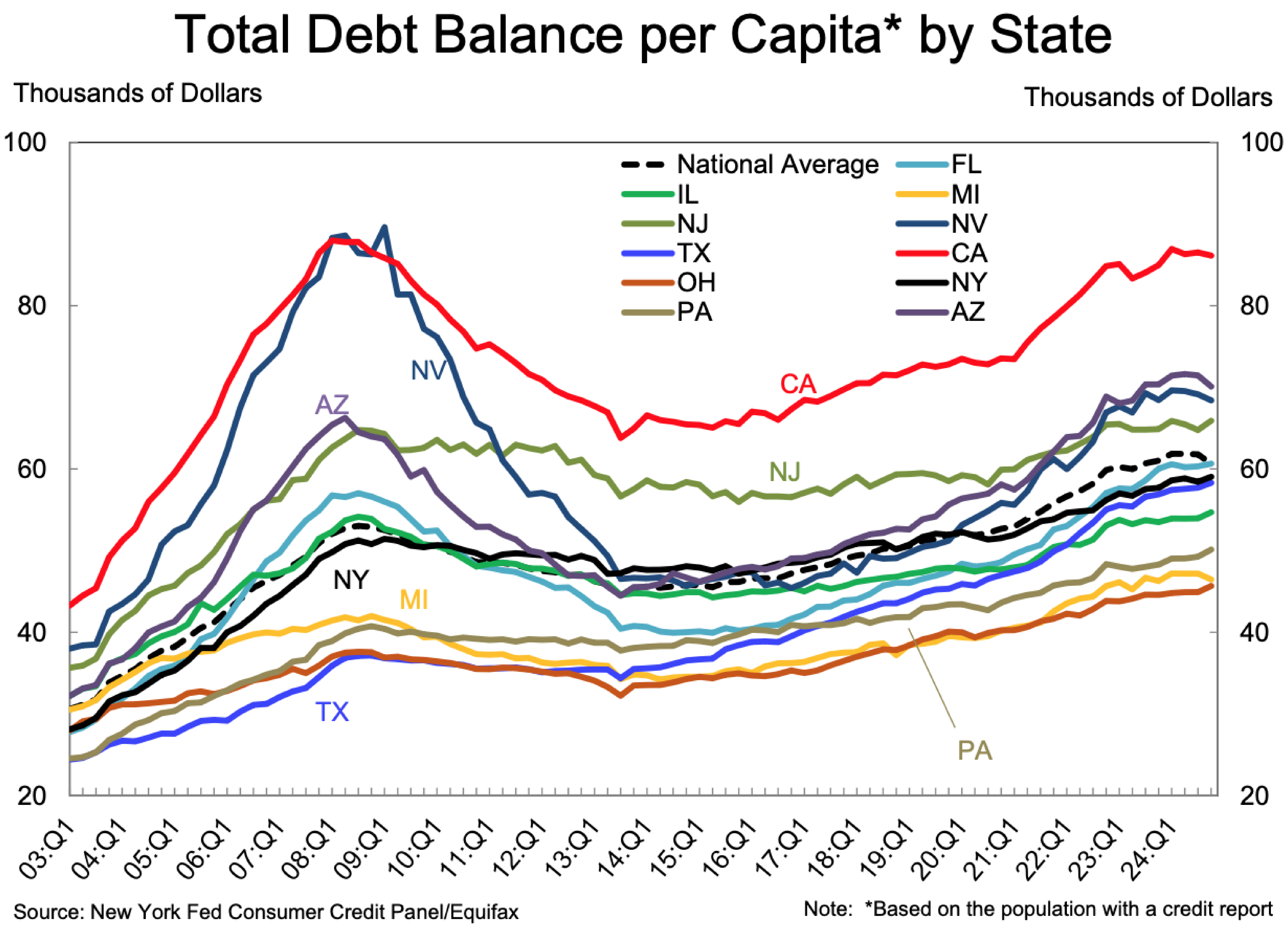
U.S. Household Debt for U.S. States



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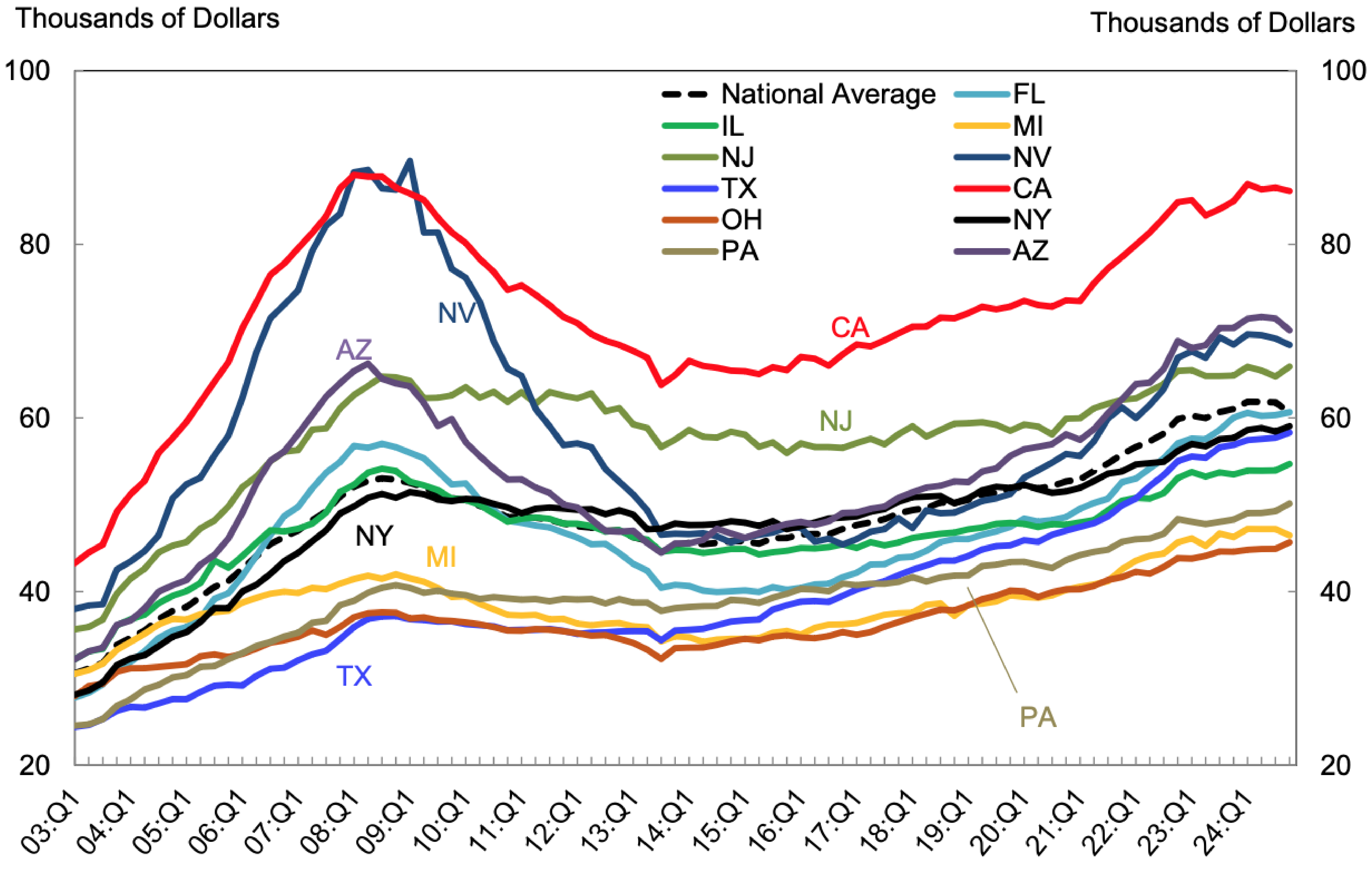
U.S. Household Debt for U.S. States

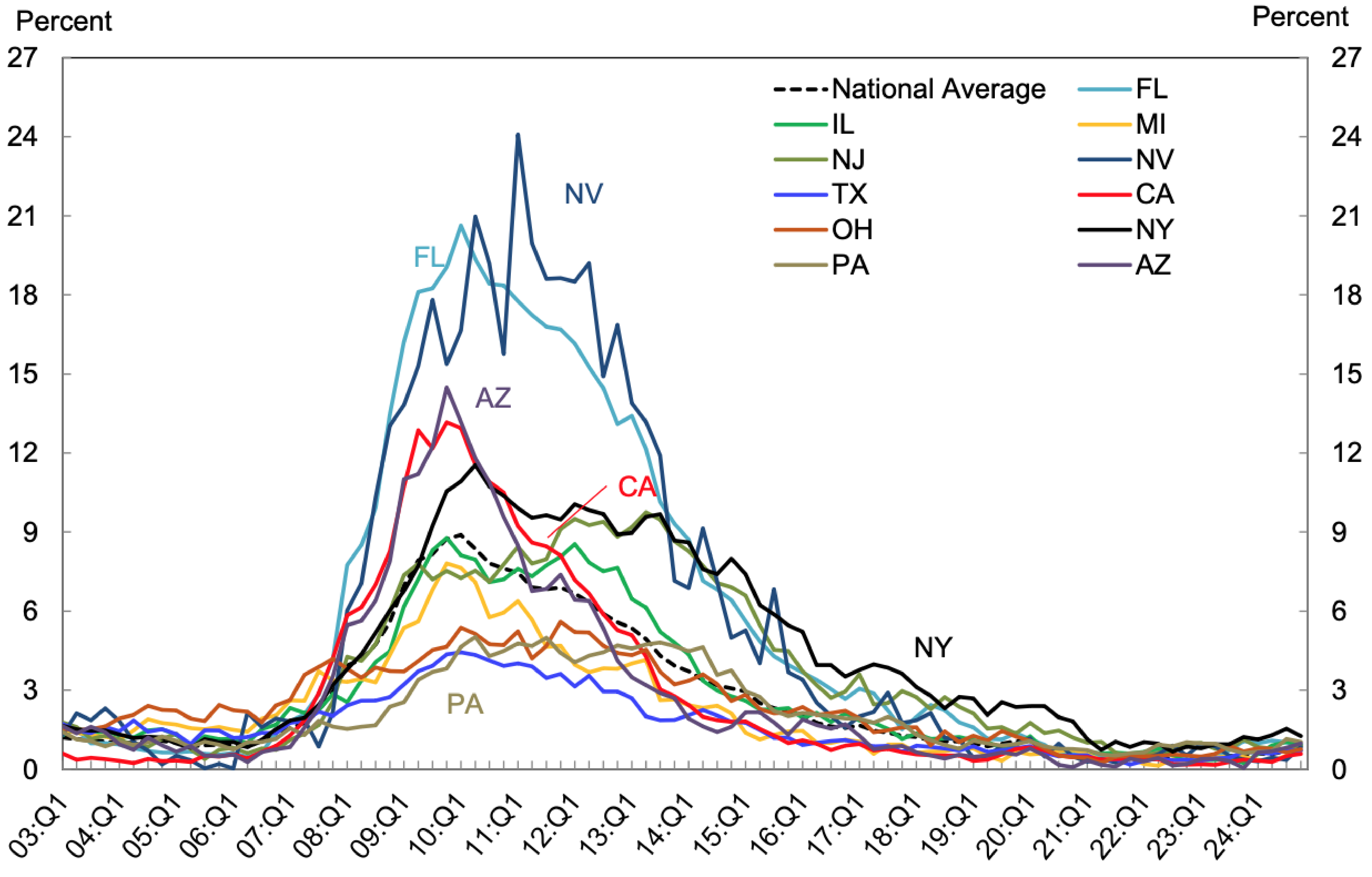
Total Debt Balance per Capita* by State



Source: New York Fed Consumer Credit Panel/Equifax

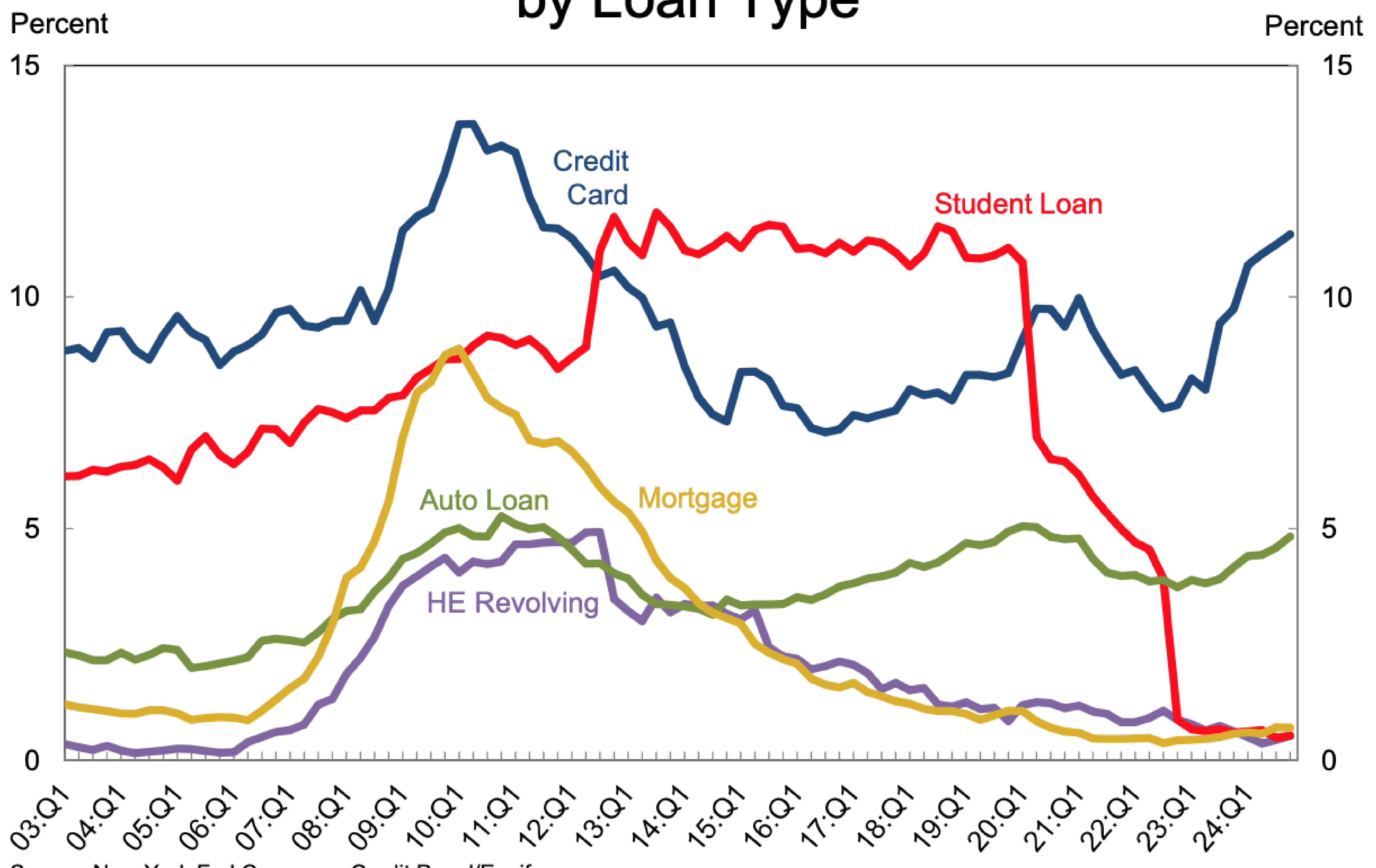
Note: *Based on the population with a credit report

Mortgage Delinquency (90+ Days Past Due)



Source: New York Fed Consumer Credit Panel/Equifax

U.S. Household Delinquencies



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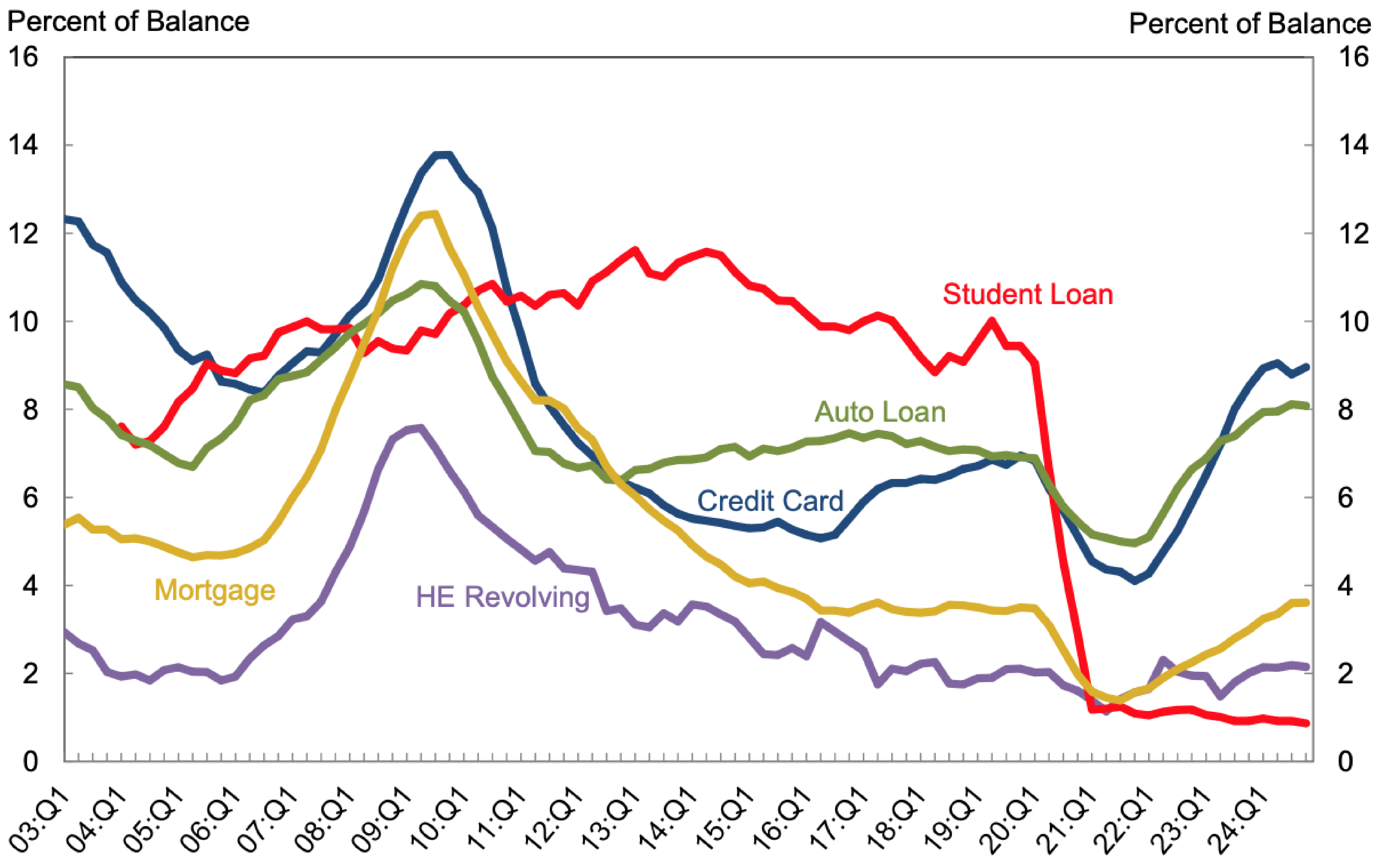

Source: New York Fed Consumer Credit Panel/Equifax

Note: 4 Quarter Moving Sum

Student loan data are not reported prior to 2004 due to uneven reporting

Economics of Borrowing

Why Borrow?

Borrowing trades-off some (“consumption”) benefit today for some costs (lower consumption as need to repay debt and interest) in the future.

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Why Borrow?

- Expecting more income/lower expenditures in the future
- Making an investment
- Making a lumpy purchase (auto, house, vacation)
- "Behavioral" reasons: Not thinking much/at all about the future

Should You Borrow $B today and Repay B(1+r) Next Year?

Next Year

Next year's income: $100,000 OR $51,000 OR $25,000

This Year

$50,000 + B

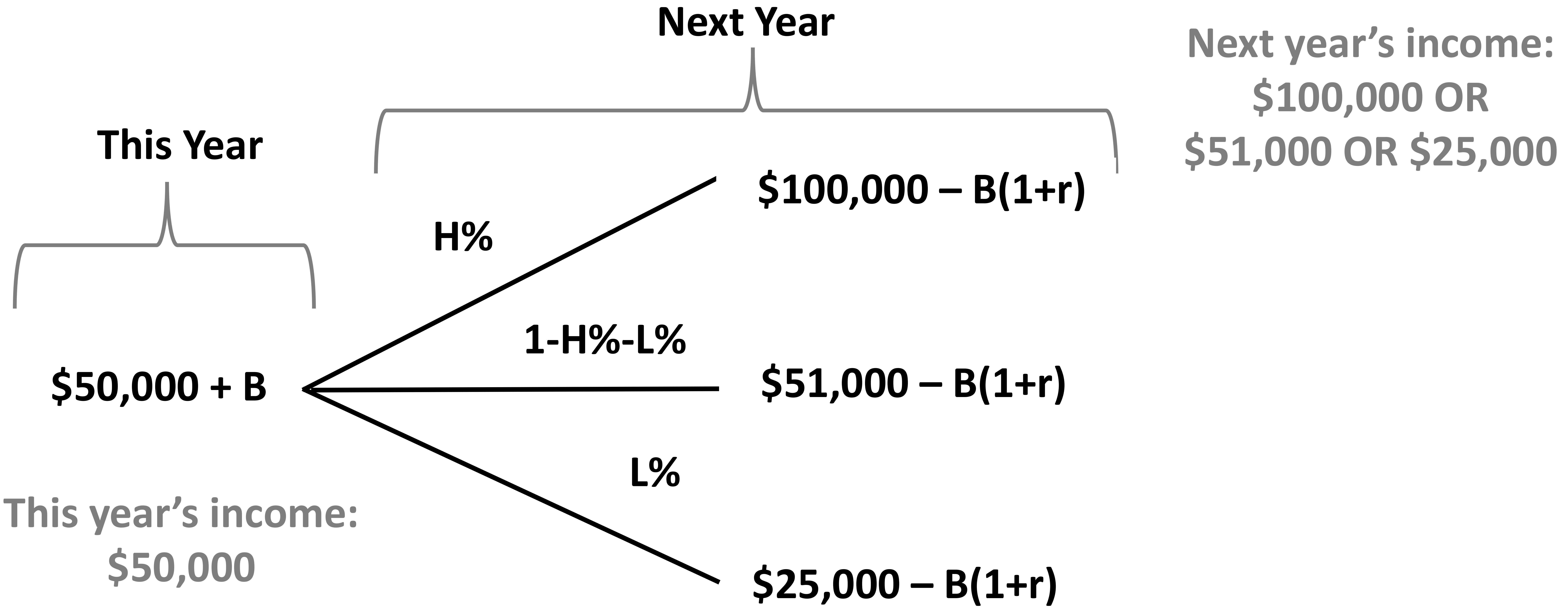
H% → **$100,000 – B(1+r)**

1-H%-L% → **$51,000 – B(1+r)**

L% → **$25,000 – B(1+r)**

This year's income: $50,000

Should You Borrow $B today and Repay B(1+r) Next Year?



$$\max_{B} u(50{,}000 + B) + \beta((H \times u(100{,}000 - B(1+r)) + ((1 - H - L) \times u(51{,}000 - B(1+r)) + (L \times u(25{,}000 - B(1+r)$$

What Do You Need to Know to Decide Borrowing?

$$\max_{B} u(50{,}000 + B) + \beta((H \times u(100{,}000 - B(1 + r)) + ((1 - H - L) \times u(51{,}000 - B(1 + r)) + (L \times u(25{,}000 - B(1 + r)$$

- How costly is borrowing? **r**
- What are the probability of income changes? **H%, L%**
- How much utility (enjoyment/happiness) extra $B today? **u(.)**
- How much utility from changes in income? **u(.)**
- How much do you discount utility in one year compared to utility today? **β**

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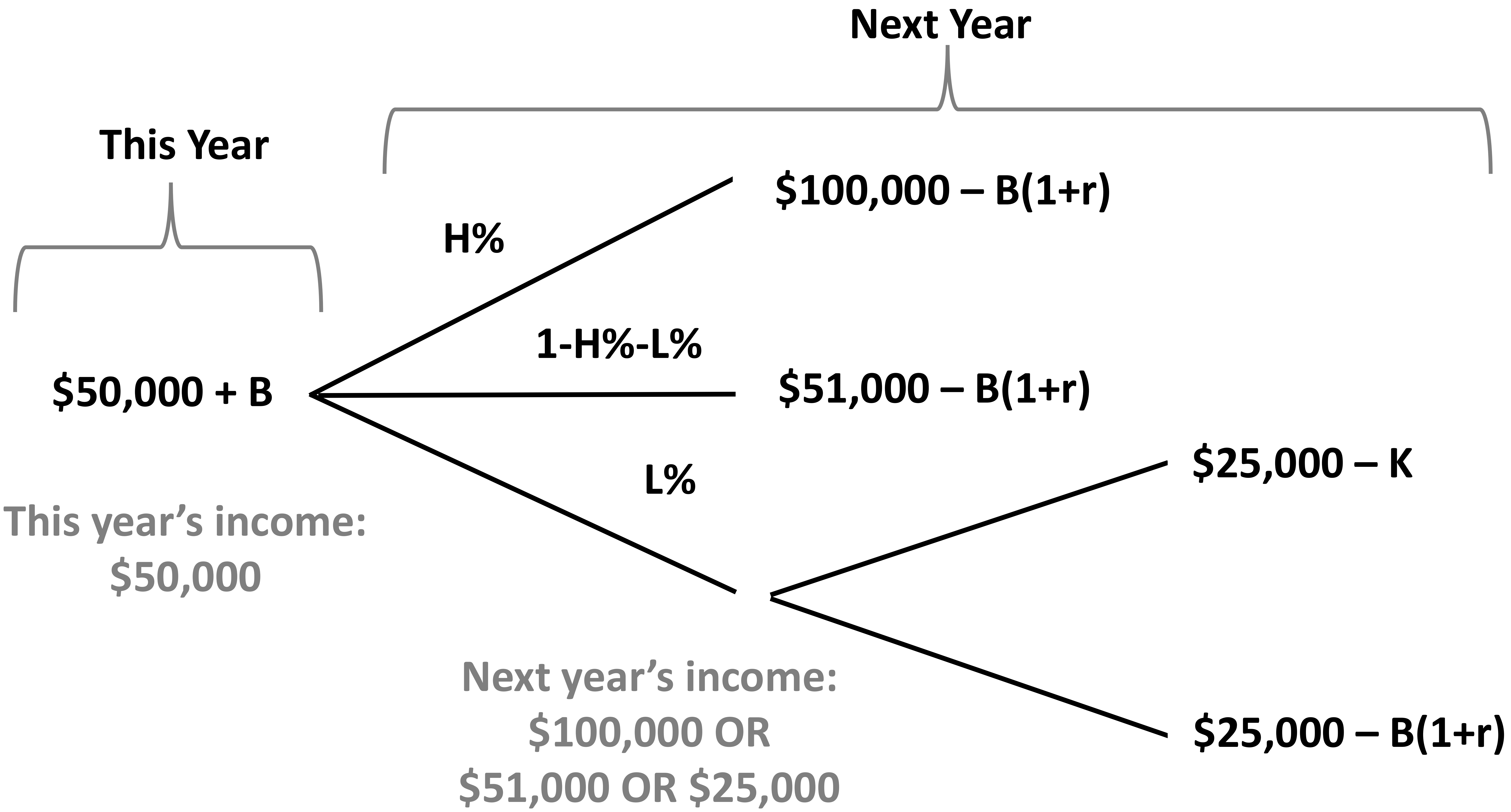
THIS IS SIMILAR TO DCF/NPV! What's different?

- **For NPV, our objective is maximize discounted cashflows ($).**
- **Whereas, consumers have different objectives: maximize utility!**
- **Discount rate for utility may differ from discount rate implied by borrowing**

Next Year Consumers Also Have A Real Option Not Repay Debt



What Do You Need to Know to Decide Borrowing & Repayment?

- How costly is borrowing? **r**
- What are the probability of income changes? **H%, L%**
- How much utility (enjoyment/happiness) extra $B today? **u(.)**
- How much utility from changes in income? **u(.)**
- How much do you discount utility in one year compared to utility today? **β**
- **How costly is defaulting? K**

- Credit histories are a way to encourage borrowers to repay debt.

- Credit scores used by lenders to predict probability of a consumer repaying, and to price that risk.

My View on Economics of Borrowing

- Clearly approximately no one is actually doing these calculations
- But these traditional economics models are still very useful & a reasonable approximation
- Consumers are sensitive to factors that the model predicts
  - if you ↑ the interest rate (r), fewer people borrow
  - if you make non-payment ("default") more costly (↑ K), more people repay loans
- Traditional economic models provide a benchmark to evaluate real-world behavior
- Traditional economic models fit some borrowing decisions better than others
  - My view is that considering both economics and psychology ("behavioral household finance") is essential to understand some borrowing decisions (e.g., credit cards)

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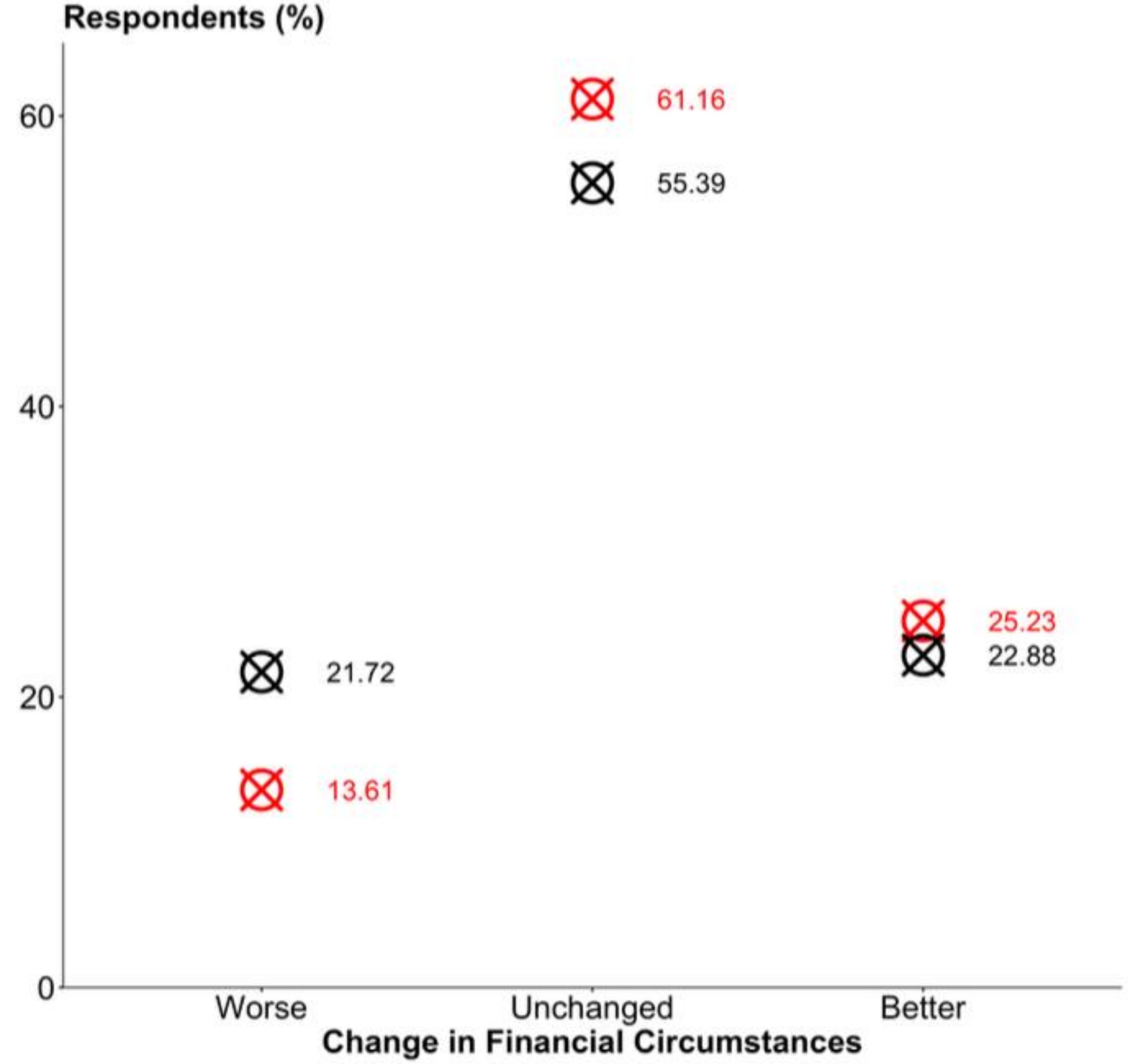
How Wise Is the British Crowd?

- Expected Change in Financial Circumstances

- Actual Change in Financial Circumstances

Overconfident expecations. Actual change symmetric: $\sim$ 22% better/worse



How Wise Is the British Crowd?
- Expected Change in Financial Circumstances
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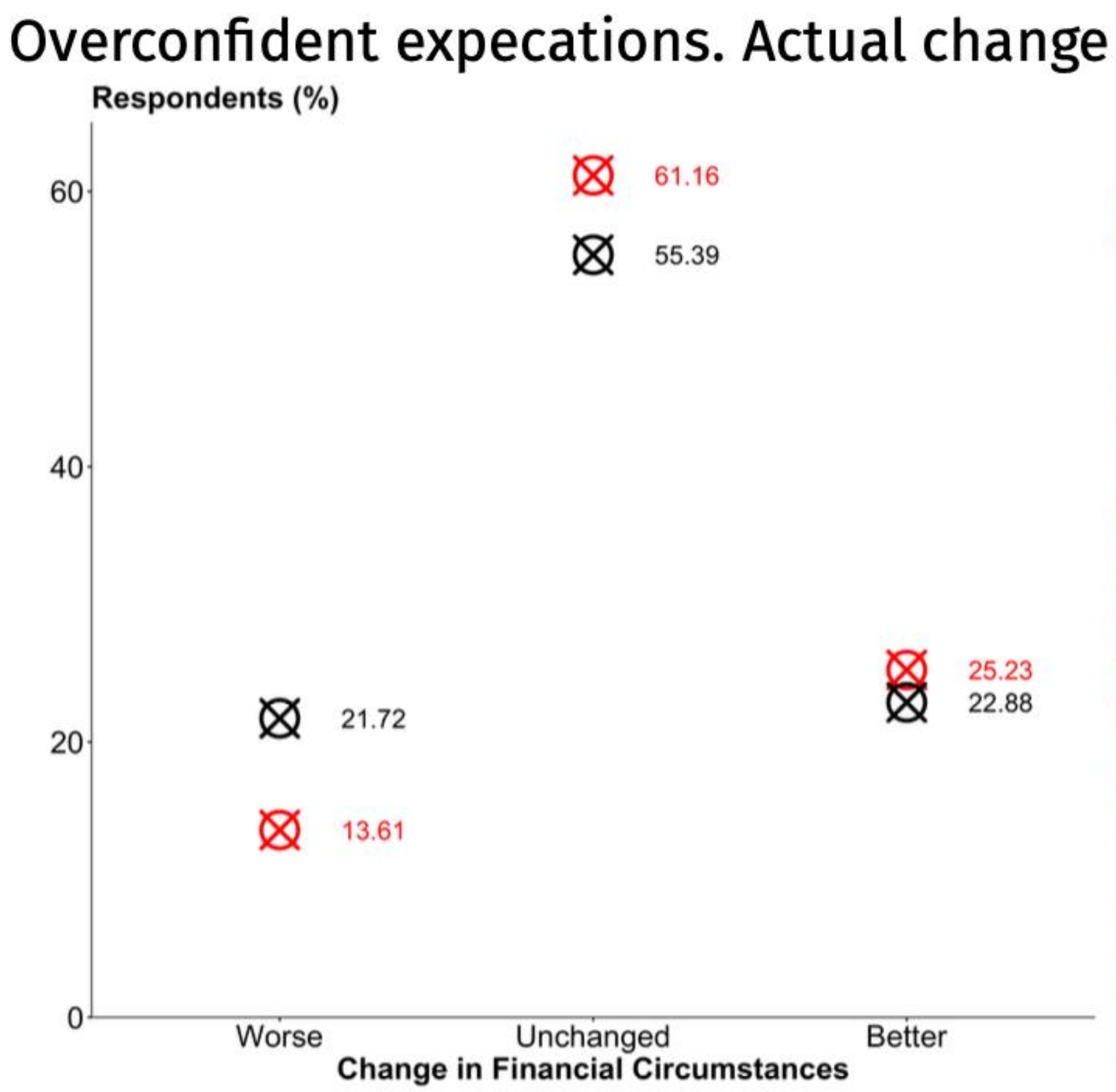

- Pattern is consistent with "loss-aversion"
- Our project is analyzing how these expectations relate to borrowing (measured in merged in linked credit bureau data)
- Role of predictable vs. unpredictable changes driving expectations & borrowing outcomes

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Retirement Savings

Should I Save for Retirement?

- Yes!

How Much to Save for Retirement?

- Try to maximize your contributions (10%+ not a bad rule-of-thumb) https://www.brookings.edu/articles/the-new-math-of-saving-for-retirement-may-boil-down-to-this-one-absurdly-simple-rule/
- Employer-matching is as close to free money as you'll ever get
- Need to have enough cash accessible (non-retirement funds) for emergencies / avoid persistent high-cost borrowing (credit cards, payday loans etc.)
- Lots of research on how employer's default retirement options affect consumer choices for how much to save and which products to save in.

What Portfolios to Save for Retirement?

- Low-fee (compounding really matters over long horizons!)
- Diversified portfolio (not just single-stocks but index)
- Lower exposure to equities as you get closer to retirement

  (exact equity share depends on your risk-preferences & your other assets)
- Target-date funds do all this automatically!

…there's a separate decision about how to draw down your funds in retirement, but you are all young so that's less relevant now!

Bank Runs

Why Do Bank Runs Happen? (Diamond & Dybvig, 1983)

- Bank deposits are more liquid than the loans they hold.
- This “maturity transforming” is part of banking!
- But it makes banks vulnerable

Why Do Bank Runs Happen? (Diamond & Dybvig, 1983)

- Bank deposits are more liquid than the loans they hold.
- This “maturity transforming” is part of banking!
- But it makes banks vulnerable
- Bank runs can be self-fulfilling. If everyone believes everyone else is going to withdraw their money, everyone will withdraw their money.
- If enough people want to withdraw their funds, the bank does not have enough liquid funds to satisfy them. The bank is illiquid NOT insolvent.
- Solutions? Government insure deposits against loss (FDIC insures up to $250,000 per depositor per insured-bank-account-category) or government lends to bank to help them survive a bank run (temporary liquidity).

Countrywide Financial

Countrywide Mortgage Book in 2007		
Pay-Option ARMs	28B	32%
Seconds	34B	39%
Total Book	87B	100%

Pay-Option Statistics		
Year	2007	2006
% No/Lite Docs	81%	81%
% Def Payment	71%	77%
% Delinquent	5.36%	0.63%

Countrywide Financial (August 2007)

- Large U.S. mortgage lender.
- Huge pre-crisis growth ("23,000%" stock because of returns 1982-2003!)
- Concern about its subprime exposure led to bank run in August 2007
- Bank failed and bought by Bank of America for $4bn in January 2008 (fire sale)
- Ultimately cost Bank of American $40+ bn due to write-downs and legal expenses
- Now called Bank of America Home Loans

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Silicon Valley Bank (SVB, March 2023)

- Lender focused on tech / venture capital
- Unusual in having many depositors above FDIC-insured limit (~80% uninsured)
- Invested deposits in government bonds but when interest rates rose sharply it shortfall.
- Largest bank run in history ~$42bn
- Bought by First Citizens Bank for ~$72bn